Nkambe Bakery & Recipe Handbook

Bread • pastry • savoury food • frozen desserts • juice & yogurt preservation

Integrated working edition | 3 October 2026 — expanded baguette workflow

Prepared from the accessible documents and bakery discussions dated or modified 4 September–3 October 2026.

Practical recipes, consistent methods and a measured route from pilot batch to daily production.

This is a development and operating handbook. Formulations, yields, equipment capabilities and commercial assumptions require the trials identified in each section. A recipe being fully written does not mean it has been physically tested or approved for sale.

The source and editorial record accompanies this handbook. It identifies the 27 discovered files, duplicate publications, retained variants, corrections, evidence limits and additional preservation references.

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1. How to use this handbook

Begin with the shared controls, select a small opening menu, and print the relevant recipe and component cards for the station. Use one approved version per product. Enter actual batch results before increasing volume.

Identifier	Meaning	How to use it
O-B01–O-B36	October shared bakery and savoury components	Make these before assembling the O-P and O-A products.
O-P01–O-P20; O-A01–O-A04	October finished products and four additions	Follow their component weights and service instructions.

Identifier	Meaning	How to use it
O-FBASE1–7; O-F01–8	October frozen bases and sale formats	Confirm machine suitability, treatment and cold chain first.
S-B/P/F/D/I/Y	September specialist recipes	Separate formulas, retained for their different product styles. S-F means fast food, whereas O-F means frozen.
T01	Your cake-donut cracking trial	A controlled test based on your actual formula.
V series	Earlier recipes and development variants	Preserve ideas and formula differences; resolve the stated gaps before release.

All principal formula quantities are grams, including milk, oil, eggs without shells, drained fish and prepared vegetables. The V-series preserves original litres, counts and ambiguous ingredients explicitly; these need weighing and specification before conversion. Only water may be approximated as 1 g/ml. A nominal egg count does not replace edible egg mass.

Baker's percentages use flour as 100%; other percentages use the stated component basis. Do not mix them. Scale every ingredient by the same factor, then verify equipment load, mixing, heat penetration and cooling afresh. Larger batches are not validated by arithmetic alone.

Raw piece mass, cooked component mass and cooled sale weight are different measurements. Modelled loss is an assumption. Whole pieces = floor(usable batch mass ÷ actual portion mass). Record leftovers, rejects and samples. Air in ice cream increases volume, not ingredient mass.

October cost examples remain illustrative FCFA assumptions. September workbook prices and operating costs are largely unfilled; a missing price is unknown, never zero. The recipe IDs retain their original suffix so the source workbook remains traceable.

Use customer feedback to choose sweetness, size and price. Keep accessible descriptions alongside familiar product names. Clearly identify wheat, egg, milk, fish, groundnut, sesame and other supplier-declared allergens; vegetarian, dairy-free and allergen-free claims require ingredient and cross-contact checks.

Completeness is relative to the sources found: 27 files across the project, bakery chat-linked working folders and relevant Downloads, plus the cake-donut discussion. Google Drive and Pages searches produced no additional in-window bakery documents. Creation dates were not always available; modification dates and documented conversation context were used. No access to every private folder or external account is claimed.

2. Shared preparation, cooking and storage controls

H01 — Hygiene, meat, eggs and controlled cooling

Use approved suppliers, potable water, cleanable surfaces and separate raw/cooked tools. Reject damaged tins, spoiled ingredients and unknown cold-chain lots. Store raw meat below and apart from ready-to-eat ingredients. Allocate different utensils and personnel tasks for raw poultry, cooked foods and bakery packing. Check flour, milk, eggs, fish, groundnut, sesame, mustard and sauce allergens; "may contain" statements do not replace cleaning or accurate supplier records [S34].

Use the common house cooking/reheating target of 75°C for 30 seconds for the meat, fish, poultry and cooked egg dishes in this handbook. Hold hot perishables at 63°C or above. These are operating targets informed by external guidance, not statements of Cameroonian law [N08]. Probe the coldest representative center with a calibrated sanitized thermometer; record time, temperature and corrective action. For this edition's internal service rule, hold hot perishables at 63°C or above and chilled ones at 4°C or below. These conservative operating targets are not presented as Cameroonian statutory limits.

For cooked perishables to be stored, use active shallow-pan cooling. FDA's reference benchmark is 57°C to 21°C within 2 hours and to 5°C or below within 6 hours total [S23, sections 3-501.14/.15]. The bakery's stricter final storage target is 4°C. Map the thickest, largest and slowest-cooling portions; an ordinary fridge filled with hot deep pots is not a validated cooler. If a limit fails or history is missing, quarantine and withhold sale; do not improvise repeated reheating as a rescue. Qualified local review must establish disposition.

At trial stage, sell hot assembled foods promptly or use a monitored hot/chilled process. Log cumulative time outside refrigeration during portioning and proofing; schedule immediate cooking. Use an internal maximum 2-hour ambient preparation allowance, reduced to 1 hour in rooms above 32°C, as a conservative action threshold —not a permission to leave every food out that long. Any product-specific formal time-as-control scheme needs its own approval and written procedure. If the filling cannot remain safe during proofing, reduce batches or defer it.

H02 — Dairy mix treatment and freezing

Use homogenized pasteurized/UHT milk and cream with certificates. Mixing powder into pasteurized milk can reintroduce contamination, so ingredient status alone does not validate the final mix. Obtain the actual STAB 2000 technical sheet and label. The proposed low-fat mixes use 0.40–0.45%, while the richer mix uses approximately 0.27%, within the supplier's differentiated range [S14]. A changed stabilizer requires a new formulation; do not substitute starch, xanthan or an unnamed premix at the same dose.

Premix stabilizer into sugar before dispersing, allow supplier-directed hydration, and use a qualified dairy treatment system. Guelph's batch reference is at least 69°C for 30 minutes throughout the mix [S09]. This edition does not translate continuous HTST seconds into an unvalidated pot process. A trained process specialist must establish the cold point, agitation, thermometer accuracy, actual hold, lid/condensate management and sanitation. Additional heating needed for supplier hydration must be specified in that approved program. If the pot cannot meet it, use a qualified pasteurizer or approved ready mix; do not sell experimental pot-treated dairy.

Cool promptly to 4°C or below using a mapped cooling method; do not leave a 10 kg mix on the bench. Age at 2–4°C only within the approved schedule. Treat post-process additions as ready-to-eat hazards. A stick blender is a dispersion tool, not a 50–200 bar homogenizer; trials must determine stability with the specified pre-homogenized dairy [S10].

Air increases ice-cream volume, not edible ingredient mass. Overrun = (mass of equal volume liquid mix ÷ mass of equal volume ice cream − 1) × 100. For a measured density of 1.08 kg/L and 30% overrun, 1 kg mix corresponds to about 1.20 L product before transfer losses; this is an illustrative calculation, not a machine result [S12]. Use actual density and mass portions for costing. Hardening and storage are separate capacities [S13]; do not put warm production into an overloaded holding freezer.

H03 — Fruit ice and infusion controls

Acidity, ginger, sugar and freezing do not sterilize fruit. Use ready-to-eat treated inputs or a qualified product-specific process. FDA's juice framework emphasizes pathogen reduction, not a universal kettle temperature for every fruit [S24]. Record pH, °Brix, fruit solids, maximum pulp size, treatment, cooling and sanitary filling. Supplier hydration guidance for Super Neutrose is not itself a pathogen-kill validation [S15]. If the treatment specification is missing, the formula is a laboratory/pilot development card only and cannot be commercially released.

Use a cleaned refractometer calibrated with water and temperature compensation. °Brix is a sucrose-equivalent refractive reading: acids and other soluble materials affect it. It is not an exact total-sugar assay. Hold fruit-source composition constant during the first trial. The old habit of writing "18–22°Brix" without calculating syrup/juice solids has been removed.

H04 — Yogurt fermentation and cooling

Yogurt uses defined lactic cultures, controlled heat treatment, incubation and refrigeration. Tetra Pak describes a 90–95°C, 5-minute milk treatment and pH-triggered cooling; the pot must be qualified for the real fortified milk [S18]. The trial starter is Yogourmet Original, provisionally 3 g per litre milk in the source formulas, containing S. thermophilus, L. bulgaricus and L. acidophilus [S19, S20]. The current manufacturer page specifies one sachet per litre and 42–44°C; verify the mass on the actual regional sachet before using the 6 g recipe entries. The manufacturer's domestic consumption window is not transferred to commercial fortified or fruit yogurt. This is an accessible named trial choice, not necessarily the least-cost commercial culture. Obtain supplier confirmation for added sugar/milk solids and planned production; switch to a commercial DVS culture only with its own dosing units and instructions.

Calibrate the pH meter with pH 4.00 and 7.00 buffers at the appropriate temperature each day. Record milk lot, culture lot, dose, inoculation temperature, readings during acidification and cooling. Start cooling at pH 4.5 as the trial endpoint; this alone does not validate safety. Establish an acceptable time-to-pH curve over repeated trials. If a batch misses the validated curve, shows gas or odor, or fails to acidify, quarantine it; do not taste or extend incubation indefinitely.

For small-cup trial engineering, aim to cool from incubation temperature to 20°C or below within 1 hour and to 4°C or below within 2 hours after endpoint. These are proposed stricter internal cooling targets to verify, not an industrial guarantee from a domestic refrigerator. Avoid disturbing set yogurt during gel formation. Stirred and drinking products need different controlled shear. No ambient shelf-stable yogurt is authorized by these recipes.

H05 — Formula, scaling and shelf-life conventions

All ingredient amounts are grams; eggs exclude shells and prepared vegetables exclude trim. Water is potable. Instant Red/Gold mean the specified instant dry type, not fresh or active dry yeast. A manufacturer-authorized conversion and a fresh trial are required when yeast type changes; no gram-for-gram interchange is allowed. Flour strength and absorption must be checked by supplier data and actual development, not by a T-number alone.

Each component table states pilot mass, baker's percent if flour is present (flour = 100), otherwise percent of that component, and a production multiplier. Dough hydration estimates include milk/egg/butter water, using declared development assumptions. For multicomponent products, filling percentages are component

percentages, not dough baker's percentages. Frozen and yogurt cards also report whole-product mix analysis where meaningful.

Mass reconciliation is explicit: component input minus handling/unretained allowance minus process loss equals retained output. Process loss applies to post-handling material. Divide by stated count; scale ingredients, losses and count by the same multiplier. Fried oil absorption and shell coatings are separate positive ingredient components. Oil-bath charge/replacement, pan grease, release aids and packaging are not silently included in edible net weight: record and cost them as process consumables/packaging. No remainder is hidden; theoretical component outputs are allocated equally across the count. Round individual weights only after choosing a production specification, and record any remainder or short count.

Loss percentages are planning hypotheses, not measurements. For cooking with fat drain and vapor, the initial combined process-loss estimate must be separated in trials by weighing drains and pre/post-cook batches. Evaporation during covered dairy treatment is initially modeled as zero; weigh before and after and update if material. Do not add untreated water after pasteurization to recover a target mass. Packaging mass is separate from edible net weight.

Every card requires first-off inspection plus at least ten representative raw/finished weights when batch count permits (all pieces in smaller pilots). Run three repeat batches before approving a working formula. Check retained plain bread/sweet products at 0, 6, 24 and 48 hours for quality; biscuits at days 0, 3, 7 and 14 with water-activity/packaging and microbiology review; refrigerated dairy at days 0, 1, 3 and 7 under a laboratory-agreed protocol; frozen products at days 0, 7, 14 and 28 with temperature records. These are study points, NOT approved sale-by periods. Spoiled or microbiologically suspect samples must never be tasted. Shelf-life and any safety-related labeling require product-specific assessment.

Commercial release and preservation

The study days in H05 are sampling times, not shelf lives. Neither a same-day policy nor a seven-day trial window makes an unvalidated process safe. Until the product-specific treatment, sanitation, storage and release procedure is established, use development batches for controlled evaluation only. Approved ready-to-use treated ingredients may simplify commissioning.

Do not use the dairy pot for raw meat. Use dedicated raw-food containers and tools, or a documented sanitation changeover validated for the equipment. A scraper is not automatically a high-shear blender, aerator, dough mixer or pasteurizer. Follow the equipment's actual working capacity and operating instructions.

Cooling benchmarks apply to cooked perishables; yogurt has its own endpoint and cooling plan. When this handbook refers to H01–H05 or F0, apply H01–H05 here and the frozen-control section. The common cold storage target is 4°C or below, with dairy aging at 2–4°C and frozen stock at −18°C or colder. The FDA reference cooling endpoint of 5°C is retained as a benchmark, followed by the house 4°C storage target.

3. Equipment, capacity and the opening menu

Item	Owner-confirmed information	Still needed
Rotary oven	16 trays; 40 × 60 cm; diesel heating; reported 3 kW electrical	Model, diesel litres/hour, maximum temperature, rack load, clearances and actual bake uniformity
Production power	Reported 20 kW inverter; 30 kWh battery storage	Continuous AC rating, phase distribution, surge rating, usable battery fraction, discharge current and PV panel kWp

Item	Owner-confirmed information	Still needed
Sales power	Reported 15 kW inverter; 30 kWh battery storage	Same details; input charging limits are not necessarily output limits
Mixing/forming	Spiral 6.5 kW; moulder 1 kW; mixer 2 kW; professional mixing pot	Minimum/maximum dough mass; pot heating/agitation specification
Heat appliances	Proofers 1 kW; fryers 5 kW; cookers 5 kW; shawarma 3.5 kW	Number of units per rating, electrical phase, fryer litres and recovery
Ice cream	Two machines: 4 kW and 1.5 kW; described as soft serve and gravity	Model and feed method of each; approved mixes, hopper cooling, minimum charge and cleaning cycle
Popsicles	Two mould sizes; one is twice the other	Working fill volumes, cavities, machine power, cycle time and brine specification
Cold chain	Owner reports substantial refrigeration/freezing at both sites	Measured pull-down capacity, temperature logs, space, connected load and backup runtime
Other	Bread slicer available	Slicer power, blender/hot plate availability, scales, probes, insulated transport equipment

“Gravity” and “soft serve” are not mutually exclusive machine categories. A manufacturer’s catalogue explicitly lists gravity-fed soft-serve machines. The plan uses one conditional soft-serve trial base; it does not infer that the second machine is a batch freezer. Obtain model plates before choosing settings or loading homemade mix. Carpigiani soft serve categories

The new 30 kWh battery figure is recorded as nominal energy at each site. Solar array capacity remains unknown. The oven’s diesel supplies the heat: count both its electrical auxiliaries and diesel consumption. The electrical rating alone is not its full energy cost.

Batch and tray capacity

A 40 × 60 cm tray has 0.24 m^2 of surface; sixteen trays have 3.84 m^2 before spacing and usable-area losses. That does not specify how many tins fit, the permissible rack weight, dough-mixer capacity or hourly output. Draw a tray layout with the actual tins. A trial of 12 milk buns per tray would mean 192 buns per rack, but it is an illustration, not a rated capacity. Do not fill all sixteen trays until airflow and bake uniformity have been checked.

The recipe library gives pilot quantities and a x5 planning production quantity. No mixer bowl or minimum machine charge has been verified. Use a smaller mixer or hand pilot if the spiral cannot mix the pilot load safely. Production quantities are for purchasing and scale-up trials, not permission to overload equipment. The professional mixing pot is not assumed to be a validated pasteurizer.

Weights and formula conventions

All formulation quantities are grams, including liquids, to avoid density errors. Water alone may be approximated as 1 g/ml. Milk, oil, condensed milk and fruit puree must be weighed. An egg is provisionally 50 g edible contents; use the weighed mass, not a fixed count. Vegetable prices are per peeled/trimmed edible kilogram; meat is boneless purchase weight; canned fish is drained weight. Convert supplier gross-pack prices before use.

Yields are trial estimates after stated losses. Cost cooked components by saleable output. Frying oil entries cover usage/disposal, not the initial fryer fill; budget the initial fill separately as working stock.

Build the opening business around six dependable core products: family bread, condensed milk buns, sugar doughnuts, puff puff, fish rolls and beef meat pies. Add chicken shawarma only when its service station and raw-meat workflow pass a live trial. Start the frozen counter with one approved soft-serve base and small milk pops; add the larger pops only after customers accept the price and freezer throughput is measured.

This is a commercial hypothesis, not a forecast of Nkambe demand. No current Nkambe customer survey, supplier quotations, competitor-weight audit, footfall count or sales history was supplied. The menu prices below are test points, not observed local market prices. Product margins use the same explicit illustrative ingredient-price list throughout.

There is a documented local sweet potato processing and bakery initiative. CLGF reported the Nkambe project in September 2025; that supports exploring a local supplier and product story, but does not establish today's supply, consumer demand or the performance of this report's 15% sweet potato formula. CLGF project report

Rank	Opening priority	Decision condition
1	O-P01 Family bread	Blind comparison against locally purchased loaves at matched weight
2	O-P02 Milk buns	Target customers accept price after tasting
3	O-P04 Sugar doughnuts	Two consecutive batches meet oil, yield and texture checks
4	O-P06 Fish rolls	Chilled filling, cooking and transport logs pass
5	O-P07 Beef pies	Filling portion and cost remain controlled
6	O-P05 Puff puff	Sell in small waves; labour and oil do not erase contribution
7	O-P13 Chicken shawarma	Only after service drill and safe cooking verification
8	O-F01 Soft serve or O-F05 milk pop	Choose one first, based on manufacturer approval and cold-chain drill

Make O-P03 sweet potato bread a small paid test, not the opening volume anchor. Keep pizza, beef shawarma, burgers, fried-chicken meals and dairy cake for later phases; they add value but also create different queues, allergens and temperature-control demands. The full 20-product library is a development range, not a recommended opening-day menu.

Decisions carried forward from earlier documents

The older divider, planetary mixer, slicer and sheeter proposals remain conditional on measured bottlenecks. The latest inventory says a slicer is already available. Confirm its condition before purchasing another. A planetary mixer supports cake creaming and pastry work; a sheeter supports lamination; neither automatically has the best return. Request working loads, minimum loads, phase, voltage, rated and starting power, spare parts, manuals, warranty and a test with your actual product. Quote delivered cost and maintenance, not purchase price alone.

Older records describe the spiral as 120 L / 50 kg flour and 2.8/5.6 kW; the October plan gives 6.5 kW. Treat these as unresolved nameplate differences. Do not size wiring, inverter or batch mass from the conflicting claims. A VFD is not a universal solution for arbitrary three-phase machinery; have compatibility checked for the exact motor and supply.

At five baguettes per tray, a 16-tray rack holds 80, not 101. The earlier 18 kg flour, 11.88 kg combined water/ice, 360 g salt and 90 g yeast batch totals 30.33 kg: 101 × 300 g before losses, requiring more than one 80-piece rack. A 300 g piece with 15% bake loss weighs 255 g, not 250 g. Starting mixes every 45 minutes cannot

continuously fill an oven cycling every 25 minutes; measure the whole line and schedule around its slowest stage.

Keep ice inside the total formula water. Choose chilled water and ice from measured dough temperature and friction, not a fixed ice percentage. Protect hot bread from road dust using clean ventilated food-grade transport; cool before sealing. Do not apply raw egg after baking. Direct combustion exhaust outdoors through a compliant installation, away from food and customers.

Offer sliced bread, household loaves and breakfast bundles as paid tests. Agege and tangzhong breads remain development opportunities. A soft crumb is a quality attribute, not proof of microbial shelf life. Avoid promises of free solar energy, guaranteed premium prices, instant market dominance or automatic profit from residual oven heat.

4. Shared bakery and savoury components

Sweetened condensed milk supplies added sugar, milk solids and water. It is different from unsweetened evaporated milk and dry milk powder. For development calculations only, assume 45% added sucrose, 27% water and 28% milk solids including about 8% fat; these fractions sum to 100%. Naturally occurring lactose is inside the milk-solids fraction, so do not add total label sugars a second time.

Check the purchased label. One manufacturer example reports 55.9 g total sugars and 8.1 g fat per 100 g; total sugars are not the same as added sucrose. No brand specification is assumed to apply to your local tin. Nestlé example product specification

In O-B02, 150 g condensed milk adds approximately 67.5 g sucrose, 40.5 g water and 42 g milk solids, in addition to the separately listed 80 g sugar and 430 g water. The trial substitution on O-P02 approximates those functions, not identical flavour. For cost: 150 g condensed milk at 3,000 FCFA/kg costs 450 FCFA; 68 g sugar + 42 g whole milk powder + 40 g water costs about 244 FCFA, saving roughly 206 FCFA per O-B02 pilot, before any quality/yield change. Test both blind.

Use condensed milk selectively in buns, banana cake, soaked cake and frozen dairy products. Keep O-P01 bread, plain puff puff and most savouries free of it in the baseline to protect entry-level cost. Added dairy glaze changes the holding requirements even if the underlying bread is normally sold at ambient temperature.

Use these component formulas with their linked finished-product cards. Costs are illustrative ingredient costs only. Each card’s production column is a planning scale, not an approved machine load. Portions and additional egg wash, coatings, frying oil and packaging appear on the finished-product cards.

O-B01 Family bread dough

Pilot input 1,715 g; target usable output 1,700 g. Planning production batch is x5: input 8,575 g and usable output 8,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	1,000	5,000	100.00%
Water	600	3,000	60.00%
Sugar	50	250	5.00%

Ingredient or prepared component	Pilot g	Production g	Baker %
Margarine	35	175	3.50%
Salt	18	90	1.80%
Yeast	12	60	1.20%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Weigh ingredients. Reserve 30 g of the listed water; add only if the flour absorbs it. The stated yield assumes all 600 g is used.
2. Mix flour, water, sugar and yeast about 3 minutes at low speed. Add salt, then mix 4–7 minutes at the mixer's dough-development speed. Add margarine during the final 2 minutes. Stop at a thin elastic window and a dough temperature near 26–28°C.
3. Rest covered 15 minutes. Divide, pre-round, rest 10 minutes, then mould and tin as specified on the product card.
4. Trial proof at 30–35°C with enough humidity to prevent skinning, approximately 45–75 minutes. Bake when a gentle indentation returns slowly; do not use the clock alone.

Yield and process note: Input 1,715 g; allow 15 g bowl/handling loss. No improver assumed. If one is used, follow its label and re-cost.

Illustrative ingredient cost: 835 FCFA per pilot batch; 491.3 FCFA/kg usable output. This excludes packaging and operating costs.

O-B02 Condensed milk bun dough

Pilot input 1,895 g; target usable output 1,870 g. Planning production batch is x5: input 9,475 g and usable output 9,350 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	1,000	5,000	100.00%
Water	430	2,150	43.00%
Condensed milk	150	750	15.00%
Sugar	80	400	8.00%
Egg	100	500	10.00%
Margarine	80	400	8.00%
Whole milk powder	20	100	2.00%
Salt	16	80	1.60%
Yeast	15	75	1.50%
Vanilla	4	20	0.40%

Baker’s percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Dissolve condensed milk in water. Mix with flour, sugar, milk powder, yeast and egg on low for 3 minutes; reserve 25 g water until dough consistency is clear.
2. Add salt, develop 4–6 minutes, then add margarine gradually and mix another 2–4 minutes to an elastic film. Target 26–28°C dough; stop before overheating.
3. Rest covered 15 minutes. Portion and round with a tight surface; use the product card for bun weights.
4. Proof around 30–35°C for approximately 50–80 minutes until light and a gentle indentation returns slowly.

Yield and process note: Input 1,895 g; 25 g handling allowance. Egg wash is listed separately on the product card.

Illustrative ingredient cost: 1,765 FCFA per pilot batch; 943.6 FCFA/kg usable output. This excludes packaging and operating costs.

O-B03 Sweet potato milk bread dough

Pilot input 1,812 g; target usable output 1,790 g. Planning production batch is x5: input 9,060 g and usable output 8,950 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	850	4,250	85.00%
Sweet potato flour	150	750	15.00%
Water	570	2,850	57.00%
Condensed milk	100	500	10.00%
Sugar	40	200	4.00%
Margarine	50	250	5.00%
Whole milk powder	20	100	2.00%
Salt	18	90	1.80%
Yeast	14	70	1.40%

Baker’s percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Blend the two flours thoroughly; baker’s percentages use their combined 1,000 g. Reserve 40 g water for absorption adjustment.
2. Mix all except margarine for 3 minutes slowly, then develop 5–8 minutes. Add margarine and mix until extensible. Expect a less strong window than wheat-only bread; excessive mixing will not restore absent gluten.

3. Rest 15 minutes, divide and mould gently. Proof around 30–33°C to a resilient loaf, stopping before the surface weakens.
4. Use the product card for baking. Record actual loaf volume, sweetness and day-of-sale softness before increasing sweet potato flour.

Yield and process note: Input 1,812 g; 22 g handling allowance. Start at 15% sweet potato flour, not an untested high replacement ratio.

Illustrative ingredient cost: 1,309 FCFA per pilot batch; 731.0 FCFA/kg usable output. This excludes packaging and operating costs.

O-B04 Yeast doughnut dough

Pilot input 1,808 g; target usable output 1,780 g. Planning production batch is x5: input 9,040 g and usable output 8,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	1,000	5,000	100.00%
Water	440	2,200	44.00%
Egg	100	500	10.00%
Sugar	120	600	12.00%
Margarine	80	400	8.00%
Whole milk powder	30	150	3.00%
Salt	15	75	1.50%
Yeast	18	90	1.80%
Nutmeg	2	10	0.20%
Vanilla	3	15	0.30%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, egg, sugar, milk powder and yeast 3 minutes slowly. Add salt and mix 4–6 minutes; finish by incorporating margarine, nutmeg and vanilla.
2. Stop at elastic dough near 26–28°C. Rest covered 15 minutes; weigh individual pieces instead of repeatedly rerolling scraps.
3. Round or form rings; proof around 30°C for approximately 40–65 minutes until light, not fragile. Uncover only briefly to make transfer manageable.
4. Fry as specified on the product card; use a thermometer and allow oil temperature to recover between loads.

Yield and process note: Input 1,808 g; 28 g handling allowance. No filling is included.

Illustrative ingredient cost: 1,417 FCFA per pilot batch; 796.2 FCFA/kg usable output. This excludes packaging and operating costs.

O-B05 Puff puff batter

Pilot input 1,846 g; target usable output 1,830 g. Planning production batch is x5: input 9,230 g and usable output 9,150 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	1,000	5,000	100.00%
Water	700	3,500	70.00%
Sugar	120	600	12.00%
Salt	12	60	1.20%
Yeast	12	60	1.20%
Nutmeg	2	10	0.20%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix dry ingredients, then add water and beat 4–6 minutes to a cohesive sticky batter without dry flour.
2. Cover and ferment near 26–30°C for approximately 45–75 minutes, until bubbly and substantially expanded. Stir once gently if bubbles are very uneven.
3. Use a calibrated scoop to deliver 30 g portions. Trial fry one piece before loading the fryer.
4. Fry at 165–175°C, turning for even colour and checking the centre is set, typically 4–6 minutes depending on size.

Yield and process note: Input 1,846 g; 16 g bowl loss. Fermentation time is conditional on temperature.

Illustrative ingredient cost: 846 FCFA per pilot batch; 462.5 FCFA/kg usable output. This excludes packaging and operating costs.

O-B06 Fish roll dough

Pilot input 1,679 g; target usable output 1,660 g. Planning production batch is x5: input 8,395 g and usable output 8,300 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	1,000	5,000	100.00%
Water	470	2,350	47.00%
Margarine	100	500	10.00%
Sugar	60	300	6.00%
Whole milk powder	20	100	2.00%

Ingredient or prepared component	Pilot g	Production g	Baker %
Salt	15	75	1.50%
Yeast	12	60	1.20%
Nutmeg	2	10	0.20%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, sugar, powder, yeast and water on low for 3 minutes. Add salt, margarine and nutmeg; mix another 4–6 minutes until smooth, with moderate gluten development.
2. Rest covered 20 minutes. Divide and roll as specified, without making the wrapper excessively thick.
3. Use fully chilled filling; seal clean dough edges firmly, with no filling trapped in the seam.
4. Rest assembled rolls 15–25 minutes before frying. They should be slightly aerated rather than fully doubled.

Yield and process note: Input 1,679 g; 19 g handling loss.

Illustrative ingredient cost: 1,068 FCFA per pilot batch; 643.6 FCFA/kg usable output. This excludes packaging and operating costs.

O-B07 Shortcrust pie pastry

Pilot input 1,773 g; target usable output 1,740 g. Planning production batch is x5: input 8,865 g and usable output 8,700 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	1,000	5,000	100.00%
Margarine	400	2,000	40.00%
Water	290	1,450	29.00%
Egg	60	300	6.00%
Salt	15	75	1.50%
Baking powder	8	40	0.80%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Keep fat and water cool. Mix flour, salt and baking powder; rub in margarine to coarse crumbs.
2. Add egg and most of the water. Fold just until cohesive; add the remainder only if needed. Do not develop a bread-like gluten network.
3. Cover and refrigerate 30–45 minutes. Roll to about 3 mm on a lightly floured surface.
4. Weigh cut portions, use chilled filling and seal. Rework clean trimmings only once; include any additional scrap in the recorded yield loss.

Yield and process note: Input 1,773 g; 33 g planned trimming/bowl loss after limited rework. Dusting flour must be recorded if material.

Illustrative ingredient cost: 1,541 FCFA per pilot batch; 885.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-B08 Fish filling

Pilot input 783 g; target usable output 700 g. Planning production batch is x5: input 3,915 g and usable output 3,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Canned fish drained	500	2,500	—
Onion trimmed	150	750	—
Carrot trimmed	80	400	—
Oil	25	125	—
Garlic peeled	10	50	—
Ginger peeled	8	40	—
Salt	5	25	—
Black pepper	2	10	—
Paprika	3	15	—

1. Check fish for bones. Weigh drained edible fish; the cost input is per drained kilogram, not unopened-can weight.
2. Soften finely chopped onion and carrot in oil 5–7 minutes. Add garlic, ginger and spices, then fish.
3. Cook while stirring until cohesive and fairly dry. Weigh the finished filling; target 700 g, with no free liquid.
4. Spread in shallow pans and actively cool under control H01–H05. Chill before filling dough; never leave a large warm pot on a counter.

Yield and process note: Input 783 g; target 700 g after evaporation. If yield differs, recompute cost per gram.

Illustrative ingredient cost: 2,322 FCFA per pilot batch; 3,317.1 FCFA/kg usable output. This excludes packaging and operating costs.

O-B09 Beef pie filling

Pilot input 1,506 g; target usable output 1,300 g. Planning production batch is x5: input 7,530 g and usable output 6,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Beef mince	500	2,500	—

Ingredient or prepared component	Pilot g	Production g	Baker %
Potato peeled	300	1,500	—
Carrot trimmed	150	750	—
Onion trimmed	200	1,000	—
Oil	30	150	—
Plain flour	35	175	—
Water	250	1,250	—
Garlic peeled	12	60	—
Ginger peeled	8	40	—
Salt	12	60	—
Black pepper	3	15	—
Thyme	2	10	—
Curry powder	4	20	—

1. Dice vegetables to about 5 mm. Brown beef in oil, breaking lumps; cook onion, garlic and ginger with it.
2. Add potato, carrot, seasonings and water. Simmer until vegetables soften; stir in flour dispersed in a little reserved water.
3. Cook until thick without raw flour taste and verify ground-meat control H01–H05. Target 1,300 g finished filling; reduce gently if wet.
4. Rapidly cool in shallow pans under H01–H05, cover after cooling and refrigerate. Use cold in pastry.

Yield and process note: Input 1,506 g; 206 g expected evaporation. Do not top up with cold untreated liquid after cooking.

Illustrative ingredient cost: 3,015 FCFA per pilot batch; 2,319.1 FCFA/kg usable output. This excludes packaging and operating costs.

O-B10 Sausage roll filling

Pilot input 1,000 g; target usable output 980 g. Planning production batch is x5: input 5,000 g and usable output 4,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Sausage meat	750	3,750	—
Onion trimmed	100	500	—
Breadcrumbs	70	350	—
Egg	60	300	—
Garlic peeled	10	50	—

Ingredient or prepared component	Pilot g	Production g	Baker %
Black pepper	2	10	—
Paprika	3	15	—
Salt	5	25	—

1. Use chilled food-grade sausage meat; verify species and allergens with supplier. Reduce the added salt if the supplied meat is already strongly seasoned.
2. Finely mince onion and garlic, then combine all ingredients just until evenly distributed.
3. Keep covered and cold; portion 40 g cylinders for assembly. Keep raw-meat tools away from cooked products.
4. Cook fully inside pastry according to H01–H05; oven colour alone does not establish meat safety.

Yield and process note: Input 1,000 g; 20 g handling loss. Yield is raw filling; cooking loss is applied in the product card.

Illustrative ingredient cost: 3,356 FCFA per pilot batch; 3,424.5 FCFA/kg usable output. This excludes packaging and operating costs.

O-B11 Creamy chicken filling

Pilot input 1,502 g; target usable output 1,300 g. Planning production batch is x5: input 7,510 g and usable output 6,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Chicken boneless	600	3,000	—
Potato peeled	150	750	—
Carrot trimmed	100	500	—
Onion trimmed	120	600	—
Pepper trimmed	80	400	—
Margarine	30	150	—
Plain flour	45	225	—
UHT milk	250	1,250	—
Water	100	500	—
Garlic peeled	10	50	—
Salt	12	60	—
Black pepper	3	15	—
Thyme	2	10	—

1. Dice raw boneless chicken separately from vegetables. Cook chicken thoroughly under H01–H05, then chop with cleaned utensils.
2. Soften onion, potato, carrot and pepper in margarine. Add garlic and flour; cook briefly, then gradually stir in milk and water.
3. Add cooked chicken and seasonings; simmer to a thick spoonable filling. Verify final centre temperature under H01–H05; target 1,300 g.
4. Cool actively in shallow pans under H01–H05 and refrigerate. This is a refrigerated filling, not a room-temperature display food.

Yield and process note: Input 1,502 g; target 1,300 g cooked. Raw chicken cost and cooking loss are included.

Illustrative ingredient cost: 3,283 FCFA per pilot batch; 2,525.0 FCFA/kg usable output. This excludes packaging and operating costs.

O-B12 Egg roll dough

Pilot input 1,674 g; target usable output 1,650 g. Planning production batch is x5: input 8,370 g and usable output 8,250 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	1,000	5,000	100.00%
Water	470	2,350	47.00%
Sugar	80	400	8.00%
Margarine	70	350	7.00%
Whole milk powder	25	125	2.50%
Salt	15	75	1.50%
Yeast	12	60	1.20%
Nutmeg	2	10	0.20%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, sugar, powder and yeast 3 minutes slowly. Add salt and fat; mix another 5–7 minutes until smooth.
2. Rest covered 15 minutes. Weigh 100 g pieces and flatten evenly.
3. Enclose a fully cooked, chilled, peeled egg; eliminate air pockets and close the seam completely.
4. Proof assembled rolls 20–35 minutes until slightly expanded, then fry according to the product card.

Yield and process note: Input 1,674 g; 24 g handling loss. Hard-boiled egg is a separate component.

Illustrative ingredient cost: 1,053 FCFA per pilot batch; 638.1 FCFA/kg usable output. This excludes packaging and operating costs.

O-B13 Hard boiled peeled eggs

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Egg	1,000	5,000	—

1. The formula uses edible egg weight. Purchase intact eggs, keep shell-handling separate, and wash hands after handling.
2. Lower into boiling water, simmer until yolks and whites are fully set; a starting trial is 10–12 minutes for medium eggs, then verify by cutting a sample.
3. Cool rapidly in potable ice water, peel hygienically and refrigerate. Weigh peeled cooked yield.
4. Portion by actual egg size; the model uses 50 g peeled eggs. Do not store peeled eggs at ambient temperature.

Yield and process note: 970 g expected cooked edible yield from 1,000 g raw edible equivalent. Shell cost is already converted into the egg price input.

Illustrative ingredient cost: 2,200 FCFA per pilot batch; 2,268.0 FCFA/kg usable output. This excludes packaging and operating costs.

O-B14 Banana condensed milk cake batter

Pilot input 3,061 g; target usable output 3,010 g. Planning production batch is x5: input 15,305 g and usable output 15,050 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	1,000	5,000	—
Banana peeled	900	4,500	—
Egg	400	2,000	—
Oil	300	1,500	—
Sugar	220	1,100	—
Condensed milk	200	1,000	—
Baking powder	20	100	—
Baking soda	5	25	—
Salt	8	40	—
Vanilla	5	25	—
Cinnamon	3	15	—

1. Mash sound ripe peeled bananas. Discard mouldy or fermented fruit; ripeness is not permission to use spoilage.

2. Whisk egg, sugar, condensed milk and oil, then banana. Sift flour, raising agents, salt and cinnamon together.
3. Fold dry into wet just until no dry flour remains. Deposit promptly into lined tins.
4. Bake under the product card. Do not leave mixed batter waiting while another oven load finishes.

Yield and process note: Input 3,061 g; 51 g bowl loss. No peanut is included in the baseline.

Illustrative ingredient cost: 3,686 FCFA per pilot batch; 1,224.5 FCFA/kg usable output. This excludes packaging and operating costs.

O-B15 Pizza dough

Pilot input 1,708 g; target usable output 1,690 g. Planning production batch is x5: input 8,540 g and usable output 8,450 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	1,000	5,000	100.00%
Water	620	3,100	62.00%
Salt	20	100	2.00%
Sugar	20	100	2.00%
Yeast	8	40	0.80%
Oil	40	200	4.00%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, sugar and yeast 3 minutes slowly; add salt and develop 5–7 minutes. Add oil near the end.
2. Cover and ferment approximately 60–90 minutes at 24–27°C until aerated. Fold once if needed.
3. Divide into 150 g balls, rest covered 20–30 minutes, then stretch to 18–20 cm without flattening every bubble at the rim.
4. Top immediately before baking; use cooked chicken and chilled sauce, not raw poultry.

Yield and process note: Input 1,708 g; 18 g handling loss.

Illustrative ingredient cost: 789 FCFA per pilot batch; 466.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-B16 Pizza tomato sauce

Pilot input 977 g; target usable output 750 g. Planning production batch is x5: input 4,885 g and usable output 3,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Tomato trimmed	800	4,000	—
Onion trimmed	100	500	—
Oil	30	150	—
Garlic peeled	15	75	—
Sugar	15	75	—
Salt	12	60	—
Oregano	3	15	—
Black pepper	2	10	—

1. Chop tomatoes and onions; soften onion and garlic in oil.
2. Add tomatoes, sugar and seasoning; simmer and stir until thick enough not to flood pizza dough.
3. Blend with sanitized equipment if a smooth sauce is wanted; finish at 750 g target yield.
4. Rapidly cool and refrigerate under H01–H05. Portion 40 g per pizza.

Yield and process note: Input 977 g; evaporation target 227 g. Tomato ripeness changes yield.

Illustrative ingredient cost: 884 FCFA per pilot batch; 1,178.4 FCFA/kg usable output. This excludes packaging and operating costs.

O-B25 Burger buns

Pilot input 1,802 g; target usable output 1,580 g. Planning production batch is x5: input 9,010 g and usable output 7,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	1,000	5,000	100.00%
Water	540	2,700	54.00%
Egg	80	400	8.00%
Sugar	65	325	6.50%
Margarine	60	300	6.00%
Whole milk powder	25	125	2.50%
Salt	18	90	1.80%
Yeast	14	70	1.40%

Baker’s percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, egg, sugar, powder and yeast 3 minutes slowly; add salt and develop 5–7 minutes. Add margarine at the end.

2. Rest 15 minutes; divide into 18 pieces of 99 g before normal handling loss. Round, rest and flatten gently.
3. Proof around 30–35°C until light. Bake at a trial 185–195°C for 14–18 minutes; cool on racks.
4. Target 1,580 g baked buns, about eighteen 87 g buns. For costing the output below is corrected to cooked usable mass.

Yield and process note: Input 1,802 g. Cooked output 1,580 g includes handling and baking losses; no egg wash or sesame included.

Illustrative ingredient cost: 1,192 FCFA per pilot batch; 754.7 FCFA/kg usable output. This excludes packaging and operating costs.

O-B26 Cooked crispy chicken pieces

Pilot input 1,772 g; target usable output 1,150 g. Planning production batch is x5: input 8,860 g and usable output 5,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Chicken boneless	1,000	5,000	—
Yogurt plain	150	750	—
Water	100	500	—
Salt	16	80	—
Garlic peeled	15	75	—
Ginger peeled	10	50	—
Paprika	8	40	—
Black pepper	3	15	—
Plain flour	280	1,400	—
Corn starch	70	350	—
Oil	120	600	—

1. Cut chicken into even boneless portions. Marinate refrigerated with yogurt, water, 10 g salt, garlic and ginger for 4–8 hours.
2. Mix flour, starch, remaining salt, paprika and pepper. Coat drained chicken, dip only into a separate working portion of marinade, then coat again. Discard all raw-contact surplus.
3. Fry in small loads at a trial 170–175°C, roughly 5–8 minutes for flattened pieces, until the thickest part meets H01–H05 poultry temperature.
4. Drain on a rack; weigh the cooked coated output. Target 1,150 g saleable product including coating. Service is immediate or controlled hot holding.

Yield and process note: Gross input includes excess coating, discarded marinade and 120 g frying-oil allowance. All are charged; 1,150 g output is a trial yield, not an absorption guarantee.

Illustrative ingredient cost: 4,963 FCFA per pilot batch; 4,315.6 FCFA/kg usable output. This excludes packaging and operating costs.

O-B27 Cooked beef burger patties

Preparation-ahead option: chapter 6 gives the full oven, cooling, portioning and reheating method. Its final-yield adjustments apply when this base is cooked, chilled and reheated.

Pilot input 1,028 g; target usable output 750 g. Planning production batch is x5: input 5,140 g and usable output 3,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Beef mince	1,000	5,000	—
Salt	12	60	—
Black pepper	3	15	—
Paprika	3	15	—
Oil	10	50	—

1. Use mince with supplier-confirmed fat content, provisionally 15–20%. Keep chilled; do not add water or breadcrumbs.
2. Mix seasoning briefly, portion ten raw patties of approximately 102 g and make a shallow centre depression.
3. Cook on a lightly oiled hot plate, turning as needed, until the centre reaches the H01–H05 ground-beef limit. Probe from the side into the thickest centre.
4. Target ten 75 g cooked patties; weigh actual yield. Assemble immediately or hold under H01–H05.

Yield and process note: Input 1,028 g; cooked yield 750 g. Oil includes pan-use allowance.

Illustrative ingredient cost: 4,572 FCFA per pilot batch; 6,095.7 FCFA/kg usable output. This excludes packaging and operating costs.

O-B28 Burger sauce

Pilot input 983 g; target usable output 975 g. Planning production batch is x5: input 4,915 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Mayonnaise pasteurized	650	3,250	—
Ketchup	220	1,100	—
Mustard	50	250	—
O-B23 Refrigerator cucumber pickles	60	300	—
Paprika	3	15	—

1. Finely chop drained O-B23 pickles. Combine with pasteurized mayonnaise, ketchup, mustard and paprika.
2. Mix with sanitized utensils and portion into small clean service containers.
3. Label and refrigerate under H01–H05; replace containers instead of topping up.
4. Use 25 g per burger and record remaining sauce at service close.

Yield and process note: Input 983 g; 8 g bowl loss. Mustard, egg and other supplier-declared allergens may apply.

Illustrative ingredient cost: 2,281 FCFA per pilot batch; 2,339.6 FCFA/kg usable output. This excludes packaging and operating costs.

O-B29 Samosa wrapper dough

Pilot input 1,538 g; target usable output 1,510 g. Planning production batch is x5: input 7,690 g and usable output 7,550 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	1,000	5,000	100.00%
Water	460	2,300	46.00%
Oil	60	300	6.00%
Salt	18	90	1.80%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour and salt with oil, then add water to make a firm smooth dough. Knead 5–7 minutes.
2. Cover and rest 30 minutes. Divide 30 g portions and roll thinly into circles; cut/fold into cones as preferred.
3. Make sealing paste from O-B30; fill with 40 g chilled O-B31 and exclude trapped air.
4. Seal all seams completely and fry promptly in small batches as specified on the product card.

Yield and process note: Input 1,538 g; 28 g handling/trimming loss. No yeast.

Illustrative ingredient cost: 756 FCFA per pilot batch; 500.3 FCFA/kg usable output. This excludes packaging and operating costs.

O-B30 Samosa sealing paste

Pilot input 100 g; target usable output 98 g. Planning production batch is x5: input 500 g and usable output 490 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	40	200	—

Ingredient or prepared component	Pilot g	Production g	Baker %
Water	60	300	—

1. Whisk water into flour to a thick smooth paste.
2. Prepare a small quantity for one assembly session; keep covered and cool.
3. Apply 2 g to each seam; cook the sealed pastry thoroughly.
4. Discard the working remainder after assembly; never use it as a ready-to-eat sauce.

Yield and process note: Input 100 g; 2 g bowl loss.

Illustrative ingredient cost: 26 FCFA per pilot batch; 268.4 FCFA/kg usable output. This excludes packaging and operating costs.

O-B31 Chicken vegetable samosa filling

Pilot input 1,141 g; target usable output 1,000 g. Planning production batch is x5: input 5,705 g and usable output 5,000 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Chicken boneless	350	1,750	—
Potato peeled	350	1,750	—
Carrot trimmed	120	600	—
Onion trimmed	150	750	—
Oil	30	150	—
Water	100	500	—
Garlic peeled	10	50	—
Ginger peeled	10	50	—
Salt	12	60	—
Curry powder	6	30	—
Cumin	3	15	—

1. Cook finely diced chicken thoroughly under H01–H05 using separate raw-meat utensils.
2. Soften onion, garlic and ginger in oil; add diced potato, carrot and water, then simmer until tender.
3. Fold in chicken and seasonings; cook to a dry filling and target 1,000 g cooked mass.
4. Rapidly cool in shallow trays under H01–H05; use cold at 40 g per samosa.

Yield and process note: Input 1,141 g; target 1,000 g after cooking.

Illustrative ingredient cost: 2,106 FCFA per pilot batch; 2,106.3 FCFA/kg usable output. This excludes packaging and operating costs.

O-B32 Milk cake sponge: batter to baked component

Pilot input 1,105 g; target usable output 950 g. Planning production batch is x5: input 5,525 g and usable output 4,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	300	1,500	—
Egg	450	2,250	—
Sugar	220	1,100	—
UHT milk	80	400	—
Oil	40	200	—
Baking powder	8	40	—
Salt	3	15	—
Vanilla	4	20	—

1. Whip eggs and sugar in the mixing machine 6–10 minutes until thick and ribboning; use a clean bowl.
2. Sift flour, baking powder and salt. Fold into the foam gently, then fold in milk, oil and vanilla pre-mixed with a little batter.
3. Deposit into a lined shallow pan. Bake at a trial 170–180°C for 22–30 minutes until springy with no wet batter on a skewer.
4. Cool the sponge on a rack, then weigh. Target 950 g baked sponge; do not soak warm cake and leave it unrefrigerated.

Yield and process note: Input 1,105 g; about 25 g bowl and 130 g bake loss. Output used for costing is 950 g baked sponge.

Illustrative ingredient cost: 1,602 FCFA per pilot batch; 1,686.5 FCFA/kg usable output. This excludes packaging and operating costs.

O-B33 Three milk soak

Pilot input 1,000 g; target usable output 990 g. Planning production batch is x5: input 5,000 g and usable output 4,950 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Condensed milk	400	2,000	—
UHT milk	400	2,000	—
Cream 35%	200	1,000	—

1. Mix cold pasteurized/UHT dairy in a sanitized jug.
2. Keep refrigerated; weigh the amount applied to each fully cooled sponge.

3. Use 800 g soak per 950 g O-B32 sponge and allow absorption under refrigeration.
4. Do not reuse liquid drained from a cake or serving tray.

Yield and process note: Input 1,000 g; 10 g handling loss. A full base leaves 190 g when one full O-B32 sponge is soaked.

Illustrative ingredient cost: 2,980 FCFA per pilot batch; 3,010.1 FCFA/kg usable output. This excludes packaging and operating costs.

O-B34 Condensed milk glaze

Pilot input 342 g; target usable output 335 g. Planning production batch is x5: input 1,710 g and usable output 1,675 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Condensed milk	200	1,000	—
Icing sugar	100	500	—
Butter	20	100	—
Water	20	100	—
Vanilla	2	10	—

1. Melt butter gently. Stir into condensed milk with sifted icing sugar and vanilla.
2. Add potable water gradually to the measured consistency; use all listed water for the model yield.
3. Apply a thin measured layer to products; prepare only the quantity needed for the session.
4. Treat the diluted glaze and glazed goods under refrigerated control H01–H05 unless a competent assessment validates another holding method.

Yield and process note: Input 342 g; 7 g bowl loss. No ambient shelf-life claim.

Illustrative ingredient cost: 904 FCFA per pilot batch; 2,698.8 FCFA/kg usable output. This excludes packaging and operating costs.

O-B17 Shawarma flatbread

Pilot input 1,681 g; target usable output 1,500 g. Planning production batch is x5: input 8,405 g and usable output 7,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Bread flour	1,000	5,000	100.00%
Water	610	3,050	61.00%
Sugar	15	75	1.50%
Salt	18	90	1.80%

Ingredient or prepared component	Pilot g	Production g	Baker %
Yeast	8	40	0.80%
Oil	30	150	3.00%

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix all ingredients 3 minutes slowly then 5–7 minutes until smooth and extensible.
2. Rest covered 30 minutes, divide into sixteen pieces of about 104 g, then rest another 15 minutes.
3. Roll each evenly to about 24–26 cm. Cook on a very hot clean plate or in a 240–260°C trial oven until set and just blistered, roughly 1–3 minutes; avoid crisping.
4. Stack briefly under a clean cover to retain flexibility, then cool without condensation. Target 1,500 g cooked bread; sixteen wraps at 90 g consume 1,440 g.

Yield and process note: Input 1,681 g; allow about 17 g handling and 164 g baking loss. Extra 60 g cooked bread is allocated to QC/weight variation in product costing.

Illustrative ingredient cost: 768 FCFA per pilot batch; 512.2 FCFA/kg usable output. This excludes packaging and operating costs.

O-B18 Cooked chicken shawarma meat

Preparation-ahead option: chapter 6 gives the full oven, cooling, portioning and reheating method. Its final-yield adjustments apply when this base is cooked, chilled and reheated.

Pilot input 1,304 g; target usable output 750 g. Planning production batch is x5: input 6,520 g and usable output 3,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Chicken boneless	1,000	5,000	—
Yogurt plain	150	750	—
Lemon juice	35	175	—
Vinegar 5%	15	75	—
Oil	30	150	—
Garlic peeled	25	125	—
Salt	16	80	—
Paprika	12	60	—
Cumin	6	30	—
Coriander	6	30	—
Black pepper	3	15	—
Cinnamon	1	5	—

Ingredient or prepared component	Pilot g	Production g	Baker %
Chili trimmed	5	25	—

1. Use trimmed boneless thigh; keep raw meat at or below 4°C. Mix marinade separately, coat meat, cover, label and marinate refrigerated 6–12 hours as a trial quality window.
2. Drain only excess free marinade. Weigh the raw load. Use the vertical machine only once its minimum cone size and safe operating instructions are established; otherwise cook small batches on a suitable hot plate.
3. Follow the dedicated shawarma service workflow in chapter 6. Shave only the cooked outer layer, then finish slices on the griddle and probe to H01–H05 poultry temperature.
4. Target 750 g saleable cooked meat per 1,000 g purchased boneless meat. All marinade cost is charged even though some is discarded. Never reuse raw marinade as sauce.

Yield and process note: Gross input 1,304 g including marinade; 750 g cooked saleable output is an assumed purchase-to-cooked yield of 75%, not 75% of the marinated total.

Illustrative ingredient cost: 4,743 FCFA per pilot batch; 6,323.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-B19 Cooked beef shawarma meat

Preparation-ahead option: chapter 6 gives the full oven, cooling, portioning and reheating method. Its final-yield adjustments apply when this base is cooked, chilled and reheated.

Pilot input 1,231 g; target usable output 700 g. Planning production batch is x5: input 6,155 g and usable output 3,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Beef boneless	1,000	5,000	—
Onion trimmed	80	400	—
Lemon juice	30	150	—
Vinegar 5%	30	150	—
Oil	30	150	—
Garlic peeled	20	100	—
Salt	16	80	—
Paprika	10	50	—
Cumin	6	30	—
Coriander	5	25	—
Black pepper	3	15	—
Cinnamon	1	5	—

1. Thinly slice trimmed beef across the grain. Combine marinade and coat in a covered food-safe container.
2. Marinate refrigerated 6–12 hours as a pilot quality window. Do not add more acid to tenderize an unsuitable tough cut indefinitely.
3. Cook using the same staged shawarma workflow as chicken; finish shaved slices thoroughly under H01–H05's shawarma house standard.
4. Target 700 g cooked saleable beef per kilogram purchased boneless meat; record actual yield and discard raw marinade.

Yield and process note: Gross input 1,231 g; cooked saleable yield assumes 70% of raw purchased beef. It is a costing assumption, not a guaranteed yield.

Illustrative ingredient cost: 4,962 FCFA per pilot batch; 7,089.1 FCFA/kg usable output. This excludes packaging and operating costs.

O-B20 Garlic mayonnaise sauce

Pilot input 987 g; target usable output 975 g. Planning production batch is x5: input 4,935 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Mayonnaise pasteurized	800	4,000	—
Lemon juice	65	325	—
Water	65	325	—
Garlic peeled	50	250	—
Salt	5	25	—
Black pepper	2	10	—

1. Use commercial pasteurized mayonnaise; do not prepare a raw-egg mayonnaise.
2. Blend garlic with lemon and water using sanitized tools, then whisk into mayonnaise and season.
3. Weigh and fill small covered service bottles; label and refrigerate under H01–H05.
4. Use a clean bottle for each replenishment rather than topping fresh sauce into old sauce.

Yield and process note: Input 987 g; 12 g bowl loss. Water and garlic addition invalidate the original unopened mayonnaise shelf life.

Illustrative ingredient cost: 2,287 FCFA per pilot batch; 2,346.0 FCFA/kg usable output. This excludes packaging and operating costs.

O-B21 Cooked hot sauce

Pilot input 880 g; target usable output 700 g. Planning production batch is x5: input 4,400 g and usable output 3,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Tomato trimmed	500	2,500	—
Chili trimmed	120	600	—
Onion trimmed	100	500	—
Garlic peeled	20	100	—
Vinegar 5%	80	400	—
Oil	30	150	—
Sugar	20	100	—
Salt	10	50	—

1. Soften onion and garlic in oil, add chopped tomato, chili, vinegar, sugar and salt.
2. Simmer 15–20 minutes, blend safely and return to a bubbling simmer; reduce to 700 g.
3. Cool actively under H01–H05, bottle hygienically and refrigerate.
4. Offer a measured 5 g serving separately to let customers choose heat.

Yield and process note: Input 880 g; 180 g evaporation. This is not a shelf-stable bottled sauce; acidity is unvalidated.

Illustrative ingredient cost: 920 FCFA per pilot batch; 1,314.3 FCFA/kg usable output. This excludes packaging and operating costs.

O-B22 Shawarma slaw

Pilot input 995 g; target usable output 975 g. Planning production batch is x5: input 4,975 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Cabbage trimmed	700	3,500	—
Carrot trimmed	180	900	—
Onion trimmed	60	300	—
Vinegar 5%	35	175	—
Sugar	15	75	—
Salt	5	25	—

1. Wash trimmed vegetables with potable water and use sanitized ready-to-eat preparation tools.
2. Shred finely, mix dressing, toss and drain excess free liquid briefly.
3. Weigh finished yield and chill covered. Portion only small quantities onto the service line.
4. Serve 45 g per wrap. Keep separate from raw poultry, its containers and its wash area.

Yield and process note: Input 995 g; 20 g draining/handling loss. Vinegar does not replace refrigeration.

Illustrative ingredient cost: 676 FCFA per pilot batch; 692.8 FCFA/kg usable output. This excludes packaging and operating costs.

O-B23 Refrigerator cucumber pickles

Pilot input 1,480 g; target usable output 650 g. Planning production batch is x5: input 7,400 g and usable output 3,250 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Cucumber trimmed	700	3,500	—
Vinegar 5%	350	1,750	—
Water	350	1,750	—
Sugar	60	300	—
Salt	20	100	—

1. Dissolve sugar and salt in the water with heat, add vinegar, and cool the brine rapidly before adding clean sliced cucumber.
2. Submerge in a clean covered food-safe container and refrigerate overnight for flavour.
3. Drain only the service quantity; target 650 g drained cucumber from the batch. Brine is discarded and fully charged to cost.
4. Label as refrigerated quick pickles. No ambient-storage or canning claim is made.

Yield and process note: Gross input 1,480 g including brine. Only drained pickles are costed as usable output.

Illustrative ingredient cost: 933 FCFA per pilot batch; 1,435.0 FCFA/kg usable output. This excludes packaging and operating costs.

O-B24 Crisp fries

Pilot input 1,128 g; target usable output 650 g. Planning production batch is x5: input 5,640 g and usable output 3,250 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Potato peeled	1,000	5,000	—
Oil	120	600	—
Salt	8	40	—

1. Cut peeled potatoes to uniform 8–10 mm sticks. Rinse with potable water, drain and dry thoroughly away from hot oil.
2. For immediate service, trial a first fry around 150–160°C until cooked but pale, then finish around 175°C until crisp. Keep batches small.

3. Drain on a rack and salt. Serve promptly; do not hold a large tray at room temperature for later frying.
4. If par-frying in advance, use the controlled cooling and refrigerated storage workflow H01–H05, then finish to order.

Yield and process note: 650 g assumed saleable fries per kg peeled potato. Oil entry is a 120 g usage/disposal cost allowance, not a claim that all oil is absorbed. Cost the fryer charge separately as working stock.

Illustrative ingredient cost: 1,095 FCFA per pilot batch; 1,684.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-B35 Beef shawarma sesame sauce

Pilot input 990 g; target usable output 975 g. Planning production batch is x5: input 4,950 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Mayonnaise pasteurized	500	2,500	—
Tahini	200	1,000	—
Lemon juice	80	400	—
Water	180	900	—
Garlic peeled	25	125	—
Salt	5	25	—

1. Whisk tahini with lemon and water until smooth, then incorporate mayonnaise, garlic and salt.
2. Use sanitized utensils and check supplier sesame and egg allergen declarations.
3. Refrigerate in small labelled service bottles under H01–H05.
4. Portion 30 g per beef wrap; keep a separate spoon for sesame sauce.

Yield and process note: Input 990 g; 15 g handling loss. A garlic-only alternative uses O-B20 at the same serving mass; re-cost.

Illustrative ingredient cost: 2,872 FCFA per pilot batch; 2,945.5 FCFA/kg usable output. This excludes packaging and operating costs.

O-B36 lemon-garlic oven fish

This is a new shared base for O-A02/O-A03. Use supplier-identified, food-grade, boneless tilapia or another firm white-fish fillet. The recipe is for trimmed edible fillet, not whole fish. Inspect carefully for pin bones before and after cooking. Keep fish utensils separate; fish is an additional allergen in the burger/wrap station.

Ingredient	Pilot g	Production trial g (×5)
Boneless trimmed white fish	1,000	5,000
Oil	25	125

Ingredient	Pilot g	Production trial g (×5)
Lemon juice	25	125
Garlic, finely crushed	15	75
Salt	10	50
Paprika	4	20
Black pepper	2	10
Total input	1,081	5,405

1. Keep fillets chilled, drain excess thaw water and pat with disposable food-safe towel. Cut pieces of similar thickness, roughly 15–20 mm; keep smaller pieces in a separate trial group. Mix the listed seasonings and oil, coat fish, and refrigerate for a short trial of 15–20 minutes. Acid does not make raw fish safe; extended lemon contact can soften it.
2. Place in a single layer in a lined raised-edge pan. Trial roast at 180–190°C, checking around 10–16 minutes; thickness and oven loading determine the actual duration. Meet H01–H05 at the thickest centre. Target moist flakes and light browning, with no raw centre. Inspect for remaining bones with clean utensils without breaking every piece unnecessarily.
3. For immediate service, portion after cooking and assemble. For preparation ahead, cool, portion and transport using the controlled route above. Trial reheating covered briefly to limit drying, then uncover if surface moisture needs removal; meet H01–H05 before assembly. Do not mix yesterday's fish with a new batch or use strongly deteriorated fish with extra spice.
4. Planning yield: 800 g chilled usable fish per pilot, then 760 g after a trial 95% reheating retention. These are unverified assumptions, not measured performance. Record cooking, chilling, trimming and reheating losses separately. At 120 g final fish, one pilot supports six sale portions and a 40 g final-equivalent balance; at 100 g, seven and a 60 g balance. Balances must have an identified same-day use or be charged as waste, not counted as full portions.

Illustrative raw fillet price: 4,500 FCFA/kg edible, newly assumed for costing only. Other inputs use chapter 12 prices. O-B36 pilot ingredients cost **4,684.0 FCFA**, or **6,163.2 FCFA/kg of final reheated output**. Whole-fish purchase cost per usable kg equals whole-fish price divided by measured fillet yield, before cooking losses. Supplier pricing and bone removal can materially change viability.

5. Bread, fried snacks and bakery meals

The finished cards use the shared components in chapter 4. Make the specified component amount, then assemble by weight. Alternatives in the S-series are separate product versions; do not blend their formulas accidentally.

O-P01 Soft family bread

Quality target: Fine even crumb, thin golden crust, clean slices and a soft but fully baked centre.

Pilot: 3 saleable units. Planning production batch: 15 units. Portion: 3 x 550 g dough; about 470–495 g baked each. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B01 Family bread dough	1,700	8,500

1. Make O-B01. Weigh three 550 g dough pieces; the 50 g surplus covers QC/test dough and is charged to these loaves.
2. Pre-round, rest 10 minutes and mould into greased tins. Proof as O-B01; leave room for oven spring rather than overproofing to the tin edge.
3. Trial bake at 195–205°C for 25–32 minutes. Check a representative loaf centre near 94–96°C as a baking-quality cue, not a universal safety limit.
4. De-tin immediately. Cool on racks until no warmth/condensation remains before slicing and bagging. Verify cooled sale weight and slicing consistency.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Dense crumb: insufficient proof/development; collapsed top: excessive proof or underbake; gummy slices: sliced too warm.

Substitution: Use the strongest consistent local wheat flour; re-trial water rather than assuming equal absorption.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
835	3	278.4	30	55	650	286.6

All values in FCFA. Contribution margin 44.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 230.9 FCFA. At 80% sell-through it falls to 195.8 FCFA per sold unit under the chapter 12 model.

O-P02 Condensed milk breakfast buns

Quality target: Rounded golden buns, shreddable soft crumb, clear milk aroma without a wet surface.

Pilot: 20 saleable units. Planning production batch: 100 units. Portion: 20 x 90 g dough; about 78–82 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B02 Condensed milk bun dough	1,870	9,350
Egg	40	200

1. Make O-B02 and portion twenty 90 g buns. Charge the remaining 70 g dough to QC/variation.
2. Round tightly, place with expansion space, proof using O-B02, then brush lightly with the 40 g weighed egg wash; discard unused raw-contact wash.
3. Bake at a trial 180–190°C for 14–18 minutes. Rotate only if safe and the oven's distribution test indicates a need.

4. Cool completely and bag. Start with plain buns; an optional 8 g O-B34 glaze adds refrigerated handling and additional cost, so it is excluded from baseline pricing.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Tight crumb: add fat too early or underproof; very dark crust: excessive sugar or heat; wrinkling: underbaking.

Substitution: Budget trial: replace each 150 g condensed milk with 68 g sugar, 42 g whole milk powder and 40 g water; sensory and cost recheck required.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,853	20	92.6	10	20	250	127.4

All values in FCFA. Contribution margin 51.0%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 108.8 FCFA. At 80% sell-through it falls to 96.7 FCFA per sold unit under the chapter 12 model.

O-P03 Nkambe sweet potato milk loaf

Quality target: Warm yellow/brown crumb depending on flour, mild sweetness and enough elasticity for clean slices.

Pilot: 3 saleable units. Planning production batch: 15 units. Portion: 3 x 580 g dough; about 495–522 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B03 Sweet potato milk bread dough	1,790	8,950

1. Make O-B03 using sound food-grade sweet potato flour. Divide three 580 g loaves; charge 50 g remaining dough to QC.
2. Mould gently, tin and proof to resilience; do not force the height of a wheat-only loaf.
3. Trial bake at 190–200°C for 27–33 minutes; verify fully baked centre and record cooled mass.
4. Cool and bag with an accurate ingredient description. Test this against O-P01 at two prices before committing a large daily volume.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Low volume: weak flour or too much sweet potato; gummy interior: underbake/high water; gritty texture: coarse flour.

Substitution: If sweet potato flour is unavailable, sell O-P01 under its own name; cooked sweet potato puree is not a direct gram-for-gram substitute.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,309	3	436.2	35	60	800	268.8

All values in FCFA. Contribution margin 33.6%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 181.6 FCFA. At 80% sell-through it falls to 136.0 FCFA per sold unit under the chapter 12 model.

O-P04 Sugar ring doughnuts

Quality target: Light crumb, an even ring, thin sugar finish and no raw centre or greasy bite.

Pilot: 25 saleable units. Planning production batch: 125 units. Portion: 25 x 70 g dough; about 66–73 g finished with sugar. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B04 Yeast doughnut dough	1,780	8,900
Oil	200	1,000
Sugar	120	600

1. Prepare O-B04, portion twenty-five 70 g rings; 30 g spare dough is charged to QC.
2. Proof until light. Fry at 170–175°C about 60–100 seconds per side as a trial, adjusting to centre doneness and colour.
3. Drain on a rack; apply 120 g sugar across the batch. Oil entry is usage/disposal allocation, not 200 g guaranteed absorption.
4. Sell in small replenishment waves. Glazed variant: replace sugar dusting with 8 g O-B34 per unit and use refrigerated handling; do not hide its extra cost in this baseline.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Dark/raw centre: oil too hot; oily heavy crumb: underproof or low oil recovery; collapse: overproof.

Substitution: Omit milk powder only after water/texture trials; do not substitute baking powder for yeast in this formula.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,833	25	73.3	10	25	250	141.7

All values in FCFA. Contribution margin 56.7%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 127.0 FCFA. At 80% sell-through it falls to 114.6 FCFA per sold unit under the chapter 12 model.

O-P05 Puff puff

Quality target: Rounded pieces with an aerated cooked centre, even colour and no sour overfermentation.

Pilot: 60 saleable units. Planning production batch: 300 units. Portion: 60 x 30 g raw portions; about 27–31 g fried. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B05 Puff puff batter	1,830	9,150
Oil	180	900

1. Make O-B05. Portion sixty 30 g scoops; the remaining 30 g is charged to trial pieces.
2. Fry at 165–175°C in small batches, turning until uniformly golden and the centre is fully set.
3. Drain and count; put five in a ventilated paper pack only when ordered. Do not seal steaming puff puff in plastic.
4. At the example price, a five-piece pack is 350–375 FCFA only if the discounted contribution remains acceptable.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Flat pieces: thin or overfermented batter; hard centre: low hydration/underproof; oily finish: low oil recovery.

Substitution: Use a 25 g raw portion for a lower-price test only after recalculating yield and clearly preserving value.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,134	60	18.9	3	8	75	45.1

All values in FCFA. Contribution margin 60.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 41.3 FCFA. At 80% sell-through it falls to 37.6 FCFA per sold unit under the chapter 12 model.

O-P06 Cameroon style fish rolls

Quality target: Crisp golden exterior, rolled seam and a clearly present dry fish filling.

Pilot: 24 saleable units. Planning production batch: 120 units. Portion: 24 x 65 g dough + 25 g filling; about 82–88 g cooked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B06 Fish roll dough	1,660	8,300
O-B08 Fish filling	600	3,000
Oil	180	900

1. Make O-B06 and O-B08 separately. Use 1,560 g dough and 600 g cold filling for 24 rolls; 100 g dough is QC allowance. A full 700 g O-B08 batch leaves 100 g allocated separately.
2. Roll 65 g dough pieces, add 25 g filling, close edges and seam, then rest briefly as O-B06.
3. Fry at a trial 165–175°C until golden and filling is thoroughly reheated under H01–H05. Check the thickest sample.
4. Drain and use holding class H. Transport finished rolls hot under control; an alternative is chilled assembled uncooked rolls for final frying at the sales site under chapter 6. Do not chill finished rolls for a second reheating of the fish filling.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under H01–H05, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in chapter 6.

Troubleshooting: Burst seams: wet filling or poor seal; gummy spiral: wrapper too thick; fishy flavour: unsuitable fish/storage.

Substitution: Use cooked deboned fresh fish only with measured purchase-to-edible yield and revised cost.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
3,347	24	139.4	10	30	300	120.6

All values in FCFA. Contribution margin 40.2%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 92.7 FCFA. At 80% sell-through it falls to 75.7 FCFA per sold unit under the chapter 12 model.

O-P07 Beef meat pies

Quality target: Short flaky bite, full corners, intact seal and visible beef/vegetable filling.

Pilot: 22 saleable units. Planning production batch: 110 units. Portion: 22 x 75 g pastry + 50 g filling; about 112–118 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B07 Shortcrust pie pastry	1,740	8,700
O-B09 Beef pie filling	1,100	5,500
Egg	45	225

1. Prepare O-B07 and O-B09. Cut twenty-two 75 g pastry portions and add 50 g chilled filling each. Pastry surplus is 90 g; full O-B09 yields 200 g unused filling.
2. Seal with clean edges, vent, brush using 45 g allocated egg wash and discard raw-contact leftovers.
3. Trial bake at 190–200°C for 22–28 minutes. Check base colour, pastry doneness and filling temperature under H01–H05.
4. Apply H holding and package without trapping excessive steam.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under H01–H05, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in chapter 6.

Troubleshooting: Tough pastry: overmixing; soggy base: wet/hot filling; leakage: overfilling or weak seam.

Substitution: Replace part of beef with potato only as a separately labelled economy formula with recalculated portions.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
4,191	22	190.5	20	35	500	254.5

All values in FCFA. Contribution margin 50.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 216.4 FCFA. At 80% sell-through it falls to 193.1 FCFA per sold unit under the chapter 12 model.

O-P08 Sausage rolls

Quality target: Crisp pastry with a fully cooked moist meat centre and clean slice.

Pilot: 24 saleable units. Planning production batch: 120 units. Portion: 24 x 65 g pastry + 40 g raw filling; about 88–96 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B07 Shortcrust pie pastry	1,740	8,700
O-B10 Sausage roll filling	960	4,800
Egg	45	225

1. Make O-B07 and keep O-B10 cold. Portion 24 rolls with 65 g pastry and 40 g raw filling each. Pastry surplus is 180 g; O-B10 batch leaves 20 g.
2. Form cylinders, enclose and seal pastry, apply 45 g allocated egg wash and vent the tops.
3. Trial bake at 190–200°C for 23–30 minutes. Probe meat to H01–H05 ground-meat limit, or poultry limit if sausage contains poultry.
4. Hold and transport under H. Record any grease leakage before adjusting meat fat or crumb content.

Holding and packaging: H — Serve immediately or hold hot under H01–H05. With the specified raw sausage filling, a fully baked roll may instead be rapidly cooled and reheated once using chapter 6. House hot-display quality target: at most 2 hours. Record the actual filling heat history.

Troubleshooting: Raw centre: rolls too thick or oven too hot; greasy pastry: high-fat filling; splitting: poor vent/seal.

Substitution: Supplier-seasoned mince can replace sausage meat only after checking salt, species and allergen details.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
4,928	24	205.3	15	35	450	194.7

All values in FCFA. Contribution margin 43.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 153.6 FCFA. At 80% sell-through it falls to 130.8 FCFA per sold unit under the chapter 12 model.

O-P09 Creamy chicken pies

Quality target: Golden short pastry, moist chicken pieces and a thick sauce that does not run.

Pilot: 21 saleable units. Planning production batch: 105 units. Portion: 21 x 80 g pastry + 60 g filling; about 125–133 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B07 Shortcrust pie pastry	1,740	8,700
O-B11 Creamy chicken filling	1,260	6,300
Egg	45	225

1. Make O-B07 and O-B11. Use twenty-one 80 g pastry portions and 60 g chilled filling each; dough spare 60 g, full O-B11 spare 40 g.
2. Seal, vent and apply allocated egg wash. Keep waiting trays refrigerated.
3. Trial bake at 190–200°C for 23–30 minutes; verify filling reaches H01–H05 reheating temperature.
4. Use H holding; limit hot display batches because pastry quality declines before safety-controlled holding necessarily ends.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under H01–H05, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in chapter 6.

Troubleshooting: Soggy pastry: loose sauce; dry chicken: prolonged hot holding; split pies: overfilled.

Substitution: Use cooked thigh from a separately costed plain preparation; do not assume raw and cooked chicken weights are interchangeable.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
4,822	21	229.6	20	40	650	360.4

All values in FCFA. Contribution margin 55.4%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 314.5 FCFA. At 80% sell-through it falls to 288.0 FCFA per sold unit under the chapter 12 model.

O-P10 Whole egg dough rolls

Quality target: A whole fully cooked egg centred inside evenly fried dough, with no wet seam.

Pilot: 16 saleable units. Planning production batch: 80 units. Portion: 16 x 100 g dough + 50 g peeled cooked egg; about 140–148 g fried. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B12 Egg roll dough	1,650	8,250
O-B13 Hard boiled peeled eggs	800	4,000
Oil	180	900

1. Prepare O-B12 and O-B13. Use sixteen 100 g dough pieces and sixteen cooked eggs averaging 50 g each; charge 50 g dough surplus to QC.
2. Wrap eggs evenly, close seams tightly and proof 20–35 minutes.
3. Trial fry at 165–170°C for 7–10 minutes, turning. Split a sample and check both dough doneness and reheated centre under H01–H05.
4. Use H holding. Full O-B13 output is 970 g: 170 g remains for separately recorded use or the next scaled batch, within the same safe production plan.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under H01–H05, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in chapter 6.

Troubleshooting: Burnt outside/raw inner dough: too high oil temperature; bursting: trapped air; uneven portions: egg sizes not checked.

Substitution: Smaller eggs require a separately priced smaller product, not silent substitution.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
3,155	16	197.2	15	35	500	252.8

All values in FCFA. Contribution margin 50.6%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 213.3 FCFA. At 80% sell-through it falls to 191.0 FCFA per sold unit under the chapter 12 model.

O-P11 Banana condensed milk cake slices

Quality target: Moist but set crumb, clear banana aroma, clean slices and no compressed wet line.

Pilot: 24 saleable units. Planning production batch: 120 units. Portion: 3 x 1,000 g batter loaves; 24 slices about 105–113 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B14 Banana condensed milk cake batter	3,010	15,050

1. Prepare O-B14 and fill three suitable loaf tins with 1,000 g batter each; 10 g is bowl/QC allocation. Choose tins with enough expansion space.
2. Trial bake at 170–180°C for 45–60 minutes; use a clean skewer and spring-back rather than time alone.

3. Cool fully, weigh each loaf, and cut each into eight equal slices; expected bake loss 10–16%.
4. Package slices only after cooling. Use moist-cake holding class M; do not assert a multi-day ambient shelf life.

Holding and packaging: M — Moist banana cake: sell on production day; protect during short display and refrigerate reserve stock at or below 4°C. Ambient multi-day shelf life requires a competent water-activity/pH assessment and validation.

Troubleshooting: Sunken centre: underbake/excess raising; gummy crumb: wet bananas; dry slice: excess bake or thin portions.

Substitution: Swap some banana for pumpkin only through a new moisture trial; optional nuts require new allergen control and cost.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
3,686	24	153.6	20	25	400	201.4

All values in FCFA. Contribution margin 50.4%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 170.7 FCFA. At 80% sell-through it falls to 151.8 FCFA per sold unit under the chapter 12 model.

O-P12 Personal chicken pizza

Quality target: Crisp underside, cooked centre, melted cheese and a a measured 50 g cooked-chicken input before baking portion.

Pilot: 11 saleable units. Planning production batch: 55 units. Portion: 11 x 150 g dough + 160 g toppings; about 275–290 g baked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B15 Pizza dough	1,690	8,450
O-B16 Pizza tomato sauce	440	2,200
O-B18 Cooked chicken shawarma meat	550	2,750
Mozzarella	550	2,750
Onion trimmed	110	550
Pepper trimmed	110	550

1. Make O-B15. Use eleven 150 g bases; 40 g dough spare is charged to QC.
2. Per base add 40 g O-B16, 50 g cooked O-B18, 50 g mozzarella, 10 g onion and 10 g pepper. Keep prepared toppings refrigerated.
3. Trial bake at 230–250°C for 7–11 minutes only if the diesel oven is rated for this temperature. Check base and reheated chicken under H01–H05; hotter baking needs a separate oven schedule.
4. Serve immediately. If the rotary oven cannot deliver a suitable base at its approved setting, postpone pizza or use a tested par-baked base workflow with controlled cooling.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under H01–H05, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in chapter 6.

Troubleshooting: Wet centre: excess sauce or low oven recovery; burnt cheese: long bake; high cost: uncontrolled cheese handfuls.

Substitution: Use less expensive melting cheese only after taste/melt testing; never replace measured chicken with extra fries.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
8,581	11	780.1	90	80	1800	849.9

All values in FCFA. Contribution margin 47.2%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 693.9 FCFA. At 80% sell-through it falls to 612.4 FCFA per sold unit under the chapter 12 model.

O-P13 Complete chicken shawarma

Quality target: Flexible wrap, hot juicy meat, crisp fries and balanced sauce with a tight clean roll.

Pilot: 16 saleable units. Planning production batch: 80 units. Portion: 1 wrap + 110 g cooked chicken + 145 g additions; about 345 g. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B17 Shawarma flatbread	1,500	7,500
O-B18 Cooked chicken shawarma meat	1,760	8,800
O-B24 Crisp fries	800	4,000
O-B22 Shawarma slaw	720	3,600
O-B23 Refrigerator cucumber pickles	240	1,200
O-B20 Garlic mayonnaise sauce	480	2,400
O-B21 Cooked hot sauce	80	400

1. Prepare the exact components in chapter 4; this card costs sixteen full standard wraps. O-B17 delivers 16 x 90 g bread with 60 g allowance.
2. For each wrap weigh 110 g cooked O-B18, 50 g O-B24 fries, 45 g O-B22 slaw, 15 g O-B23 pickles, 30 g O-B20 garlic sauce and optional 5 g O-B21 hot sauce; hot sauce cost is included.
3. Assemble using separate hot and ready-to-eat utensils. Fold sides, roll firmly and toast about 30–60 seconds per side on the clean hot plate without crushing.
4. Wrap in food-contact paper and hand over immediately. Use fresh safe service stock; a filled wrap is not a long-hold product.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under H01–H05 until assembly. Hand over promptly and recommend immediate consumption; do not build an

ambient ready-wrapped stock.

Troubleshooting: Wet wrap: undrained slaw or excess sauce; dry bread: overbaking; chewy meat: wrong cut or excessive holding.

Substitution: Use thigh as default; breast needs separate moisture/yield tests. A garlic-only mild option omits hot sauce without changing meat.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
15,321	16	957.5	80	60	2000	902.5

All values in FCFA. Contribution margin 45.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 711.0 FCFA. At 80% sell-through it falls to 628.1 FCFA per sold unit under the chapter 12 model.

O-P14 Complete beef shawarma

Quality target: Tender sliced beef with a balanced sesame sauce, no gristle and an intact wrap.

Pilot: 16 saleable units. Planning production batch: 80 units. Portion: 1 wrap + 110 g cooked beef + 145 g additions; about 345 g. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B17 Shawarma flatbread	1,500	7,500
O-B19 Cooked beef shawarma meat	1,760	8,800
O-B24 Crisp fries	800	4,000
O-B22 Shawarma slaw	720	3,600
O-B23 Refrigerator cucumber pickles	240	1,200
O-B35 Beef shawarma sesame sauce	480	2,400
O-B21 Cooked hot sauce	80	400

1. Use the same sixteen-wrap bread allocation as O-P13, substituting cooked O-B19 beef and O-B35 sesame sauce.
2. Assemble each with 110 g cooked beef, 50 g fries, 45 g slaw, 15 g drained pickles, 30 g sesame sauce and optional 5 g hot sauce.
3. Toast and wrap as O-P13. Check the raw-purchase quantity using 70% beef yield before purchasing.
4. Serve immediately and disclose sesame. Keep sauce utensils separate from the garlic-only option.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under H01–H05 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Tough beef: wrong cut or slicing direction; bitter sauce: poor tahini; wet wrap: excess liquid.

Substitution: O-B20 can replace O-B35 at 30 g per wrap for a sesame-free recipe only when cross-contact can also be controlled.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
16,955	16	1,059.7	80	65	2400	1,195.3

All values in FCFA. Contribution margin 49.8%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 983.4 FCFA. At 80% sell-through it falls to 894.1 FCFA per sold unit under the chapter 12 model.

O-P15 Crispy chicken burger

Quality target: Toasted soft bun, crisp fully cooked chicken and fresh measured garnish.

Pilot: 10 saleable units. Planning production batch: 50 units. Portion: 87 g bun + 110 g cooked coated chicken + 75 g garnish; about 272 g. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B25 Burger buns	870	4,350
O-B26 Cooked crispy chicken pieces	1,100	5,500
O-B28 Burger sauce	250	1,250
Lettuce trimmed	150	750
Tomato trimmed	250	1,250
O-B23 Refrigerator cucumber pickles	100	500

1. Make or allocate O-B25 buns; a full batch supplies eighteen buns, of which ten are used here.
2. Use ten 110 g cooked coated O-B26 portions. Toast split buns on a clean hot plate.
3. Per burger add 25 g O-B28 sauce, 15 g lettuce, 25 g tomato and 10 g drained pickles. Keep garnish cold until assembly.
4. Serve immediately in a vented box. Optional cheese is 20 g and costs extra; fries are not included.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under H01–H05 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Soggy crust: covered hot chicken or early assembly; raw middle: chicken too thick; slippery burger: watery tomato.

Substitution: A grilled O-B18 chicken burger is a separate 110 g cooked-meat option with recalculated cost.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
6,602	10	660.2	70	65	1800	1,004.8

All values in FCFA. Contribution margin 55.8%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 872.8 FCFA. At 80% sell-through it falls to 806.0 FCFA per sold

unit under the chapter 12 model.

O-P16 Beef cheeseburger

Quality target: Fully cooked juicy patty, melted cheese, toasted bun and a balanced bite.

Pilot: 10 saleable units. Planning production batch: 50 units. Portion: 87 g bun + 75 g cooked patty + 105 g cheese/garnish; about 267 g. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B25 Burger buns	870	4,350
O-B27 Cooked beef burger patties	750	3,750
Cheese slices	200	1,000
O-B28 Burger sauce	250	1,250
Lettuce trimmed	150	750
Tomato trimmed	250	1,250
Onion trimmed	100	500
O-B23 Refrigerator cucumber pickles	100	500

1. Prepare ten O-B27 patties from the full base batch, verifying H01–H05 ground-meat temperature.
2. Toast ten O-B25 buns. Melt 20 g cheese on each cooked patty briefly.
3. Add 25 g sauce, 15 g lettuce, 25 g tomato, 10 g onion and 10 g pickles per burger.
4. Serve immediately. Advertise raw patty mass only if clearly labelled; the cooked target is 75 g, not 100 g.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under H01–H05 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Dry patty: lean mince or overholding; domed patty: poor shaping; cold centre: heat too high/short cook.

Substitution: Do not add breadcrumbs to conceal a beef reduction; a lower-cost patty needs a disclosed new formula.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
7,807	10	780.7	70	60	1900	989.3

All values in FCFA. Contribution margin 52.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 833.2 FCFA. At 80% sell-through it falls to 761.6 FCFA per sold unit under the chapter 12 model.

O-P17 Crispy chicken and fries box

Quality target: Crisp chicken pieces, hot fries, sauce in a separate cup and no trapped steam.

Pilot: 5 saleable units. Planning production batch: 25 units. Portion: 220 g cooked coated chicken + 150 g fries + 30 g sauce. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B26 Cooked crispy chicken pieces	1,100	5,500
O-B24 Crisp fries	750	3,750
O-B28 Burger sauce	150	750

1. Prepare O-B26 as smaller even strips, applying the same full cooking checks. Allocate 220 g cooked coated chicken per box.
2. Finish 150 g O-B24 fries per box near handover; salt consistently.
3. Place 30 g O-B28 sauce in a separate cup. Use a vented food box.
4. Serve promptly. Do not describe this boneless-strip portion as a specified number of bone-in pieces.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under H01–H05 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Soft crust: steaming sealed box; uneven cook: mixed piece sizes; low contribution: unweighed chicken.

Substitution: Bone-in chicken needs a separate raw-weight, yield and cook validation; this cost is for boneless portions.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
6,362	5	1,272.3	100	100	2500	1,027.7

All values in FCFA. Contribution margin 41.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 773.2 FCFA. At 80% sell-through it falls to 659.6 FCFA per sold unit under the chapter 12 model.

O-P18 Chicken vegetable samosas

Quality target: Thin crisp triangular shell, generous dry filling and fully sealed corners.

Pilot: 50 saleable units. Planning production batch: 250 units. Portion: 50 x 30 g wrapper + 40 g filling + 2 g paste; about 65–72 g cooked. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B29 Samosa wrapper dough	1,510	7,550
O-B31 Chicken vegetable samosa filling	2,000	10,000
O-B30 Samosa sealing paste	100	500
Oil	200	1,000

1. Make O-B29 and two O-B31 batches. Weigh fifty 30 g wrappers, 40 g chilled filling and 2 g sealing paste each; wrapper spare 10 g.
2. Fold into cones, fill without air pockets and seal firmly. O-B30 must be scaled to supply 100 g usable paste, slightly above one base batch.
3. Fry at a trial 170–175°C for 4–6 minutes until crisp and filling meets H01–H05 reheating control.
4. Drain and use H holding. For catering, agree a timed handover and holding method rather than leaving trays at ambient temperature.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under H01–H05, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in chapter 6.

Troubleshooting: Blisters/leaks: poor seal or wet filling; tough wrapper: too thick; greasy shell: low oil temperature.

Substitution: Vegetable version can replace chicken with cooked lentils only as a separately costed and labelled new filling.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
5,315	50	106.3	10	25	300	158.7

All values in FCFA. Contribution margin 52.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 137.4 FCFA. At 80% sell-through it falls to 123.4 FCFA per sold unit under the chapter 12 model.

O-P19 Loaded chicken shawarma fries

Quality target: Hot crisp fries, evenly distributed chicken and restrained sauces.

Pilot: 5 saleable units. Planning production batch: 25 units. Portion: 180 g fries + 80 g chicken + 25 g garlic sauce + 5 g hot sauce + 15 g pickle. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B24 Crisp fries	900	4,500
O-B18 Cooked chicken shawarma meat	400	2,000
O-B20 Garlic mayonnaise sauce	125	625
O-B21 Cooked hot sauce	25	125
O-B23 Refrigerator cucumber pickles	75	375

1. Finish five portions of 180 g fries close to service.
2. Add 80 g freshly cooked or safely held O-B18 chicken to each, then 25 g O-B20, 5 g O-B21 and 15 g drained O-B23.
3. Use a wide shallow tray; sauce immediately before handover.
4. Never use this product to dispose of meat outside its safe time/temperature controls.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under H01–H05 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Soggy base: over-saucing or deep closed box; uneven value: unweighed chicken; lukewarm fries: early assembly.

Substitution: Offer mild with sauce on the side; removing hot sauce alone does not materially reduce cost.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
4,480	5	895.9	90	60	1800	754.1

All values in FCFA. Contribution margin 41.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 574.9 FCFA. At 80% sell-through it falls to 492.6 FCFA per sold unit under the chapter 12 model.

O-P20 Condensed milk soaked cake portions

Quality target: Moist evenly soaked sponge that holds its shape, with no large pool of liquid.

Pilot: 12 saleable units. Planning production batch: 60 units. Portion: 950 g baked sponge + 800 g soak; 12 portions about 140 g after trim. Scale all components x5; validate capacity and losses before adoption.

Prepared component or ingredient	Pilot g	Production g
O-B32 baked milk cake sponge	950	4,750
O-B33 Three milk soak	800	4,000

1. Bake a full O-B32 sponge and cool. Make or allocate 800 g O-B33; a full soak base leaves 190 g.
2. Prick sponge evenly, apply the weighed cold soak gradually, cover and refrigerate immediately.
3. Chill 4 hours as a quality trial; trim and divide into twelve portions. Gross cake mass 1,750 g less 70 g planned trim leaves 1,680 g, or 140 g each.
4. Sell from a refrigerator under holding class C. No whipped topping is included; add one only with a full formula, cost and cold-chain plan.

Holding and packaging: C — Chilled dairy cake: display and transport at or below 4°C. House pilot policy is prepare for same-day sale and discard unsold portions at close. This is a conservative operating policy, not a validated shelf life.

Troubleshooting: Wet puddles: too much soak or weak sponge; dry centre: uneven perforation; collapse: underbaked sponge.

Substitution: Use UHT milk and cream as specified; evaporated milk substitutions need solids/sweetness recalculation.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
4,010	12	334.2	45	35	650	235.8

All values in FCFA. Contribution margin 36.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 169.0 FCFA. At 80% sell-through it falls to 132.3 FCFA per sold unit under the chapter 12 model.

O-A01 Roast chicken burger — ten-unit pilot

Use O-B25 buns and O-B28 sauce exactly as formulated earlier. This is an oven-roasted alternative to O-P15's crispy chicken.

Component	Final amount per burger g	Ten-unit pilot g
O-B25 baked bun	87	870
O-B18 chicken, after final reheat	110	1,100
O-B28 burger sauce	25	250
Lettuce	15	150
Tomato	25	250
O-B23 drained pickles	10	100
Finished assembly	272	2,720

At 95% reheating retention, prepare 1,157.9 g chilled O-B18 chicken: multiply every ingredient in the O-B18 750 g base by 1.54386 (raw boneless chicken 1,543.9 g). Weigh ten chilled portions about 115.8 g, then confirm 110 g after reheating. Reheat chicken under H01–H05, toast bun, spread sauce, add lettuce/tomato/pickles and assemble promptly. Choose a bun diameter that holds the chicken without excessive stacking. Keep sauce and vegetables refrigerated until assembly. For 50 burgers multiply every table quantity and base scaling factor by five after the pilot passes. Allergens: wheat/gluten, milk, egg and mustard in the specified bases; check purchased sauce labels.

O-A02 Lemon-garlic fish burger — ten-unit pilot

Component	Final amount per burger g	Ten-unit pilot g
O-B25 baked bun	87	870
O-B36 fish, after final reheat	120	1,200
O-B28 burger sauce	25	250
Lettuce	15	150
Tomato	20	200
O-B23 drained pickles	10	100
Finished assembly	277	2,770

Multiply every O-B36 ingredient by 1,200/760 = 1.57895 (raw boneless fish 1,578.9 g). Chilled fish needed is 1,263.2 g, or about 126.3 g per portion, to target 120 g after reheating. Reheat gently but completely under H01–H05; use a broad clean spatula to keep pieces intact. Toast bun, spread sauce, add fish and cold garnish, and hand over promptly. For cooking to order, use measured first-cook yield instead of adding an unnecessary

reheat allowance. For 50 burgers multiply all pilot quantities by five after validation. Allergens: fish, wheat/gluten, milk, egg and mustard; segregate fish from the beef/chicken service tools.

O-A03 Lemon fish wrap — ten-unit pilot

Component	Final amount per wrap g	Ten-unit pilot g
O-B17 cooked flatbread	90	900
O-B36 fish, after final reheat	100	1,000
O-B22 cabbage carrot slaw	45	450
O-B23 drained pickles	15	150
O-B20 garlic sauce	25	250
O-B24 freshly cooked fries	35	350
O-B21 hot sauce, optional but costed	5	50
Finished assembly	315	3,150

Multiply all O-B36 ingredients by 1,000/760 = 1.31579 (raw fillet 1,315.8 g); target 1,052.6 g chilled fish before reheating. Scale every other base by the pilot quantity divided by its listed usable yield. Warm the bread, reheat fish under H01–H05, cook fries fresh, then add weighed slaw, pickles and sauces. Fold gently so fish does not crush the bread; a short final toast is for the bread after the protein has met H01–H05. Serve immediately. The ten portions consume exactly the listed prepared component amounts; any whole-base leftovers require planned same-day use or a waste charge. Scale to 50 wraps by multiplying every pilot quantity by five. Allergens: fish, wheat/gluten and egg/mustard as present in sauces; check milk in purchased ingredients.

O-A04 Beef mini pizza — ten-unit pilot

Component	Amount per pizza g	Ten-unit pilot g
O-B15 pizza dough	150	1,500
O-B16 pizza sauce	40	400
O-B19 beef, final target after pizza bake	50	500
Mozzarella	50	500
Onion	10	100
Sweet pepper	10	100

Use 526.3 g chilled O-B19 beef at the illustrative 95% final-bake retention: scale every O-B19 ingredient by 526.3/700 = 0.75188 (raw beef 751.9 g). Do not separately reheat the beef and then store it again. Prepare ten 150 g dough rounds using O-B15, add 40 g sauce, 52.6 g cold cooked beef, 50 g cheese and 10 g each onion/pepper. Trial bake using O-P12's oven method and check that the meat topping reaches H01–H05 during that same bake. The bake is the beef's one reheating step. Serve immediately or hold hot under H01–H05; do not cool unsold pizzas for another reheat. Target approximately 275–290 g finished pizza, verified by weighing; moisture losses in dough and vegetables are separate from the meat allowance. Scale to 50 pizzas by

multiplying all ingredients by five. Allergens: wheat/gluten and milk, plus anything declared in purchased sauces.

6. Shawarma service and preparation ahead

Component	Standard chicken g	Standard beef g	Small chicken g	Small beef g
Cooked flexible bread	90	90	70	70
Cooked meat	110	110	80	80
Cooked fries	50	50	35	35
Slaw	45	45	30	30
Drained pickles	15	15	10	10
Garlic or sesame sauce	30	30	20	20
Optional hot sauce, included in cost	5	5	3	3
Approximate finished weight	345	345	248	248

Small bread: portion O-B17 dough into approximately 80 g pieces and cook to about 70 g flexible wraps; validate diameter and measured bake loss. Use the same component recipes for both sizes. A standard chicken portion requires 110/0.75 = 146.7 g raw boneless purchase weight; standard beef requires 110/0.70 = 157.1 g. The smaller portion needs 106.7 g raw chicken or 114.3 g raw beef. These assumptions include no bone yield.

Small chicken wrap example: ingredient cost 686.7 FCFA; packaging 65 and other variable allowance 45 FCFA; price 1,500 FCFA; contribution 703.3 FCFA before fixed costs. This assumes exactly usable component weights and must add measured bread/assembly rejects in practice.

Small beef wrap example: ingredient cost 759.9 FCFA; packaging 65 and other variable allowance 45 FCFA; price 1,800 FCFA; contribution 930.1 FCFA before fixed costs. This assumes exactly usable component weights and must add measured bread/assembly rejects in practice.

Vertical spit service workflow

1. Before service: verify raw storage, refrigerated garnish station, handwashing, cleaned probes, clean knives, ventilation and machine guards. Make a cone only as large as forecast sales and the manufacturer’s minimum allow; switch to small-batch pan cooking if those quantities conflict.
2. Keep raw marinated meat cold until loading. Cook continuously according to the machine instructions; do not turn off heat while leaving a partly cooked cone warm for later service. Never add raw meat on top of a cooking cone.
3. Shave thin slices only from a thoroughly cooked exterior using the designated knife. Finish sliced meat immediately on the hot plate; verify the H01–H05 house centre-temperature target. Exterior colour alone is insufficient.
4. Transfer cooked meat with clean utensils into a controlled hot holding pan or directly into a wrap. Never use the raw-meat tray or tongs for finished slices.

5. At close, use the conservative one-day cone policy: discard unsold cone and shaved meat. Do not refrigerate a partly cooked cone for the next day. Reduce tomorrow's starting cone size if waste is excessive. Local-authority doner safety guidance
6. Clean and sanitize raw station, knife, drip areas, hot plate and service surfaces according to equipment and chemical instructions; record disposal and close temperatures.

Station and handover

Flow: refrigerated raw meat → dedicated cooker → clean finishing/hot-holding zone → wrap assembly → toast → packaging → customer. Beside assembly, keep slaw, pickles and sauces refrigerated with separate utensils. Put raw-meat prep physically away from ready-to-eat vegetables. One person handles money or washes hands and changes tasks before touching food. Use measured scoops verified on scales, not free pouring.

Service target after training: a standard wrap in 3–5 minutes when safe components are ready. During low demand, cook to order in smaller batches rather than maintaining an oversized cone. Optional 20 g cheese adds 130 FCFA ingredient cost at the assumed price before other costs; trial an add-on price of 250–300 FCFA. Do not add optional components without a portion and allergen instruction.

Yes: fully cook suitable meat at the production site, cool it rapidly, weigh and label chilled portions, transport them cold, and reheat only the portions ordered before assembly. This can suit the two-site business and avoids keeping a large shawarma cone running during quiet periods. It is a complete cooking and controlled cooling process; briefly warming cooked meat or toasting an assembled cold burger is insufficient.

This revision retains the original 20 core products and eight frozen formats, and adds four optional savoury products O-A01–O-A04. They are development options, not a recommendation to launch 32 products together. H01–H05 and this section govern preparation ahead wherever an older recipe refers to cooking, cooling, transport or reheating. All cooking times, tray loads and yields below require a small measured trial on the actual oven.

Product-by-product service decision

Product	What to prepare ahead	What happens when ordered or in a small service batch
O-P13/O-P14 chicken/beef shawarma	Fully oven-cooked strips, cooled and portioned; bread, sauces and garnish stored separately	Reheat meat thoroughly, cook fries, assemble and toast bread briefly
O-P16 beef cheeseburger	Fully cooked individual patties, rapidly cooled separately	Reheat patty, melt cheese, toast bun, then add cold garnish and sauce
O-P15 crispy chicken burger; O-P17 chicken box	Chilled raw portions and separate coating; or fully cooked chilled chicken after a texture trial	Prefer a full fry to order; the alternative is one complete reheat in a dry oven or suitable fryer, with temperature and crispness checks
O-P06 fish roll; O-P07 beef pie; O-P09 chicken pie; O-P18 chicken samosa	Cooked chilled filling and separate dough; assemble small chilled batches	Final fry/bake heats the filling again: this is its one reheating step. Serve or hold hot; do not chill finished leftovers for another reheat
O-P08 sausage roll	Chilled raw sausage filling enclosed in pastry	Bake completely as the first cook; a finished roll can instead be cooled and reheated once under the documented route

Product	What to prepare ahead	What happens when ordered or in a small service batch
O-P10 egg roll	Cooked chilled egg and separate dough	Final fry is the egg's reheating step; serve or hold hot, with no later cool-and-reheat cycle
O-P12 chicken pizza; O-A04 beef pizza	Chilled cooked topping, dough and sauce kept separate	Final pizza bake is the meat's reheating step; verify the topping temperature and serve
O-P19 loaded chicken fries	Chilled cooked chicken; sauces and pickles separate	Reheat chicken once, cook fries fresh, assemble immediately
O-A01 roast chicken burger	Oven-cooked chicken portion	Reheat, then assemble with toasted bun and chilled garnish
O-A02 fish burger; O-A03 fish wrap	Deboned raw chilled fish for cooking to order, or fully cooked chilled fish for the measured oven route	Cook or reheat as applicable, then assemble; avoid repeated handling of fragile fish

The single-reheat house rule follows UK food business guidance. It is an operating choice for this plan, not a statement of Cameroon law. Track the heat history of the ingredient: putting reheated chicken into a different product does not reset its history. Prefer fully cooked patties and roast chicken for the initial prepare-ahead trial; compare fish and crispy products with freshly cooked controls before adoption.

One controlled route from production to the customer

1. Receive sound ingredients and keep raw proteins chilled under H01–H05. Thaw covered in the refrigerator, below ready-to-eat food. Weigh raw portions with dedicated utensils; wash hands and clean/sanitize the area before handling cooked food. Marinade is not a preservation treatment. Discard raw-contact marinade after use; never use it as a finishing sauce.

2. Fully cook using the appropriate method below. Use the H01–H05 house cooking target throughout; check the thickest portions at the oven's slowest-heating positions. Record batch identity, time and measured food temperature. Browning is a quality cue, not evidence of core doneness. Never partly cook meat or chicken, refrigerate it, then finish later. USDA guidance on partial cooking

3. Start active cooling immediately after cooking. Arrange patties or fillets in a single layer; spread strips/fillings in shallow food-safe pans with circulation around them. Use a blast chiller if available, or another method demonstrated to meet H01–H05. A large refrigerator/freezer does not establish sufficient cooling capacity. Protect loosely covered food during cooling, then cover fully when chilled. Log the cooling checkpoints; do not leave trays on the counter to reach room temperature. FDA employee cooling guidance

4. Weigh chilled usable output with clean utensils, quickly enough to maintain temperature control. Portion into shallow labelled containers; do not pack hot food into stacked deep takeaway boxes. Label recipe, species, batch, cook time, cooling completion, weight, use-by and reheat count. Patties may be formed before cooking; weigh after chilling to check shrinkage. Discard bones, gristle and unsuitable pieces before calculating saleable yield.

5. Transport covered portions with sufficient ice packs in clean insulated carriers, or suitable refrigerated transport. Record dispatch and arrival food temperatures; maintain the H01–H05 cold limit during the journey and sales-site storage. Keep raw and cooked foods separate. Do not rely on journey length alone.

Quarantine and discard food with an unknown unsafe temperature history; reheating cannot repair every form of temperature abuse.

6. Reheat only the ordered portion, directly from refrigeration, using a suitable induction pan with a lid, griddle or small oven. Reheat rapidly to the H01–H05 house target, confirmed at the thickest centre; a holding cabinet is not a reheater. Avoid using the large rotary oven for individual orders. Set a timed method only after testing multiple portions and starting temperatures. Add any measured drinking water early enough for the complete portion to reach the target; do not add cold juices afterwards.

7. Toast the bun or bread separately. Add the hot protein, then refrigerated sauce, washed garnish and drained pickles with clean utensils. Serve promptly. A brief final wrap toast is a bread-quality step after the meat is already safely reheated. Never prepare a stock of warm assembled burgers or wraps on the counter.

8. For a small rush, maintain finished hot protein under H01–H05; monitor actual food temperature. Use the report's two-hour hot-display quality target and reduce batches if drying occurs. Reheated unsold portions are discarded, not cooled again. Initial launch policy remains same-day preparation and sale with disposal at close. Do not infer a refrigerated multi-day shelf life from this process. Planned frozen production needs its own documented storage, thawing and quality trial before rollout; freezing must never be used to rescue mishandled food.

Oven method for O-B18 chicken and O-B19 beef shawarma

Use the complete marinades already specified in O-B18/O-B19. For a pilot use their 1 kg raw boneless meat formulas; multiply every ingredient by five only after proving the smaller load. Bone-in purchasing requires an additional measured deboning yield. Food-grade solid roasting pans with raised edges are needed inside the 40 × 60 cm trays; do not let raw juices drip through racks. Do not bake bread or ready-to-eat pastries underneath or alongside raw proteins.

Chicken: use boneless thigh for a forgiving reheating trial, or test breast separately because it dries more easily. Cut even pieces about 15–20 mm thick, marinate covered under refrigeration for a trial 2–12 hours within the same-day workflow, and spread in one layer without crowding. Trial a preheated 190°C oven, checking early and typically around 15–25 minutes depending on cut and loading. Turn or redistribute only with raw/cooked utensil control. Release by the H01–H05 probe result, not time. A lightly browned surface with moist interior is the quality target. Slice using a clean knife after cooking, then cool promptly; include slicing time in the cooling record.

Beef: choose a tender boneless cut and slice across the grain into even thin strips. Use O-B19 marinade and a short refrigerated trial of 2–6 hours. Trial 190–200°C in a shallow single layer, starting checks around 8–15 minutes. The cook may be shorter or longer; do not hold it in the oven merely to reach an assumed time. Tough stew cuts are not interchangeable with tender strips: use a separately tested covered braise, then cool and portion the shredded result with a revised yield. Do not promise shawarma texture from an unidentified cheap cut.

The original O-B18/O-B19 targets of 750/700 g per 1 kg raw meat are initial cooking assumptions. For the new route, weigh output after cooling and again after reheating. Use only the final usable meat mass for the advertised serving. If the chilled yield matches those assumptions and reheating retains 95%, final yields become 712.5 g chicken and 665 g beef. Do not add raw marinade weight to saleable meat yield. Reheat with only a small measured splash of water when needed; avoid boiling strips in a large pool of liquid.

Beef hamburger patties: oven, portion and finish

Keep O-B27's full formula: beef mince 1,000 g; salt 12 g; black pepper 3 g; paprika 3 g; oil allowance 10 g. Mix the mince and dry seasoning briefly; use the oil for pan release. The meat mixture is 1,018 g before the pan oil. Form ten equal patties approximately 102 g each, about 12–15 mm thick. Do not heavily compress or press them during cooking. Keep raw patties cold until loading.

Trial cook on lightly oiled raised-edge pans at 190°C, turning once if needed, with initial checks around 10–15 minutes. Probe sideways into the centre, reach H01–H05, then cool patties separately in one layer. This fully cooked oven route is an alternative to O-B27's fresh hot-plate method. At service, reheat the patty in a suitable pan/oven, check the centre, then melt the 20 g cheese portion and assemble O-P16. A quick surface sear alone does not adequately reheat the centre.

O-P16 promises 75 g final cooked beef. Its original 750 g batch assumes ten portions directly after cooking. At an illustrative 95% reheating retention, a 75 g chilled patty yields only 71.25 g after reheating. To preserve the 75 g final claim, target 78.95 g chilled per patty and scale the O-B27 batch by 1/0.95: beef 1,052.6 g; salt 12.63 g; pepper 3.16 g; paprika 3.16 g; oil 10.53 g. Form ten larger raw patties from that mixture and validate the final yield. Do not silently describe a smaller reheated portion as 75 g. Whole patties cannot be topped up neatly with scraps; adjust raw forming weight from measured losses.

An alternative is to chill safely prepared raw patties and cook fully to order. Compare that route with prepare-ahead patties for juiciness, waiting time, saleable yield and total cost. Never pre-brown raw patties for later completion.

Crispy chicken, pies, rolls and samosas

For O-B26/O-P15/O-P17, the best starting quality trial is refrigerated raw marinated portions followed by coating and a full fry to order. Keep coating contaminated by raw chicken away from clean dry stock and discard the used portion. Do not partly fry chicken and store it for later finishing. A fully cooked, rapidly chilled portion may be reheated once in a dry oven or suitable fryer, but the centre must meet H01–H05; reject a method that burns the outside, dries the meat or softens the coating. Do not assume the existing 1,150 g coated yield remains unchanged after this extra process.

For O-P06/O-P07/O-P09/O-P18, cool the prepared filling before assembly. The final bake/fry reheats that filling. If production and sales are separate, either transport the finished product hot under control, or transport properly chilled assembled uncooked pastries for final cooking in suitable equipment at the sales site. The latter route needs validated same-day handling and final cooking capacity; no sales-site oven has been confirmed. If neither route is possible, postpone that product rather than chilling a finished twice-heated filling for a third heat cycle. O-P10 cooked egg follows the same principle.

O-P08 is different because O-B10 uses raw sausage meat. Baking the assembled sausage roll is its first complete cook; it may then be rapidly cooled and reheated once before sale, with both stages checked under H01–H05. Do not substitute an already cooked sausage without revising that history. Reheated pastry should be heated in an oven appropriate to the sales site for texture; a warm display unit cannot perform this step.

All new prices below are test assumptions in FCFA, not observed Nkambe prices or promised profit. Base cost includes all its ingredient inputs divided by usable yield. The additional 5% reheating loss is an illustrative planning allowance; replace it with measured chilled-to-final retention for each product. Prepared components in these ten-unit pilots are scaled proportionately; if staff make whole batches and discard balances, charge that extra waste separately.

Product	Ingredients/unit	Pack	Other variable	Test price	Contribution	Margin
O-A01 Roast chicken burger	917.7	60	100	2,200	1,122.3	51.0%
O-A02 Fish burger	921.1	60	100	2,300	1,218.9	53.0%
O-A03 Fish wrap	839.3	65	110	2,200	1,185.7	53.9%
O-A04 Beef mini pizza	835.3	45	90	1,500	529.7	35.3%

The variable allowance covers a provisional combined amount for cooking/reheating energy, cold handling, cleaning and incremental service work. It is not a measured quotation and excludes fixed payroll, rent, financing and taxes. Fixed overhead still has to be paid from contribution. Sales prices must be tested against portion size and local willingness to pay; the fish menu may be unsuitable at the assumed fillet price.

For the existing products, update rather than reuse the original margin unaltered. At unchanged final portions and 95% reheat retention, the protein ingredient surcharge alone is about **24.1 FCFA per O-P16 burger**, **36.6 per standard O-P13 chicken wrap**, **41.0 per standard O-P14 beef wrap**, and **26.6 per O-P19 loaded fries**. Add measured refrigeration, packaging, transport, energy and labour differences as well. These surcharges assume the original cooking yield equals the measured chilled yield; if it does not, recalculate from raw purchase weight. Pies/fillings and coated chicken need their own measured yields rather than a universal 5% deduction.

Cost per final kg of meat = total raw batch ingredient cost ÷ chilled usable kilograms ÷ reheating retention. Reheat retention = final edible weight ÷ chilled portion weight. Yield percentages multiply: chicken at 75% chilled yield and 95% reheat retention gives 71.25% final raw-to-service yield. Margin is not markup, and contribution is not net profit.

Staff release sheet and equipment implications

Record for every production batch	Entry needed
Identity and quantities	Date, recipe, supplier lot, raw kg, operator and intended products
Cooking	Oven/pan setting, tray positions, thickest-centre probe results and time
Cooling	Start and checkpoint times/temperatures against H01–H05; pan depth and batch size
Portioning	Chilled usable kg, portion g, number packed, rejects and label/use-by
Transfer	Dispatch and receipt times/food temperatures, carrier and receiving operator
Service	Reheat result, final portion weight, reheat count, orders sold and disposal
Quality/cost	Juiciness or crispness, assembly time, measured retention, energy and total unit cost

Begin with one or two trays, map cooking and cooling performance, and increase only when the slowest portions meet the controls. The 16-tray oven's physical capacity is not evidence that refrigeration can safely cool 16 trays at once. Separate production of different species where needed for handling, allergens and temperature checks. Use oven-safe pans, cleaned probe thermometers, accurate scales, shallow cooling containers, insulated transport carriers and a suitable sales-site reheating pan or oven. Confirm what is already owned before buying anything.

Cooking in the diesel oven uses diesel plus its listed 3 kW electrical auxiliaries. Refrigeration after cooking and reheating at the sales point still consume power. The sales inverter remains 15 kW with 30 kWh nominal battery storage; replacing the 3.5 kW shawarma appliance with a 5 kW induction cooker can increase instantaneous demand if operated at full power. Rebuild the simultaneous-load schedule with both ice-cream machines, fryers and refrigeration before service. Keep cold storage supplied continuously; this workflow does not justify switching it off to run another appliance.

Launch acceptance: a documented safe process, a successful cold-transport trial, repeatable final weights, acceptable taste against fresh controls, and a revised contribution calculation using actual quotations and losses. If cooling capacity or service reheating fails, reduce the batch or cook fully to order. Do not extend storage time to make an oversized batch commercially convenient.

7. Frozen desserts and portion formats

Gravity machines need a smooth mix that flows reliably. Use the actual manufacturer's approved range, hopper refrigeration requirement, minimum charge and cleaning instructions. The 1 kg formula is a composition pilot, not necessarily a sufficient operating charge. Separate hopper cooling and approved standby functions are model-specific; an unknown machine must not inherit another brand's operating procedure. Taylor gravity soft-serve example

For on-site dairy-mix processing, follow H02. Its batch reference is at least 69°C for 30 minutes throughout the mix in a qualified system. A continuous-process holding time cannot be transferred to an ordinary kettle. Use an approved ready mix if treatment and rapid chilling have not been demonstrated. [N04]

After a validated heat step, chill promptly under H01–H05 and target 1–4°C before aging/loading. Never put warm mix into the hopper. Keep pre-freeze mix covered and chilled, process it in small batches, and do not return dispensed ice cream to the hopper. Use the actual manual's dismantling, detergent, sanitizer concentration/contact time, rinsing and lubrication directions. The safe default launch routine is fresh mix each day, no carryover/rerun, and daily dismantle-clean-sanitize unless a documented approved cycle explicitly differs.

For frozen stock, adopt a storage target of −18°C or colder and log temperatures. This is an operating target, not a validated shelf life; a storage freezer's nameplate does not establish its ability to rapidly harden new bars. Do not assign a commercial life from a generic seven-day trial; validate the actual product, process and storage. Soft serve is dispensed for immediate consumption, not stored as prefilled cups at its serving temperature.

Measuring overrun and mould volume

Overrun measures added volume, not extra mass. Use: overrun % = 100 × (weight of a fixed cup of liquid mix / weight of the same cup of frozen product − 1). At assumed mix density 1.09 kg/L and 30% overrun, frozen density is 1.09/1.30 = 0.838 kg/L. Measure rather than assuming either machine reaches that value. University of Guelph overrun calculations

The owner confirms the large pop is twice the small one, but no absolute volume is known. Calibrate each mould with water to its intended fill line: tared water grams approximate ml. Repeat three times and average. Then weigh a known volume of actual mix for its density. Fill mass = working volume × mix density. Units = floor(usable mix mass / fill mass); remainder = usable mix mass − units × fill mass. Leave manufacturer-required expansion space; do not fill to the brim.

Illustration only	Small	Large
Working fill volume	50 ml	100 ml
Mix density assumed	1.10 g/ml	1.10 g/ml
Fill mass	55 g	110 g
Usable mass from 1 kg formula	970 g	970 g
Full units	17	8
Unallocated remainder charged to batch	35 g	90 g

For a x5 batch, 4,850 g usable mix would fill 88 small or 44 large example moulds, leaving 10 g; it does not have to preserve the inefficient pilot's 17/8 yields. The costing table deliberately uses conservative pilot counts. Recalculate production contribution using actual cavity count, startup residue and integer mould yields; do not mix the two bases when reporting profit.

Frozen mix composition assumptions

For trial calculations, UHT whole milk is assumed 3.5% fat and 13% total solids; cream 35% fat and 40% solids; skim powder 97% solids; condensed milk uses the condensed-milk assumptions in chapter 4; glucose is dry glucose powder, not an 80% solids syrup. FBASE1 is approximately 6.2% fat and 35% total solids. These are engineering estimates to check against labels, not laboratory analyses or legal "ice cream" classifications.

Glucose form, fruit sugar, milk solids and stabilizer affect freezing and viscosity. Do not replace dry glucose with equal-weight syrup or omit stabilizer without rebalancing. FBASE3 uses pasteurized fruit ingredients and supplier-defined handling rather than borrowing a dairy pasteurization schedule. FBASE4's yogurt and pasteurized puree are assembled hygienically cold; the final mix has no claimed kill step. Keep both out of an unapproved dairy hopper.

O-FBASE1 Gravity soft serve vanilla trial mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
UHT milk	660	3,300	—
Cream 35%	100	500	—
Skim milk powder	55	275	—
Sugar	100	500	—
Glucose powder	30	150	—
Condensed milk	50	250	—
Ice cream stabilizer blend	3	15	—
Vanilla	2	10	—

1. Confirm each machine model, feed system and permitted fat, solids, stabilizer and viscosity range before loading. The owner describes one machine as soft serve and the other as gravity; these categories overlap, so the second is provisionally treated as gravity-fed soft serve, with its actual product mode still to be verified. If either manual rejects this mix, use its approved commercial premix until a compatible reformulation is tested.
2. Dry-blend sugar, glucose powder, milk powder and an ice-cream-specific stabilizer/emulsifier blend. The 3 g dose is provisional: follow the actual blend label, offset any dose change with milk and recheck solids.
3. Warm milk and cream, whisk in dry blend, condensed milk and vanilla. Apply the controlled dairy processing protocol F0 before use. Reconstituted milk powder must not be treated as sterile.
4. Cool rapidly, keep covered at 1–4°C and age 4–8 hours as a texture trial if the stabilizer instructions require it. Load only sanitized equipment and cold mix; record input and output weights.
5. Freeze and dispense following the manufacturer's start-up settings. Begin testing at 30% overrun as a calculation assumption, not a machine guarantee. Never add chunks, coarse nut paste or fruit fibre to a gravity feed hopper.

Yield and process note: 1,000 g mix input; 970 g saleable frozen mass after assumed 3% loss. Estimate 6.2% fat and 35% total solids using stated ingredient compositions. The real machine's hold-up may exceed this allowance; a 1 kg pilot is a formulation trial, not necessarily a sufficient machine charge.

Illustrative ingredient cost: 2,100 FCFA per pilot batch; 2,164.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-FBASE2 Vanilla condensed milk popsicle mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
UHT milk	640	3,200	—
Condensed milk	180	900	—
Cream 35%	60	300	—
Sugar	70	350	—
Skim milk powder	45	225	—
Ice cream stabilizer blend	3	15	—
Vanilla	2	10	—

1. Dry-blend sugar, skim powder and stabilizer suitable for frozen bars; verify the blend's label dose.
2. Warm milk and cream, whisk in dry ingredients and condensed milk, then use F0 dairy processing and rapid chilling.
3. Fill cleaned moulds to their measured working volume, leaving the manufacturer-required expansion space. Insert food-safe sticks and freeze in the popsicle machine according to its manual.

4. Demould with potable-water contact controlled by the machine instructions; bag immediately and transfer to the dedicated hardening/storage freezer. Track yield by weight.

Yield and process note: 1,000 g input; 970 g usable mix after loss. Approximate 6% fat, 35% solids and 15% added sucrose before naturally occurring lactose; recheck labels. No overrun assumed.

Illustrative ingredient cost: 2,057 FCFA per pilot batch; 2,120.6 FCFA/kg usable output. This excludes packaging and operating costs.

O-FBASE3 Pineapple fruit ice mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Pineapple puree pasteurized	500	2,500	—
Water	310	1,550	—
Sugar	140	700	—
Glucose powder	45	225	—
Lemon juice	3	15	—
Ice cream stabilizer blend	2	10	—

1. Use commercially pasteurized puree and potable water. Confirm the stabilizer is suitable for acidic water ices, not simply any dairy stabilizer.
2. Heat water with sugar, glucose and stabilizer according to its hydration instructions, then combine with puree using sanitized equipment. Select a supplier-approved fruit processing schedule; dairy time/temperature guidance is not a fruit-process validation.
3. Chill promptly under H01–H05. Measure soluble solids with a refractometer; a trial target is 22–25 degrees Brix, adjusted for the actual fruit. Reweigh any formula adjustment and re-cost.
4. Fill measured moulds, insert sticks, freeze, demould hygienically, bag and harden. This formula is for moulded ices, not automatically for the soft-serve hopper.

Yield and process note: 1,000 g input; 970 g usable mix. At 12 degrees Brix puree, calculated soluble solids are roughly 24.5% before small acid/stabilizer contributions. Zero milk fat; not dairy ice cream.

Illustrative ingredient cost: 1,017 FCFA per pilot batch; 1,048.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-FBASE4 Mango yogurt popsicle mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Yogurt plain	500	2,500	—
Mango puree pasteurized	250	1,250	—
Condensed milk	120	600	—
Sugar	70	350	—
Water	58	290	—
Ice cream stabilizer blend	2	10	—

1. Use commercially pasteurized fruit puree and yogurt made from pasteurized milk, kept cold. Use a stabilizer intended for acidic frozen yogurt bars.
2. Hydrate stabilizer with sugar and water according to its instructions, then cool rapidly before blending into the yogurt, puree and condensed milk.
3. Blend with sanitized tools, keep at or below 4°C and fill moulds promptly. Do not claim live-culture counts or probiotic benefits.
4. Freeze, demould, bag and harden as for milk pops. This is a cold-assembled process requiring hygienic ingredients and validated local handling, not a claim of on-site pasteurization.

Yield and process note: 1,000 g input; 970 g usable mix. Provisional solids around 26% and fat around 2.7% using yogurt at 12% solids and 3.5% fat, mango at 15% solids, and the stated condensed milk assumption; actual labels control the final balance.

Illustrative ingredient cost: 1,902 FCFA per pilot batch; 1,961.1 FCFA/kg usable output. This excludes packaging and operating costs.

O-FBASE5 Chocolate sundae sauce

Pilot input 955 g; target usable output 900 g. Planning production batch is x5: input 4,775 g and usable output 4,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Water	400	2,000	—
Sugar	250	1,250	—
Cocoa	100	500	—
Condensed milk	200	1,000	—
Salt	2	10	—
Vanilla	3	15	—

1. Whisk cocoa and sugar, gradually add water and bring to a simmer while stirring.
2. Add condensed milk and salt; cook gently to 900 g finished mass without scorching, then add vanilla.
3. Cool under H01–H05 and keep refrigerated in labelled small containers.

4. Serve 20 g cold sauce per sundae; do not return used sauce to the storage batch.

Yield and process note: Input 955 g; about 55 g evaporation. This milk-containing sauce requires refrigeration.

Illustrative ingredient cost: 1,539 FCFA per pilot batch; 1,709.8 FCFA/kg usable output. This excludes packaging and operating costs.

O-FBASE6 Soft sandwich cookies

Pilot input 1,086 g; target usable output 900 g. Planning production batch is x5: input 5,430 g and usable output 4,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Plain flour	500	2,500	—
Margarine	220	1,100	—
Sugar	220	1,100	—
Egg	100	500	—
Cocoa	35	175	—
Baking powder	4	20	—
Salt	4	20	—
Vanilla	3	15	—

1. Cream margarine and sugar just until smooth; mix in egg and vanilla.
2. Fold in sifted flour, cocoa, baking powder and salt. Chill 20 minutes, roll 3–4 mm and cut evenly.
3. Bake at 170–180°C for a trial 9–12 minutes until set but not rock-hard. Cool fully; target 900 g baked cookies.
4. Match by diameter. Use two 20 g cookies per ice cream sandwich; record breakage separately.

Yield and process note: Input 1,086 g; expected cooked usable cookies 900 g after bake/trim loss. Verify freezer bite before launch.

Illustrative ingredient cost: 1,416 FCFA per pilot batch; 1,572.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-FBASE7 Chocolate dipping coating

Pilot input 1,000 g; target usable output 980 g. Planning production batch is x5: input 5,000 g and usable output 4,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

Ingredient or prepared component	Pilot g	Production g	Baker %
Compound chocolate coating	950	4,750	—
Oil	50	250	—

1. Use a commercially supplied compound coating intended for frozen confectionery. Melt only within its manufacturer's temperature range.
2. Add the stated oil only if its instructions allow thinning; otherwise use 1,000 g coating and update cost.
3. Dip fully hardened dry milk bars briefly using clean tools. Keep moisture out of the coating bowl.
4. Return coated bars immediately to frozen storage; do not put unused dipped-product crumbs back into clean coating stock.

Yield and process note: Input 1,000 g; 20 g bowl/handling loss. Remaining coating working stock must be recorded; do not charge it repeatedly if still usable.

Illustrative ingredient cost: 4,355 FCFA per pilot batch; 4,443.9 FCFA/kg usable output. This excludes packaging and operating costs.

O-F01 Vanilla soft serve cup or cone

Quality target: Smooth fine texture, clean vanilla flavour and stable shape for immediate eating.

Portion and pilot yield: 10 x 95 g; 20 g serving remainder. Costed saleable units: 10.

Prepared component	Pilot g	Planning x5 g
O-FBASE1 Gravity soft serve vanilla trial mix	970	4,850

1. Prepare O-FBASE1 under F0 and dispense using the machine manual. Weigh ten 95 g servings; a 20 g residual is charged to the batch.
2. At the illustrative 1.09 g/ml mix density and 30% overrun, 95 g occupies about 113 ml; choose a cup with adequate headroom. A cone is an alternative package, not a second product.
3. Serve immediately. Keep hopper mix cold, log temperatures and follow the specific daily dismantling/sanitation or approved heat-treatment cycle.

Compatibility and yield: Prototype for the reported soft serve and gravity machine arrangement, conditional on both manuals. Chocolate alternative per 1 kg input: replace 25 g milk with 25 g cocoa; recheck viscosity, freezing and cost. No chunk inclusions.

Troubleshooting: Icy: unstable temperature/low solids; weak swirl: formulation or machine setting; sandy: excessive milk solids or storage abuse.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
2,100	10	210.0	35	30	600	325.0

All values in FCFA. Contribution margin 54.2%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 283.0 FCFA. At 80% sell-through it falls to 256.2 FCFA per sold unit under the chapter 12 model.

O-F02 Chocolate peanut sundae

Quality target: Clean layers, measured topping and crunch without a flooded cup.

Portion and pilot yield: 120 g soft serve + 20 g sauce + 10 g nuts. Costed saleable units: 5.

Prepared component	Pilot g	Planning x5 g
O-FBASE1 Gravity soft serve vanilla trial mix	600	3,000
O-FBASE5 Chocolate sundae sauce	100	500
Peanut roasted	50	250

1. Use O-FBASE1 and fully chilled O-FBASE5; use reputable commercially roasted peanuts, inspect and store dry.
2. Dispense 120 g ice cream into each cup, add 20 g sauce and 10 g nuts with dedicated utensils.
3. Serve immediately. Offer a nut-free recipe only where cross-contact can be controlled; removing visible nuts is insufficient.

Compatibility and yield: A service format from existing soft serve, not an extra freezer formulation. Cost uses five servings, 750 g total.

Troubleshooting: Rapid melt: warm sauce/cup; excessive cost: unmeasured nuts; stale taste: old nuts.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,560	5	312.0	60	35	1000	593.0

All values in FCFA. Contribution margin 59.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 530.6 FCFA. At 80% sell-through it falls to 491.3 FCFA per sold unit under the chapter 12 model.

O-F03 Chocolate milkshake

Quality target: Drinkable thick consistency, no ice lumps and consistent fill.

Portion and pilot yield: 150 g soft serve + 120 g milk + 20 g sauce. Costed saleable units: 5.

Prepared component	Pilot g	Planning x5 g
O-FBASE1 Gravity soft serve vanilla trial mix	750	3,750
UHT milk	600	3,000
O-FBASE5 Chocolate sundae sauce	100	500

1. Confirm a food-safe high-speed blender is available; the mixing machine/pot alone is not evidence of blending capability.

2. Blend one 150 g ice cream portion with 120 g cold UHT milk and 20 g chilled sauce for 15–25 seconds, until just smooth.
3. Use a roughly 350 ml cup after measuring actual aeration volume. Serve immediately; wash and sanitize blender parts between production sessions and allergens.

Compatibility and yield: Conditional on blender availability. No raw egg. 1,450 g input for five portions; mass 290 g each before trivial blender residue, charged via the loss model.

Troubleshooting: Thin: overblending/warm milk; excessive sweetness: unmeasured syrup; lumps: inadequate blender.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
2,515	5	502.9	75	40	1200	582.1

All values in FCFA. Contribution margin 48.5%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 481.5 FCFA. At 80% sell-through it falls to 427.6 FCFA per sold unit under the chapter 12 model.

O-F04 Chocolate cookie ice cream sandwich

Quality target: Biteable cookie with a firm clean filling, no large ice crystals.

Portion and pilot yield: 2 x 20 g cookies + 70 g frozen filling. Costed saleable units: 10.

Prepared component	Pilot g	Planning x5 g
O-FBASE6 Soft sandwich cookies	400	2,000
O-FBASE1 Gravity soft serve vanilla trial mix	700	3,500

1. Bake O-FBASE6, cool fully and select twenty 20 g cookies.
2. Dispense or spread O-FBASE1 into clean shallow moulds, portioning 70 g per filling. Harden quickly in suitable freezing equipment; storage-freezer capacity alone does not prove hardening throughput.
3. Sandwich each hardened filling between two cookies, wrap individually and return to frozen storage immediately. Pilot freezer bite and condensation resistance.

Compatibility and yield: Uses soft-serve output hardened in shallow forms; no batch freezer is assumed. Start with small production, and approve only after a recorded hardening test.

Troubleshooting: Hard cookie: overbake; crumbly assembly: unmatched diameter; icy filling: slow hardening.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
2,145	10	214.5	25	35	700	425.5

All values in FCFA. Contribution margin 60.8%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 382.6 FCFA. At 80% sell-through it falls to 356.9 FCFA per sold unit under the chapter 12 model.

O-F05 Condensed milk popsicles

Quality target: Creamy clean release and securely centred stick.

Portion and pilot yield: Example small 50 ml mould at density 1.10 g/ml: 55 g each. Costed saleable units: 17.

Prepared component	Pilot g	Planning x5 g
O-FBASE2 Vanilla condensed milk popsicle mix	970	4,850

1. Prepare O-FBASE2 and use the popsicle machine manual for mould immersion, stick timing and freeze completion.
2. At example 50 ml working fill and 1.10 g/ml density, make 17 x 55 g bars, with 35 g remainder. At example 100 ml, make 8 x 110 g bars with 90 g remainder. Actual moulds are unconfirmed.
3. Bag each bar, include the stick cost, and move immediately to hardening/frozen storage. Use F0 storage policy.

Compatibility and yield: One product in two sizes. Larger example price 500 FCFA (test 450–650); separate cost row supplied. Densities and volumes must be measured.

Troubleshooting: Icy: low solids/slow freezing; too soft: excessive sugar; pulled-out stick: incomplete freezing.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
2,057	17	121.0	20	20	300	139.0

All values in FCFA. Contribution margin 46.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 114.8 FCFA. At 80% sell-through it falls to 98.8 FCFA per sold unit under the chapter 12 model.

O-F06 Pineapple fruit ice pops

Quality target: Distinct fruit flavour and biteable ice texture, not plain coloured sugar water.

Portion and pilot yield: Example small 50 ml x 1.10 g/ml = 55 g. Costed saleable units: 17.

Prepared component	Pilot g	Planning x5 g
O-FBASE3 Pineapple fruit ice mix	970	4,850

1. Make O-FBASE3; measure fruit solids and density before filling.
2. Use the same illustrative 17 small or 8 large bar yield calculation as O-F05 only if measured density is 1.10 g/ml. Freeze, demould and bag under the machine procedure.
3. Offer the large 100 ml example at 400 FCFA (test 350–500). Keep a separate dairy-contact workflow if marketing as dairy-free.

Compatibility and yield: One product in two sizes. Do not feed this fruit formula into gravity soft serve unless its manual expressly permits it.

Troubleshooting: Rock-hard: too little sugar/solids; sticky soft: excessive Brix; weak fruit: diluted puree.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,017	17	59.9	20	18	250	152.2

All values in FCFA. Contribution margin 60.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 140.2 FCFA. At 80% sell-through it falls to 127.7 FCFA per sold unit under the chapter 12 model.

O-F07 Mango yogurt frozen bar

Quality target: Balanced mango acidity, smooth bite and firm shape.

Portion and pilot yield: Example 70 g bars: 12 units; 130 g remainder. Costed saleable units: 12.

Prepared component	Pilot g	Planning x5 g
O-FBASE4 Mango yogurt popsicle mix	970	4,850

1. Make O-FBASE4 with hygienic ready-to-eat ingredients. Portion 70 g each into a mould with sufficient verified working volume.
2. Freeze promptly, demould hygienically and wrap. Charge the 130 g remainder to the twelve bars until a measured mould/batch combination reduces it.
3. Run a sourness and freezer-bite test; declare milk and do not imply health or live-culture benefits.

Compatibility and yield: Moulded yogurt dessert, not a gravity-machine frozen-yogurt recipe. Approximate 64 ml at 1.10 g/ml density, subject to measurement.

Troubleshooting: Curdled/grainy: acid imbalance; icy: low solids; weak mango: poor puree.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
1,902	12	158.5	25	25	450	241.5

All values in FCFA. Contribution margin 53.7%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 209.8 FCFA. At 80% sell-through it falls to 189.3 FCFA per sold unit under the chapter 12 model.

Planning clarification: 12 bars is the source's conservative costing count. At 970 g usable mix and 70 g per bar, 13 whole bars are arithmetically possible with 60 g remaining; use measured mould capacity, losses and saleable count before changing costs.

O-F08 Chocolate dipped peanut milk bar

Quality target: Thin crisp coating, creamy core and clean wrapper.

Portion and pilot yield: 66 g milk bar + 18 g coating allocation + 5 g peanut. Costed saleable units: 10.

Prepared component	Pilot g	Planning x5 g
O-FBASE2 Vanilla condensed milk popsicle mix	660	3,300
O-FBASE7 Chocolate dipping coating	180	900
Peanut roasted	50	250

1. Freeze ten 66 g O-FBASE2 cores using an appropriately sized mould and harden fully.
2. Use O-FBASE7 following coating supplier temperature instructions; dip dry cold cores quickly. Target 15 g adhering chocolate per bar; the 18 g allocation includes 3 g drips/handling.
3. Sprinkle 5 g roasted peanut on each before coating sets, then wrap and freeze immediately. Keep nuts segregated.

Compatibility and yield: Approximate sale weight 86 g; 30 g coating loss across ten bars is charged. This is an added-value process and allergen burden, not simply a new flavour.

Troubleshooting: Coating falls off: surface frost; thick shell: wrong viscosity; cracking: excessive coating.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

Batch ingredient cost	Saleable units	Ingredients/unit	Pack/unit	Other variable/unit	Test price	Contribution/unit
2,290	10	228.9	30	35	600	306.1

All values in FCFA. Contribution margin 51.0%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 260.3 FCFA. At 80% sell-through it falls to 232.6 FCFA per sold unit under the chapter 12 model.

8. Specialist bread, pastry, meals and cultured dairy

These 28 September formulas preserve meaningful differences: true-butter brioche, baguettes, cinnamon rolls, biscuits, cake-style beignets, vegetable samosas, cultured yogurt and other formats. They use their own yields and batch multiples. Generic preparation and costing instructions have been consolidated in H01–H05 and the commercial chapter; product-specific formulas, methods, defects, variants and packaging remain.

Every S-series card is a trial specification. The original fit classes mean: 1 = close fit with the stated equipment; 2 = operational adjustments/accessories; 3 = additional equipment or process qualification; 4 = unsuitable for the stated setup. These are judgments, not measured commissioning results. Recipe references [S01]–[S34] point to the inherited research register in the final chapter.

S-B01 — Daily French-style baguette

Long loaf 35–40 cm; one loaf/retail unit. Steam capability must be confirmed; without it trial a different crust expectation.

Stage: Launch • Fit class 1 • Pilot 5 units • Production example ×10 = 50 units.

Commercial case: Households and breakfast vendors; existing bread sector supported historically by S32. Correct weight and two reliable daily windows are the hypothesis.

Technical basis: [S02], [S25], [S32]. All exact formulas and operating adaptations here require controlled trials.

Formula — Baguette dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	10,000
Water	660	66.000%	6,600
Salt	20	2.000%	200
Instant yeast Red	5	0.500%	50
TOTAL	1,685	168.500%	16,850

Effective ingredient-water estimate: 66.00% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,685 g input − 33.7 g handling/unretained (2%) − 247.695 g process loss (15% of post-handling mass) = 1,403.605 g retained. Allocation: 280.721 g finished component per unit. Production retained mass: 14,036.05 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,685 g; modeled losses 281.395 g; net edible output 1,403.605 g = 5 × 280.721 g expected finished units, with no modeled remainder. Production example: 16,850 g input → 14,036.05 g output across 50 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 330.26 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Weigh all water, including any ice, within the formula. Mix flour and water 3 minutes slowly; rest 15 minutes before adding instant yeast and salt.
2. Develop for an initial 3–6 minutes at second speed, stopping at a thin elastic membrane and 24–26°C dough. Record friction; never extend mixing just to meet a clock.
3. Bulk ferment covered 45–60 minutes at 25–27°C; fold once if slack. Divide the usable dough equally into the stated count; rest preshapes 15–20 minutes.
4. Mold gently to 35–40 cm, within actual tray length. Proof 45–75 minutes at 27°C, 75–80% RH, until expanded with slowly returning fingertip indentation.
5. Score 3–5 overlapping cuts at about 30° blade angle. Trial 230°C loading and 215–220°C bake for 20–25 minutes. Use only the oven's approved steam system at loading; vent near the end.
6. Confirm a set crumb, dry sidewalls and a trial center of 96–98°C. Cool uncovered on protected racks until no condensation occurs in the selected paper sleeve. Weigh after cooling.

Quality, defects and controlled variants

Burst sides → short proof or poor scoring: correct proof/end cuts. Flat loaf → overproof/weak dough: shorten wait and assess flour. Leathery crust → inadequate final drying: check approved vent.

For a 250 g finished specification, calculate raw weight from measured bake loss; do not print 250 g based on this theoretical pilot.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-B02 — Kumba-style soft block loaf

Four pan loaves per pilot; shape and sweetness must be taste-tested with Nkambe customers.

Stage: Launch • Fit class 2 • Pilot 4 units • Production example ×8 = 32 units.

Commercial case: Familiar Cameroonian name (S05), but there is no single verified universal Kumba formula. Test soft block versus dense block preference.

Technical basis: [S05], [S03]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	8,000
Water	550	55.000%	4,400
Sugar	100	10.000%	800
Margarine 80%	60	6.000%	480
Skim milk powder	30	3.000%	240
Salt	18	1.800%	144
Instant yeast Gold	12	1.200%	96
Nutmeg	2	0.200%	16
TOTAL	1,772	177.200%	14,176

Effective ingredient-water estimate: 56.08% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,772 g input − 35.44 g handling/unretained (2%) − 208.387 g process loss (12% of post-handling mass) = 1,528.173 g retained. Allocation: 382.043 g finished component per unit. Production retained mass: 12,225.382 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,772 g; modeled losses 243.827 g; net edible output 1,528.173 g = 4 × 382.043 g expected finished units, with no modeled remainder. Production example: 14,176 g input → 12,225.382 g output across 32 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 434.14 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Blend flour, water, milk powder, sugar and yeast slowly for 4 minutes. Add salt and margarine; develop 4–7 minutes to smooth extensible dough at 25–27°C.
2. Bulk rest 40–50 minutes, covered. Divide equally, degas evenly and rest 10–15 minutes. Mold a close crumb with progressive roller reduction, never force stiff dough through a narrow gap.
3. Place in greased open loaf tins around 20 × 10 × 8 cm as a first trial; record tin volume. Proof at 28–30°C, 75–80% RH for 60–90 minutes until puffy and approximately approaching the rim.
4. Bake without added steam at 185–195°C for 28–38 minutes. Seek uniform brown sidewalls and 94–96°C crumb; adjust time to measured pan load.
5. Unmold after 5–10 minutes, cool on racks and bag only without condensation. Cut a trial loaf to check fine, soft crumb rather than a compressed wet band.

Quality, defects and controlled variants

Dense gummy center → underproof/underbake. Tearing → insufficient rest. Greasy crumb → fat type or underdevelopment. Correct one variable per trial.

Do not call margarine bread butter bread; replace fat only by a separately reconciled butter trial.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, milk.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-B03 — True-butter pain beurre

Six short loaves; no added coloring and no post-bake raw egg.

Stage: Phase 2 • Fit class 1 • Pilot 6 units • Production example ×8 = 48 units.

Commercial case: Butter aroma is a premium hypothesis. Test its price premium against S-B02 rather than assuming butter cost is recoverable.

Technical basis: [S01], [S03]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	8,000
Water	550	55.000%	4,400
Butter 82%	120	12.000%	960
Sugar	80	8.000%	640
Salt	20	2.000%	160
Instant yeast Gold	12	1.200%	96
TOTAL	1,782	178.200%	14,256

Effective ingredient-water estimate: 56.92% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,782 g input − 35.64 g handling/unretained (2%) − 209.563 g process loss (12% of post-handling mass) = 1,536.797 g retained. Allocation: 256.133 g finished component per unit. Production retained mass: 12,294.374 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,782 g; modeled losses 245.203 g; net edible output 1,536.797 g = 6 × 256.133 g expected finished units, with no modeled remainder. Production example: 14,256 g input → 12,294.374 g output across 48 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 291.06 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix flour, water, yeast and sugar 3–4 minutes slowly; add salt and develop until the dough begins to stretch cleanly.
2. Add pliable butter in three additions, mixing until each disappears. Finish at 24–26°C with a smooth membrane, approximately 5–8 additional minutes.
3. Bulk ferment 45–60 minutes at 26°C. Divide, rest 15 minutes and mold six short cylinders about 20–22 cm.
4. Proof 45–75 minutes at 28°C and 75–80% RH until puffy. Make shallow decorative scores without crushing.
5. Bake at 185–195°C, no steam, for 20–28 minutes. Seek 93–96°C crumb, no wet seam and even color; cool fully before bagging.

Quality, defects and controlled variants

Butter leakage → warm or underdeveloped dough. Dark outside/raw inside → reduce setpoint and extend bake. Dry next day → measure loss and pack barrier.

Margarine creates a differently named soft loaf; never use a butter claim for that substitution.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, milk.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-B04 — Sandwich pan loaf / pain de mie

Three loaves; slicer is not assumed. Supply whole loaves or hand-sliced preorder quantities.

Stage: Launch • Fit class 2 • Pilot 3 units • Production example ×8 = 24 units.

Commercial case: Breakfast shops and institutional sandwiches: consistent slice dimensions and delivery matter more than elaborate shapes. Test five prospective buyers, no contract assumed.

Technical basis: [S04], [S03]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dough

Percent basis: total wheat flour in this component = 100% (baker’s percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	8,000
Water	610	61.000%	4,880
Sugar	50	5.000%	400
Oil	50	5.000%	400
Skim milk powder	30	3.000%	240
Salt	20	2.000%	160
Instant yeast Red	10	1.000%	80
TOTAL	1,770	177.000%	14,160

Effective ingredient-water estimate: 61.00% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,770 g input − 35.4 g handling/unretained (2%) − 173.46 g process loss (10% of post-handling mass) = 1,561.14 g retained. Allocation: 520.38 g finished component per unit. Production retained mass: 12,489.12 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,770 g; modeled losses 208.86 g; net edible output 1,561.14 g = 3 × 520.38 g expected finished units, with no modeled remainder. Production example: 14,160 g input → 12,489.12 g output across 24 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 578.2 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix dry ingredients and water 4 minutes slowly. Add oil and finish to full windowpane at 25–27°C, initial 5–8 minutes faster.
2. Bulk ferment 45–60 minutes; divide into three equal pieces and bench-rest 15 minutes.
3. Mold tight even cylinders without tearing and put in greased tins around 22 × 11 × 10 cm. Trial lid clearance; pan dimensions are not confirmed equipment.
4. Proof at 28°C, 75–80% RH for about 60–85 minutes; for a lidded loaf close when near 1.5–2 cm below the lid and only after trialing actual pan volume.
5. Bake at 190–200°C for 32–40 minutes without steam. Check 94–96°C crumb, set sidewalls, then unmold and cool completely before manual slicing with a guarded work procedure.

Quality, defects and controlled variants

Waisted sides → early unmolding/underbake. Dense top → lid overfill. Ragged slices → slicing warm or unsuitable knife.

Open-top pan route avoids buying lids immediately; define a distinct pan/load specification.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, milk.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-B05 — Soft burger buns

20 buns; no wash or seed in the costed base, simplifying allergens and pickup.

Stage: Launch • Fit class 1 • Pilot 20 units • Production example ×5 = 100 units.

Commercial case: B2B or in-house burger platform. Reserve a part of the same dough for S-F03 only under separate lot/quantity records.

Technical basis: [S03], [S04]. All exact formulas and operating adaptations here require controlled trials.

Formula — Soft bun dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	5,000
Water	560	56.000%	2,800
Sugar	80	8.000%	400
Oil	80	8.000%	400
Skim milk powder	40	4.000%	200
Salt	18	1.800%	90
Instant yeast Gold	12	1.200%	60
Whole egg	50	5.000%	250
TOTAL	1,840	184.000%	9,200

Effective ingredient-water estimate: 59.80% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,840 g input − 36.8 g handling/unretained (2%) − 216.384 g process loss (12% of post-handling mass) = 1,586.816 g retained. Allocation: 79.341 g finished component per unit. Production retained mass: 7,934.08 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,840 g; modeled losses 253.184 g; net edible output 1,586.816 g = 20 × 79.341 g expected finished units, with no modeled remainder. Production example: 9,200 g input → 7,934.08 g output across 100 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 90.16 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix all ingredients except oil for 4 minutes slowly. Add oil and develop to a smooth windowpane, keeping dough at 25–27°C.
2. Bulk ferment covered 45–60 minutes. Divide the usable dough equally into 20 pieces, round with a sealed base and rest 10 minutes.
3. Flatten to approximately 9 cm; place with at least 3 cm expansion clearance. Proof at 28–30°C, 75–80% RH for 45–70 minutes to a soft slow-return indentation.
4. Bake 13–18 minutes at 185–195°C without steam. Check a set center near 94–96°C. Cool on racks completely before bagging or splitting.
5. Check 10 raw and cooled weights. Test one bun with a hot patty and sauce for 15 minutes; approve softness without base collapse.

Quality, defects and controlled variants

Flat crown → overproof or careless transfer. Splits → tight skin/underproof. Dry bun → too much bake loss.

For sesame variant, add and cost 1 g seed per bun before baking; seeds introduce sesame and may shed. Base recipe excludes them.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-B06 — Professional 50%-butter brioche

Five plain loaves; original brioche retained and deepened. Not Brioche Vendéenne or a protected-origin claim.

Stage: Phase 2 • Fit class 3 • Pilot 5 units • Production example ×5 = 25 units.

Commercial case: Premium weekend preorder or festive braid. Butter cost and chilling capacity, not maximum mixer size, determine feasibility.

Technical basis: [S01], [S03]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	5,000
Whole egg	500	50.000%	2,500
Butter 82%	500	50.000%	2,500

Ingredient / specification	Pilot g	Percent	Production g
Sugar	150	15.000%	750
Whole milk 3.5%	100	10.000%	500
Salt	20	2.000%	100
Instant yeast Gold	15	1.500%	75
Vanilla	5	0.500%	25
TOTAL	2,290	229.000%	11,450

Effective ingredient-water estimate: 54.75% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 2,290 g input − 45.8 g handling/unretained (2%) − 269.304 g process loss (12% of post-handling mass) = 1,974.896 g retained. Allocation: 394.979 g finished component per unit. Production retained mass: 9,874.48 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,290 g; modeled losses 315.104 g; net edible output 1,974.896 g = 5 × 394.979 g expected finished units, with no modeled remainder. Production example: 11,450 g input → 9,874.48 g output across 25 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 448.84 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Use strong extensible flour (trial 11.5–13% protein), shell-free weighed chilled egg, and block butter. A 2.29 kg pilot needs a suitable small mixer or manual kneading, not an unverified minimum spiral load.
2. Start eggs/milk at 2–5°C and butter pliable at 16–18°C. Mix everything except butter 4–5 minutes slowly then develop until elastic.
3. Add butter in 4–6 portions on low speed; finish only once glossy with a thin windowpane. Aim for 24–26°C dough. Stop and chill if butter oils out; do not add flour to hide melting.
4. Bulk rest 45–60 minutes at 25–26°C. Flatten to 3–4 cm in covered shallow trays and refrigerate at 2–4°C for 8–16 hours. Log center cooling; never chill a deep 11 kg ball.
5. Divide chilled dough into five loaves; divide each loaf into five equal balls. Fit an open pan around 20–22 × 10 × 8 cm. Keep waiting dough chilled.
6. Proof at 26–28°C and 75–80% RH for an initial 2–3 hours until clearly puffy with a slow-return indentation. No free butter may pool below pieces.
7. Bake at 165–175°C, no steam, for about 30–40 minutes; aim for 90–94°C center plus fully set crumb. Color alone is unreliable. Rest 5–10 minutes, unmold and cool.
8. Approve a cut loaf showing fine pull-apart strands and clean butter aroma. Weigh raw and cooled units, record refrigeration occupancy and cost before accepting the premium format.

Quality, defects and controlled variants

Greasy → butter too warm/added too early. Dense → weak gluten or short proof. Burnt top/gummy middle → bake too hot. Collapsed sides → underbake or overproof.

Optional wash: prepare 50 g egg + 5 g milk per pilot, apply before bake only; weigh actual pickup and leftover separately. This is excluded from base mass and cost; base needs no wash.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-B07 — Milk bread finger rolls / pain au lait

24 finger rolls; milk is not interchangeable with the same mass of water without reformulation.

Stage: Launch • Fit class 1 • Pilot 24 units • Production example ×5 = 120 units.

Commercial case: Child/family breakfast and school preorders, subject to school authorization and allergen information. Milk flavour is a hypothesis to test against price.

Technical basis: [S03], [S05]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	5,000
Whole milk 3.5%	640	64.000%	3,200
Sugar	120	12.000%	600
Butter 82%	80	8.000%	400
Salt	18	1.800%	90
Instant yeast Gold	12	1.200%	60
TOTAL	1,870	187.000%	9,350

Effective ingredient-water estimate: 57.28% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,870 g input − 37.4 g handling/unretained (2%) − 219.912 g process loss (12% of post-handling mass) = 1,612.688 g retained. Allocation: 67.195 g finished component per unit. Production retained mass: 8,063.44 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,870 g; modeled losses 257.312 g; net edible output 1,612.688 g = 24 × 67.195 g expected finished units, with no modeled remainder. Production example: 9,350 g input → 8,063.44 g output across 120 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 76.358 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix flour, milk, sugar and yeast 4 minutes slowly. Add salt then butter; develop to a supple windowpane at 25–27°C.
2. Ferment covered 45–60 minutes. Divide equally, round, rest 10–15 minutes and roll to 10–12 cm fingers.
3. Proof at 28°C, 75–80% RH for 45–70 minutes until visibly light. Make three shallow snips without deflating.
4. Bake 12–17 minutes at 180–190°C without steam. Check an evenly set 94–96°C center and a thin golden crust.
5. Cool protected on racks; bag without condensation. Weigh a three-roll family pack as well as individual pieces.

Quality, defects and controlled variants

Tough crumb → dusting flour or underhydration. Dark crust → too hot for sugar. Weak proof → yeast not suited to sweet dough.

Vanilla or coconut toppings belong to separately costed variants, not extra strategic products.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, milk.

Cool fully before food-grade bagging; use paper/ventilated protection for crisp baguettes. Label lot and allergens. Covered dry transport; no unvalidated multi-day life. Assess at 0/6/24/48 hours; safety/storage approval remains H05. No reheating is required for the plain product.

S-P01 — Yeast donuts: plain, glazed and beignets sucrés

30 dough-divided donuts; oil bath charge is not absorbed oil. Record all discarded/filter oil in process consumables.

Stage: Launch • Fit class 1 • Pilot 30 units • Production example ×4 = 120 units.

Commercial case: Plain base with two low-complexity finishes. Glaze creates a visual premium; beignets sucrés here means a rich solid yeast dough, not a second invented dough.

Technical basis: [S03], [S22]. All exact formulas and operating adaptations here require controlled trials.

Formula — Yeast dough

Percent basis: total wheat flour in this component = 100% (baker’s percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	4,000
Water	490	49.000%	1,960
Whole egg	100	10.000%	400
Sugar	100	10.000%	400
Butter 82%	80	8.000%	320
Skim milk powder	30	3.000%	120
Salt	18	1.800%	72
Instant yeast Gold	15	1.500%	60
Vanilla	5	0.500%	20
TOTAL	1,838	183.800%	7,352

Effective ingredient-water estimate: 57.88% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,838 g input − 36.76 g handling/unretained (2%) − 216.149 g process loss (12% of post-handling mass) = 1,585.091 g retained. Allocation: 52.836 g finished component per unit. Production retained mass: 6,340.365 g. All losses are unmeasured planning allowances.

Formula — Absorbed frying oil

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Oil	120	100.000%	480
TOTAL	120	100.000%	480

Mass route, pilot: 120 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 120 g retained. Allocation: 4 g finished component per unit. Production retained mass: 480 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,958 g; modeled losses 252.909 g; net edible output 1,705.091 g = 30 × 56.836 g expected finished units, with no modeled remainder. Production example: 7,832 g input → 6,820.365 g output across 120 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 60.041 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix dry ingredients with water and eggs slowly 4 minutes; develop briefly then add butter. Finish smooth at 24–26°C with a thin membrane.

2. Bulk ferment 45–60 minutes. Divide equally into 30 pieces and form rings by opening the center; this avoids uncosted cutter scraps. For solid beignets sucrés, flatten rounds to 10–12 mm.
3. Proof at 28–30°C, 70–75% RH for about 35–55 minutes on transferable screens until light but strong enough to lift.
4. Fry rings at 175°C initial trial, approximately 90–120 seconds per side; solid rounds may need 2–3 minutes per side at 170–175°C. Log recovery between loads.
5. Drain on a rack. Split a first-off sample: no raw flour line or wet center. Verify egg-containing product reaches 75°C for 30 seconds while developing a separate full cooked-crumb quality target.
6. For the costed plain format, pack after steam dissipates. For glaze or sugar finishing use the exact variant allowances below, record pickup and maintain separate SKU costs.

Quality, defects and controlled variants

Greasy → cool oil/overproof/long fry. Black crust/raw center → hot oil or large pieces. Glaze runs → hot donut or thin glaze. Fry offcuts separately, never hide them in yield.

Glaze for 30: icing sugar 240 g + potable water 40 g + glucose syrup 80% solids 15 g + vanilla 2 g = 297 g. Model 20% bowl/drip loss = 237.6 g pickup, 7.92 g/donut; no milk. Sugar finish: charge 120 g caster sugar, model 25% unretained = 90 g pickup, 3 g/donut. Choose one finish; do not add both. Workbook contains separate finish rows.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Pack only after steam/condensation is controlled. Dry sleeves for immediate fried sales, sealed barrier packs for fully cooled biscuits/cake. During pilots use fresh same-day sales; H05 retained samples and relevant water-activity/microbial review set any later life. Reheating is not a preservation method.

S-P02 — Yeast puff-puff / beignets farine

60 small beignets = 20 three-piece portions. Pilot raw piece is calculated, not guessed from scoop volume.

Stage: Launch • Fit class 1 • Pilot 60 units • Production example ×4 = 240 units.

Commercial case: S29/S31 show flour-beignet/bean breakfast formats. Test a small breakfast bundle after a separate bean/porridge process is approved.

Technical basis: [S29], [S31]. All exact formulas and operating adaptations here require controlled trials.

Formula — Batter

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	4,000
Water	650	65.000%	2,600
Sugar	120	12.000%	480
Salt	12	1.200%	48

Ingredient / specification	Pilot g	Percent	Production g
Instant yeast Gold	12	1.200%	48
Nutmeg	2	0.200%	8
TOTAL	1,796	179.600%	7,184

Effective ingredient-water estimate: 65.00% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,796 g input − 35.92 g handling/unretained (2%) − 246.411 g process loss (14% of post-handling mass) = 1,513.669 g retained. Allocation: 25.228 g finished component per unit. Production retained mass: 6,054.675 g. All losses are unmeasured planning allowances.

Formula — Absorbed frying oil

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Oil	150	100.000%	600
TOTAL	150	100.000%	600

Mass route, pilot: 150 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 150 g retained. Allocation: 2.5 g finished component per unit. Production retained mass: 600 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,946 g; modeled losses 282.331 g; net edible output 1,663.669 g = 60 × 27.728 g expected finished units, with no modeled remainder. Production example: 7,784 g input → 6,654.675 g output across 240 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 29.335 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix water, flour, sugar, yeast, salt and nutmeg to a stretchy dropping batter; no dry pockets. Target 26–28°C.
2. Cover and ferment 60–90 minutes at 27–29°C until well aerated and increased about 75% in volume. Do not continuously degas the batter.
3. Use a weighed release scoop to divide equally into the stated count. Trial oil at 170–175°C; fry small loads for 3–5 minutes, turning until evenly browned.
4. Cut samples to verify a fully set elastic center. Drain and weigh oil uptake/loss; keep sweet frying before savory products.
5. Serve near production in a dry paper sleeve, ideally three pieces per purchase. No overnight ambient carryover during trials.

Quality, defects and controlled variants

Sour/flat → overfermentation. Dense → cold batter/inactive yeast. Hollow wet shell → excessive oil heat. Irregular weights → scoop calibration.

BHB bundle is retained as a commercial concept, not counted as a fully developed recipe here; do not add uncosted beans and porridge to the three-piece base.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten.

Pack only after steam/condensation is controlled. Dry sleeves for immediate fried sales, sealed barrier packs for fully cooled biscuits/cake. During pilots use fresh same-day sales; H05 retained samples and relevant water-activity/microbial review set any later life. Reheating is not a preservation method.

S-P03 — Cameroonian cake-style beignets soufflés

40 cake-like fritters; the name is grounded in S06, but local naming should be checked at the counter.

Stage: Phase 2 • Fit class 1 • Pilot 40 units • Production example ×4 = 160 units.

Commercial case: An additional familiar bakery-style snack with no proofing delay. Egg and milk costs require a distinct price test.

Technical basis: [S06], [S22]. All exact formulas and operating adaptations here require controlled trials.

Formula — Batter

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Plain flour	1,000	100.000%	4,000
Whole egg	400	40.000%	1,600
Whole milk 3.5%	480	48.000%	1,920
Sugar	220	22.000%	880
Oil	100	10.000%	400
Baking powder	20	2.000%	80
Salt	10	1.000%	40
Vanilla	6	0.600%	24
Nutmeg	2	0.200%	8
TOTAL	2,238	223.800%	8,952

Effective ingredient-water estimate: 72.40% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 2,238 g input − 44.76 g handling/unretained (2%) − 307.054 g process loss (14% of post-handling mass) = 1,886.186 g retained. Allocation: 47.155 g finished component per unit. Production retained mass: 7,544.746 g. All losses are unmeasured planning allowances.

Formula — Absorbed frying oil

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Oil	160	100.000%	640
TOTAL	160	100.000%	640

Mass route, pilot: 160 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 160 g retained. Allocation: 4 g finished component per unit. Production retained mass: 640 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,398 g; modeled losses 351.814 g; net edible output 2,046.186 g = 40 × 51.155 g expected finished units, with no modeled remainder. Production example: 9,592 g input → 8,184.746 g output across 160 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 54.831 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Whisk eggs, milk, sugar, oil and vanilla by hand in a suitable bowl; do not assume a spiral mixer can aerate cake batter.
2. Sieve flour with double-acting baking powder, salt and nutmeg. Fold into liquids only until smooth. No yeast fermentation or proofer is used.
3. Deposit promptly with a release scoop into 170–175°C oil. The trial piece count requires equal weighed scoops; fry about 4–6 minutes while turning.
4. Check the center 75°C for 30 seconds and a fully set cakey texture, not simply a dark exterior. If the surface colors too soon, reduce portion size or heat on the next trial.
5. Drain, cool until steam dissipates and pack in a ventilated sleeve. Compare fresh texture with S-P02 in a blind tasting.

Quality, defects and controlled variants

Tough → overmix after flour. Bitter → excess or unsuitable leavener. Raw inside → large scoops/high heat. Flat → exhausted baking powder or delayed frying.

Use only specified double-acting baking powder at the recorded dose; supplier change triggers leavening test. Do not substitute yeast.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Pack only after steam/condensation is controlled. Dry sleeves for immediate fried sales, sealed barrier packs for fully cooled biscuits/cake. During pilots use fresh same-day sales; H05 retained samples and relevant water-activity/microbial review set any later life. Reheating is not a preservation method.

S-P04 — Coconut butter biscuits

80 biscuits; sell four-packs. Budget manual labour; spiral hook is not a creaming paddle.

Stage: Launch • Fit class 2 • Pilot 80 units • Production example ×3 = 240 units.

Commercial case: Portable small-ticket treat and event packs; coconut is a signature hypothesis, not a proved bestseller.

Technical basis: [S07], [S34]. All exact formulas and operating adaptations here require controlled trials.

Formula — Biscuit dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Plain flour	1,000	100.000%	3,000
Butter 82%	550	55.000%	1,650
Icing sugar	350	35.000%	1,050
Whole egg	100	10.000%	300
Desiccated coconut	150	15.000%	450
Salt	5	0.500%	15
Vanilla	5	0.500%	15
TOTAL	2,160	216.000%	6,480

Effective ingredient-water estimate: 16.40% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 2,160 g input − 43.2 g handling/unretained (2%) − 211.68 g process loss (10% of post-handling mass) = 1,905.12 g retained. Allocation: 23.814 g finished component per unit. Production retained mass: 5,715.36 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,160 g; modeled losses 254.88 g; net edible output 1,905.12 g = 80 × 23.814 g expected finished units, with no modeled remainder. Production example: 6,480 g input → 5,715.36 g output across 240 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 26.46 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Use dry fine desiccated coconut, not fresh coconut at the same weight. Cream pliable butter and icing sugar manually for a small trial; use a paddle mixer only if separately available.
2. Add egg gradually and vanilla. Fold in flour, salt and coconut briefly until just bound. Keep dough about 20–23°C; chill briefly if too soft, with safe refrigeration.
3. Divide into 80 portions; press to discs about 5 cm diameter and 6–8 mm thick with 2 cm spacing.
4. Bake at 160–170°C without steam for 14–20 minutes; rotate rack only according to machine safety procedure. Seek evenly golden edges and a dry cooled snap.
5. Cool fully, then seal in moisture-barrier food packs. Conduct water-activity and mold/packaging studies before any multi-day retail shelf-life claim.

Quality, defects and controlled variants

Spread → warm fat or inaccurate flour. Tough → too much mixing. Soft after packing → cooling/poor barrier. Rancid → coconut/butter age.

Fresh coconut introduces water and spoilage risk: reformulate and validate rather than substitute directly.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk, coconut (declare ingredient; verify local allergen classification).

Pack only after steam/condensation is controlled. Dry sleeves for immediate fried sales, sealed barrier packs for fully cooled biscuits/cake. During pilots use fresh same-day sales; H05 retained samples and relevant water-activity/microbial review set any later life. Reheating is not a preservation method.

S-P05 — Banana snack cake

32 slices; whole-tray or individually wrapped sale. Use the weighed finished tray for exact cut yield.

Stage: Launch • Fit class 2 • Pilot 32 units • Production example ×3 = 96 units.

Commercial case: Familiar fruit-cake direction with locally testable banana supply. Plain slices reduce frosting and refrigeration complexity, not automatically safety risk.

Technical basis: [S07], [S22]. All exact formulas and operating adaptations here require controlled trials.

Formula — Cake batter

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Plain flour	1,000	100.000%	3,000
Banana pulp	900	90.000%	2,700
Sugar	650	65.000%	1,950
Oil	350	35.000%	1,050
Whole egg	500	50.000%	1,500

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	150	15.000%	450
Baking powder	25	2.500%	75
Salt	10	1.000%	30
Cinnamon	5	0.500%	15
Vanilla	5	0.500%	15
TOTAL	3,595	359.500%	10,785

Effective ingredient-water estimate: 51.12% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 3,595 g input − 71.9 g handling/unretained (2%) − 422.772 g process loss (12% of post-handling mass) = 3,100.328 g retained. Allocation: 96.885 g finished component per unit. Production retained mass: 9,300.984 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 3,595 g; modeled losses 494.672 g; net edible output 3,100.328 g = 32 × 96.885 g expected finished units, with no modeled remainder. Production example: 10,785 g input → 9,300.984 g output across 96 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 110.097 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Prepare peeled ripe but sound banana; reject moldy fruit, and do not use a baking step to rescue spoiled ingredients. Mash to uniform pulp.
2. Whisk egg, sugar, oil, milk, banana and vanilla. Sieve flour, baking powder, salt and cinnamon; fold briefly into the wet mixture. Avoid intentional aeration with a spiral hook.
3. Distribute into lined shallow tins to about 3 cm raw depth; record actual tin area and fill mass. Do not pour the whole production batch into one deep pan.
4. Bake at 160–170°C, no steam, for about 30–45 minutes. Seek a set resilient center, clean skewer and initial quality target 95–98°C.
5. Cool protected, remove liners and divide the measured finished cake into 32 equal slices. The mass balance includes a 2% combined bowl/cutting reject allowance; record actual trimming separately.

Quality, defects and controlled variants

Gummy line → too deep/underbaked/excess banana moisture. Tunnel → overmix. Crumbling → hot cutting or overbake.

No cream cheese or custard icing in the base. Sugar reductions change structure and need a new trial.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Pack only after steam/condensation is controlled. Dry sleeves for immediate fried sales, sealed barrier packs for fully cooled biscuits/cake. During pilots use fresh same-day sales; H05 retained samples and relevant water-activity/microbial review set any later life. Reheating is not a preservation method.

S-P06 — Cinnamon breakfast rolls

24 unglazed rolls. Filling loss is modeled separately from dough loss.

Stage: Phase 2 • Fit class 1 • Pilot 24 units • Production example ×4 = 96 units.

Commercial case: Premium breakfast/event preorder rather than another daily staple. Cinnamon supply and premium acceptance need testing.

Technical basis: [S08], [S03]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	1,000	100.000%	4,000
Whole milk 3.5%	520	52.000%	2,080
Whole egg	100	10.000%	400
Sugar	120	12.000%	480
Butter 82%	120	12.000%	480
Salt	18	1.800%	72
Instant yeast Gold	15	1.500%	60
TOTAL	1,893	189.300%	7,572

Effective ingredient-water estimate: 55.02% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 1,893 g input − 37.86 g handling/unretained (2%) − 222.617 g process loss (12% of post-handling mass) = 1,632.523 g retained. Allocation: 68.022 g finished component per unit. Production retained mass: 6,530.093 g. All losses are unmeasured planning allowances.

Formula — Cinnamon filling

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Butter 82%	150	40.984%	600
Sugar	200	54.645%	800
Cinnamon	16	4.372%	64

Ingredient / specification	Pilot g	Percent	Production g
TOTAL	366	100.000%	1,464

Mass route, pilot: 366 g input − 7.32 g handling/unretained (2%) − 43.042 g process loss (12% of post-handling mass) = 315.638 g retained. Allocation: 13.152 g finished component per unit. Production retained mass: 1,262.554 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,259 g; modeled losses 310.838 g; net edible output 1,948.162 g = 24 × 81.173 g expected finished units, with no modeled remainder. Production example: 9,036 g input → 7,792.646 g output across 96 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 77.297 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix and develop dough, adding butter after initial gluten formation, to 24–26°C. Bulk ferment covered 45–60 minutes.
2. Blend filling butter, sugar and cinnamon until spreadable. Rest dough 10–15 minutes then roll to about 6 mm, using the mini sheeter only after thickness and safety checks.
3. Spread the filling evenly, leaving a clean 1 cm edge. Roll a 48 cm log and cut 24 pieces approximately 2 cm wide. Divide into two logs if bench size requires.
4. Proof at 28°C, 75–80% RH for 45–65 minutes until rolls touch lightly and spring back slowly. Do not squeeze out filling.
5. Bake at 175–185°C for 20–28 minutes without steam. Check center rolls are fully set, near 93–96°C, not wet around the spiral. Cool protected before packing.

Quality, defects and controlled variants

Loose spiral → poor seam or uneven filling. Burnt sugar bottom → excess filling/lower heat issue. Dense center → short proof or shallow browning mistaken for doneness.

Optional water glaze uses S-P01 exact glaze recipe scaled to desired pickup; cost as an extra. No cream icing is included.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Pack only after steam/condensation is controlled. Dry sleeves for immediate fried sales, sealed barrier packs for fully cooled biscuits/cake. During pilots use fresh same-day sales; H05 retained samples and relevant water-activity/microbial review set any later life. Reheating is not a preservation method.

S-F01 — Complete beef hamburger

10 complete burgers, not ten buns. Cheese upgrade: 20 g separately costed per unit; no fries in base.

Stage: Phase 2 • Fit class 3 • Pilot 10 units • Production example ×5 = 50 units.

Commercial case: S30 identifies burgers in Douala; local serving size and price need paid trials. Pan throughput and beef yield drive contribution.

Technical basis: [S22], [S30]. All exact formulas and operating adaptations here require controlled trials.

Formula — Soft bun dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	500	100.000%	2,500
Water	280	56.000%	1,400
Sugar	40	8.000%	200
Oil	40	8.000%	200
Skim milk powder	20	4.000%	100
Salt	9	1.800%	45
Instant yeast Gold	6	1.200%	30
Whole egg	25	5.000%	125
TOTAL	920	184.000%	4,600

Effective ingredient-water estimate: 59.80% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 920 g input − 18.4 g handling/unretained (2%) − 108.192 g process loss (12% of post-handling mass) = 793.408 g retained. Allocation: 79.341 g finished component per unit. Production retained mass: 3,967.04 g. All losses are unmeasured planning allowances.

Formula — Seasoned beef patty

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Raw beef mince	1,000	98.814%	5,000
Salt	10	0.988%	50
Black pepper	2	0.198%	10
TOTAL	1,012	100.000%	5,060

Mass route, pilot: 1,012 g input − 0 g handling/unretained (0%) − 253 g process loss (25% of post-handling mass) = 759 g retained. Allocation: 75.9 g finished component per unit. Production retained mass: 3,795 g. All losses are unmeasured planning allowances.

Formula — Burger sauce

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Pasteurized-egg mayonnaise	120	60.000%	600
Ketchup	50	25.000%	250
Mustard	20	10.000%	100
Drained pickle	10	5.000%	50
TOTAL	200	100.000%	1,000

Mass route, pilot: 200 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 200 g retained. Allocation: 20 g finished component per unit. Production retained mass: 1,000 g. All losses are unmeasured planning allowances.

Formula — Prepared salad

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Lettuce	100	20.000%	500
Tomato	200	40.000%	1,000
Onion	100	20.000%	500
Drained pickle	100	20.000%	500
TOTAL	500	100.000%	2,500

Mass route, pilot: 500 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 500 g retained. Allocation: 50 g finished component per unit. Production retained mass: 2,500 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,632 g; modeled losses 379.592 g; net edible output 2,252.408 g = 10 × 225.241 g expected finished units, with no modeled remainder. Production example: 13,160 g input → 11,262.04 g output across 50 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 90.16 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Make the listed half-size S-B05 dough and divide into ten buns; follow S-B05 proof and bake. Cool completely. If using stock buns, transfer their actual batch cost and do not charge fresh dough too.
2. Keep beef at 4°C or below. Shape ten equal patties about 10–11 cm wide; do not knead heavily. Season just before cooking.
3. Mix the 200 g sauce with clean utensils in the ready-to-eat zone and refrigerate; prepare washed, drained vegetables separately from raw beef.
4. In a suitable induction pan, trial a 190–210°C pan surface and 3–4 minutes each side. Probe the patty sideways to verify 75°C for 30 seconds; measured temperature overrides minutes.

5. Toast bun faces. For each burger add one patty, 20 g sauce, 10 g lettuce, 20 g tomato, 10 g onion and 10 g pickle. Serve immediately.
6. Weigh cooked patties: the 25% cook loss is a planning estimate. If the output differs, record the actual portion and recost; do not present raw 100 g as cooked 100 g.

Quality, defects and controlled variants

Wet base → undrained vegetables/excess sauce. Tough patty → overworking/pressing. Slow queue → too many cooked-to-order products for one pan.

Serve a smaller verified-price format only after recalculating bun/patty/salad portions; never change advertised meat weight silently.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, mustard, milk.

Serve promptly in a vented food-grade pack or use H01 monitored hot/chilled holding and delivery. Cold components stay at 4°C or below; reheated prepared fillings reach 75°C for 30 seconds. Test eating quality at 15/30 minutes. No ambient overnight sale and no unvalidated chilled shelf-life claim.

S-F02 — Pain omelette with sardines

Ten compact sandwiches; bread size is smaller than the old 125 g half-loaf. Re-price and label accordingly.

Stage: Phase 2 • Fit class 3 • Pilot 10 units • Production example ×5 = 50 units.

Commercial case: S29 lists sardine omelette at 1,000 XAF, not this sandwich. Breakfast adaptation must be tested independently.

Technical basis: [S22], [S29]. All exact formulas and operating adaptations here require controlled trials.

Formula — Baguette dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	500	100.000%	2,500
Water	330	66.000%	1,650
Salt	10	2.000%	50
Instant yeast Red	2.5	0.500%	12.5
TOTAL	842.5	168.500%	4,212.5

Effective ingredient-water estimate: 66.00% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 842.5 g input − 16.85 g handling/unretained (2%) − 123.847 g process loss (15% of post-handling mass) = 701.803 g retained. Allocation: 70.18 g finished component per unit. Production retained mass: 3,509.012 g. All losses are unmeasured planning allowances.

Formula — Omelette

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole egg	500	52.356%	2,500
Drained sardines	200	20.942%	1,000
Onion	80	8.377%	400
Tomato	100	10.471%	500
Sweet pepper	40	4.188%	200
Oil	30	3.141%	150
Salt	4	0.419%	20
Black pepper	1	0.105%	5
TOTAL	955	100.000%	4,775

Mass route, pilot: 955 g input − 0 g handling/unretained (0%) − 114.6 g process loss (12% of post-handling mass) = 840.4 g retained. Allocation: 84.04 g finished component per unit. Production retained mass: 4,202 g. All losses are unmeasured planning allowances.

Formula — Sandwich sauce

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Pasteurized-egg mayonnaise	70	70.000%	350
Ketchup	30	30.000%	150
TOTAL	100	100.000%	500

Mass route, pilot: 100 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 100 g retained. Allocation: 10 g finished component per unit. Production retained mass: 500 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,897.5 g; modeled losses 255.297 g; net edible output 1,642.202 g = 10 × 164.22 g expected finished units, with no modeled remainder. Production example: 9,487.5 g input → 8,211.012 g output across 50 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 82.565 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Bake the listed half-size S-B01 dough as five short baguettes and cool. Split each into two portions; the card models ten bread portions.

2. Open intact tins, drain and weigh sardines; cost drained fish, not gross tin weight. Refrigerate opened fish. Sweat onion and pepper in some of the listed oil, add tomato and cook off free water.
3. Per order use 50 g shell-free egg, 20 g fish and one-tenth of cooked vegetables, seasoning and remaining oil. Keep raw egg mix chilled and away from finished sandwiches.
4. Cook in a clean induction pan, fold to fit bread and verify 75°C for 30 seconds at the egg center. A starting 3–5 minutes is only a trial timing.
5. Toast bread lightly, apply 10 g sauce and add one omelette. Serve immediately; do not store warm assembled sandwiches in an ambient display.

Quality, defects and controlled variants

Soggy bread → watery vegetables. Fishy pastry cross-odor → poor separation. Rubbery egg → excessive pan time after safe endpoint.

Egg-only version omits fish and changes total weight/cost/allergen declarations; use a separate SKU record.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, fish.

Serve promptly in a vented food-grade pack or use H01 monitored hot/chilled holding and delivery. Cold components stay at 4°C or below; reheated prepared fillings reach 75°C for 30 seconds. Test eating quality at 15/30 minutes. No ambient overnight sale and no unvalidated chilled shelf-life claim.

S-F03 — Beef-and-cheese filled bread

Ten sealed filled buns; raw beef is included, so there is no hidden cooked-meat purchasing assumption.

Stage: Phase 2 • Fit class 3 • Pilot 10 units • Production example ×5 = 50 units.

Commercial case: Portable office/event lunch adaptation; no verified exact-product sales ranking. Compare with S-F01 on contribution per operator minute.

Technical basis: [S22], [S03]. All exact formulas and operating adaptations here require controlled trials.

Formula — Unbaked bun dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	500	100.000%	2,500
Water	280	56.000%	1,400
Sugar	40	8.000%	200
Oil	40	8.000%	200
Skim milk powder	20	4.000%	100
Salt	9	1.800%	45
Instant yeast Gold	6	1.200%	30

Ingredient / specification	Pilot g	Percent	Production g
Whole egg	25	5.000%	125
TOTAL	920	184.000%	4,600

Effective ingredient-water estimate: 59.80% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 920 g input − 18.4 g handling/unretained (2%) − 108.192 g process loss (12% of post-handling mass) = 793.408 g retained. Allocation: 79.341 g finished component per unit. Production retained mass: 3,967.04 g. All losses are unmeasured planning allowances.

Formula — Beef filling

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Raw beef mince	500	56.754%	2,500
Onion	100	11.351%	500
Carrot	80	9.081%	400
Sweet pepper	80	9.081%	400
Oil	20	2.270%	100
Tomato paste	30	3.405%	150
Water	50	5.675%	250
Garlic	10	1.135%	50
Salt	8	0.908%	40
Paprika	3	0.341%	15
TOTAL	881	100.000%	4,405

Mass route, pilot: 881 g input − 0 g handling/unretained (0%) − 220.25 g process loss (25% of post-handling mass) = 660.75 g retained. Allocation: 66.075 g finished component per unit. Production retained mass: 3,303.75 g. All losses are unmeasured planning allowances.

Formula — Cheese

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Low-moisture cheese	100	100.000%	500
TOTAL	100	100.000%	500

Mass route, pilot: 100 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 100 g retained. Allocation: 10 g finished component per unit. Production retained mass: 500 g. All losses are

unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 1,901 g; modeled losses 346.842 g; net edible output 1,554.158 g = 10 × 155.416 g expected finished units, with no modeled remainder. Production example: 9,505 g input → 7,770.79 g output across 50 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 90.16 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Brown beef and verify 75°C for 30 seconds. Add the diced vegetables, oil, garlic, paste, water and seasonings; cook until cohesive with no free liquid. Weigh yield, then actively cool under H01.
2. Mix the bun dough using S-B05, but do not prebake. Bulk ferment 45–60 minutes, divide into ten, rest 10–15 minutes and flatten to 12–14 cm discs.
3. Keep filling and cheese at 4°C or below until needed. Fill with one-tenth of the measured chilled filling plus 10 g cheese; leave a clean border and seal seam down.
4. Proof only a timed small load, initially 30–45 minutes at 28°C. Track cumulative filling time outside refrigeration; have an oven slot ready.
5. Bake immediately at 180–190°C for 20–28 minutes, no steam, and verify 75°C for 30 seconds in the filling as well as a set bread shell.
6. Use monitored hot holding or active cooling, never the ordinary bread display. The model assigns 25% filling cook/evaporation and 12% shell bake loss; measure filling loss again in the final bake and revise.

Quality, defects and controlled variants

Burst seam → dirty edge/wet filling. Gummy collar → filling too wet or underbake. Dark shell/cold filling → lower bake heat and verify center.

Cooked chicken substitution requires a new raw-to-cooked yield and poultry process; not a gram-for-gram raw meat swap.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Serve promptly in a vented food-grade pack or use H01 monitored hot/chilled holding and delivery. Cold components stay at 4°C or below; reheated prepared fillings reach 75°C for 30 seconds. Test eating quality at 15/30 minutes. No ambient overnight sale and no unvalidated chilled shelf-life claim.

S-F04 — Chicken shawarma wrap

Ten wraps. No unlisted grill, fries, purchased tortilla or shared raw-meat tank is assumed.

Stage: Phase 2 • Fit class 3 • Pilot 10 units • Production example ×5 = 50 units.

Commercial case: S29/S31 show shawarma availability; smaller Nkambe wrap is an adaptation. Fast assembly may suit lunch preorders after cold-chain gates.

Technical basis: [S22], [S29], [S31]. All exact formulas and operating adaptations here require controlled trials.

Formula — Flatbread

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	600	100.000%	3,000
Water	330	55.000%	1,650
Oil	30	5.000%	150
Salt	12	2.000%	60
TOTAL	972	162.000%	4,860

Effective ingredient-water estimate: 55.00% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 972 g input − 19.44 g handling/unretained (2%) − 95.256 g process loss (10% of post-handling mass) = 857.304 g retained. Allocation: 85.73 g finished component per unit. Production retained mass: 4,286.52 g. All losses are unmeasured planning allowances.

Formula — Marinated chicken

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Raw chicken thigh	1,200	80.808%	6,000
Plain pasteurized yogurt	100	6.734%	500
Oil	30	2.020%	150
Onion	60	4.040%	300
Lime juice	30	2.020%	150
Garlic	18	1.212%	90
Ginger	10	0.673%	50
Paprika	10	0.673%	50
Cumin	6	0.404%	30
Black pepper	3	0.202%	15
Salt	18	1.212%	90
TOTAL	1,485	100.000%	7,425

Mass route, pilot: 1,485 g input − 0 g handling/unretained (0%) − 445.5 g process loss (30% of post-handling mass) = 1,039.5 g retained. Allocation: 103.95 g finished component per unit. Production retained mass: 5,197.5 g. All losses are unmeasured planning allowances.

Formula — Garlic sauce

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Pasteurized-egg mayonnaise	140	70.000%	700
Plain pasteurized yogurt	50	25.000%	250
Cooked garlic puree	10	5.000%	50
TOTAL	200	100.000%	1,000

Mass route, pilot: 200 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 200 g retained. Allocation: 20 g finished component per unit. Production retained mass: 1,000 g. All losses are unmeasured planning allowances.

Formula — Salad

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Cabbage	150	42.857%	750
Carrot	100	28.571%	500
Tomato	100	28.571%	500
TOTAL	350	100.000%	1,750

Mass route, pilot: 350 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 350 g retained. Allocation: 35 g finished component per unit. Production retained mass: 1,750 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 3,007 g; modeled losses 560.196 g; net edible output 2,446.804 g = 10 × 244.68 g expected finished units, with no modeled remainder. Production example: 15,035 g input → 12,234.02 g output across 50 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 95.256 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix flatbread to medium development, 24–26°C, then rest covered 30 minutes. Divide into ten equal portions and rest 15 minutes. Roll to 2 mm and about 22–25 cm using measured thickness.
2. Cook on a heavy induction pan with a trial surface 220–240°C, approximately 45–75 seconds each side, until brown spots and no raw layer. Stack briefly under a clean cover for flexibility.
3. Marinate chicken in the listed ingredients in a dedicated raw-meat container at 4°C or below, 4–12 hours. Never use the dairy pot for raw marinade.
4. Use the shawarma machine only after verifying its minimum cone load and operating method. A small trial may be below that minimum: cook chicken in a pan instead. Probe 75°C for 30 seconds; shave only

cooked layers and finish shavings on the pan when required. Never partially cook and save a raw-core cone for the next day.

5. Mix sauce only in the clean zone. Assemble one wrap with one-tenth cooked chicken, 20 g sauce and 35 g dry salad; fold ends inward and roll firmly.
6. Serve immediately in food paper. Log chicken yield: the 30% combined meat/marinade loss is a trial allowance, not measured meat-only shrinkage.

Quality, defects and controlled variants

Cracked wrap → overcooked/thick bread. Watery filling → poor drainage. Undercooked shavings → cone/shaving rate mismatch; stop and finish cook.

Shawarma baguette variant uses S-B01 and changes bread mass/cost. Keep fries as a separate option, not part of the base claim.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, egg, milk.

Serve promptly in a vented food-grade pack or use H01 monitored hot/chilled holding and delivery. Cold components stay at 4°C or below; reheated prepared fillings reach 75°C for 30 seconds. Test eating quality at 15/30 minutes. No ambient overnight sale and no unvalidated chilled shelf-life claim.

S-F05 — Vegetable samosas

30 medium samosas with fully quantified shell/filling. Pilot oil absorption is 3 g each, to be weighed.

Stage: Phase 2 • Fit class 3 • Pilot 30 units • Production example ×4 = 120 units.

Commercial case: Fryer-compatible portable snack; Nkambe demand is unresolved. Compare vegetable filling value against meat-filled bread.

Technical basis: [S22], [S23]. All exact formulas and operating adaptations here require controlled trials.

Formula — Pastry

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Plain flour	600	100.000%	2,400
Water	250	41.667%	1,000
Oil	60	10.000%	240
Salt	9	1.500%	36
TOTAL	919	153.167%	3,676

Effective ingredient-water estimate: 41.67% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 919 g input − 18.38 g handling/unretained (2%) − 108.074 g process loss (12% of post-handling mass) = 792.546 g retained. Allocation: 26.418 g finished component per unit. Production retained mass: 3,170.182 g. All losses are unmeasured planning allowances.

Formula — Vegetable filling

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Peeled potato	800	62.598%	3,200
Peas	160	12.520%	640
Carrot	100	7.825%	400
Onion	140	10.955%	560
Oil	40	3.130%	160
Garlic	10	0.782%	40
Ginger	10	0.782%	40
Cumin	6	0.469%	24
Salt	12	0.939%	48
TOTAL	1,278	100.000%	5,112

Mass route, pilot: 1,278 g input − 0 g handling/unretained (0%) − 127.8 g process loss (10% of post-handling mass) = 1,150.2 g retained. Allocation: 38.34 g finished component per unit. Production retained mass: 4,600.8 g. All losses are unmeasured planning allowances.

Formula — Absorbed frying oil

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Oil	90	100.000%	360
TOTAL	90	100.000%	360

Mass route, pilot: 90 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 90 g retained. Allocation: 3 g finished component per unit. Production retained mass: 360 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,287 g; modeled losses 254.254 g; net edible output 2,032.746 g = 30 × 67.758 g expected finished units, with no modeled remainder. Production example: 9,148 g input → 8,130.982 g output across 120 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 30.021 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Rub oil into flour and salt; add water, knead to a firm smooth dough and rest covered 30 minutes. Avoid full bread-like gluten development.
2. Steam or boil potato pieces to complete tenderness and drain. Sauté onion, carrot, garlic and ginger in the listed oil; add peas, potato, cumin and salt. Reduce until dry. Treat cooking water as a separately measured process input/drain, not saleable mass.
3. Weigh filling and cool safely if not used immediately. Divide pastry into 30 portions; sheet to about 1.5–2 mm. Form cones or folded triangles.
4. Allocate one-thirtieth of the filling per shell. Seal using water from a small clean brush, weighing any added sealing water in the actual batch log. Pinch a 1 cm clean seam with no trapped air.
5. Fry at 170–175°C for about 4–6 minutes, turn, drain on racks and check the filling reheats to 75°C for 30 seconds when previously chilled.
6. Pack in a vented sleeve for prompt sale or use H01 hot/chilled controls. A potato filling is not automatically safe at room temperature because it has no meat.

Quality, defects and controlled variants

Blisters/burst → wet filling/air/weak seam. Greasy → low oil temperature. Hard pastry → overkneaded or too thick.

A baked version is a new trial (about 195–205°C); remove absorption allowance and add measured surface oil. Do not carry fried yield into baked claims.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten.

Serve promptly in a vented food-grade pack or use H01 monitored hot/chilled holding and delivery. Cold components stay at 4°C or below; reheated prepared fillings reach 75°C for 30 seconds. Test eating quality at 15/30 minutes. No ambient overnight sale and no unvalidated chilled shelf-life claim.

S-F06 — Mini chicken pizzas

Twelve snack pizzas; all toppings are included. A deck-style char is not promised from the rotary oven.

Stage: Phase 2 • Fit class 3 • Pilot 12 units • Production example ×4 = 48 units.

Commercial case: Urban-inspired snack with an oven rather than pan bottleneck. Test group preorders; direct Nkambe pizza sales evidence unavailable.

Technical basis: [S22], [S02]. All exact formulas and operating adaptations here require controlled trials.

Formula — Pizza dough

Percent basis: total wheat flour in this component = 100% (baker's percentage).

Ingredient / specification	Pilot g	Percent	Production g
Bread flour	500	100.000%	2,000
Water	300	60.000%	1,200

Ingredient / specification	Pilot g	Percent	Production g
Oil	15	3.000%	60
Sugar	10	2.000%	40
Salt	10	2.000%	40
Instant yeast Red	5	1.000%	20
TOTAL	840	168.000%	3,360

Effective ingredient-water estimate: 60.00% of flour, assuming egg 76%, milk 87.5%, butter 16% and margarine 18% water. Flour moisture is excluded; supplier data supersedes assumptions. This is not the volume of liquid to add.

Mass route, pilot: 840 g input − 16.8 g handling/unretained (2%) − 82.32 g process loss (10% of post-handling mass) = 740.88 g retained. Allocation: 61.74 g finished component per unit. Production retained mass: 2,963.52 g. All losses are unmeasured planning allowances.

Formula — Tomato sauce

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Tomato	240	83.916%	960
Onion	30	10.490%	120
Oil	8	2.797%	32
Garlic	4	1.399%	16
Salt	3	1.049%	12
Oregano	1	0.350%	4
TOTAL	286	100.000%	1,144

Mass route, pilot: 286 g input − 0 g handling/unretained (0%) − 57.2 g process loss (20% of post-handling mass) = 228.8 g retained. Allocation: 19.067 g finished component per unit. Production retained mass: 915.2 g. All losses are unmeasured planning allowances.

Formula — Chicken

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Raw chicken thigh	500	96.525%	2,000
Oil	10	1.931%	40
Salt	5	0.965%	20
Paprika	3	0.579%	12

Ingredient / specification	Pilot g	Percent	Production g
TOTAL	518	100.000%	2,072

Mass route, pilot: 518 g input − 0 g handling/unretained (0%) − 145.04 g process loss (28% of post-handling mass) = 372.96 g retained. Allocation: 31.08 g finished component per unit. Production retained mass: 1,491.84 g. All losses are unmeasured planning allowances.

Formula — Cheese topping

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Low-moisture cheese	300	100.000%	1,200
TOTAL	300	100.000%	1,200

Mass route, pilot: 300 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 300 g retained. Allocation: 25 g finished component per unit. Production retained mass: 1,200 g. All losses are unmeasured planning allowances.

Formula — Vegetable topping

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Sweet pepper	100	50.000%	400
Onion	100	50.000%	400
TOTAL	200	100.000%	800

Mass route, pilot: 200 g input − 0 g handling/unretained (0%) − 20 g process loss (10% of post-handling mass) = 180 g retained. Allocation: 15 g finished component per unit. Production retained mass: 720 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,144 g; modeled losses 321.36 g; net edible output 1,822.64 g = 12 × 151.887 g expected finished units, with no modeled remainder. Production example: 8,576 g input → 7,290.56 g output across 48 units. Display rounding is not label-weight verification.

Raw dough/batter allocation before bake/fry: 68.6 g per final unit, excluding other components. Filling/glaze/oil are separate above.

Numbered production method

1. Mix dough 3 minutes slowly then 4–6 minutes to elasticity at 24–26°C. Bulk ferment 45–60 minutes, divide into twelve and rest 15–20 minutes.
2. Simmer tomato sauce ingredients to a thick spoon-coating consistency; weigh actual output. Cook seasoned chicken to 75°C for 30 seconds, weigh cooked output, cool both by H01.

3. Lightly sauté/roast topping vegetables and drain. Keep components at 4°C or below. Stretch each dough piece to 12–14 cm and dock the center.
4. Allocate one-twelfth of each component, including 25 g cheese, to each base. Bake immediately at 210–220°C for 10–15 minutes, no steam.
5. Verify reheated chicken topping reaches 75°C for 30 seconds and base is cooked, not wet. Serve promptly in a vented box; cost actual cheese and chicken yield.

Quality, defects and controlled variants

Wet base → thin sauce/too much topping/weak bottom heat. Bubbles → undocked base. Tough rim → excess dusting flour.

Vegetarian variant removes chicken and must have a new mass/cost card, not the same advertised serving weight.

Allergens, packing, life study and costing

Allergens/ingredients to declare: wheat/gluten, milk.

Serve promptly in a vented food-grade pack or use H01 monitored hot/chilled holding and delivery. Cold components stay at 4°C or below; reheated prepared fillings reach 75°C for 30 seconds. Test eating quality at 15/30 minutes. No ambient overnight sale and no unvalidated chilled shelf-life claim.

S-D01 — Vanilla dairy soft serve

20 mass-based portions from a 2 kg mix pilot. Cup volume depends on actual overrun and density.

Stage: Cold-line phase • Fit class 3 • Pilot 20 units • Production example ×5 = 100 units.

Commercial case: S29/S31 support urban frozen-dessert availability, not guaranteed Nkambe demand. Prefer approved ready-made mix for machine commissioning before own-mix trials.

Technical basis: [S09], [S10], [S11], [S12], [S14], [S21]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dairy mix

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	1,400	70.000%	7,000
Cream 35%	100	5.000%	500
Skim milk powder	110	5.500%	550
Sugar	280	14.000%	1,400
Dextrose anhydrous	60	3.000%	300
STAB 2000	9	0.450%	45
Vanilla	8	0.400%	40

Ingredient / specification	Pilot g	Percent	Production g
Water	33	1.650%	165
TOTAL	2,000	100.000%	10,000

Mass route, pilot: 2,000 g input − 40 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 1,960 g retained. Allocation: 98 g finished component per unit. Production retained mass: 9,800 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,000 g; modeled losses 40 g; net edible output 1,960 g = 20 × 98 g expected finished units, with no modeled remainder. Production example: 10,000 g input → 9,800 g output across 100 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 4.26%, milk solids-not-fat 11.82%, total solids 33.52%, unfermented lactose 6.38%; added sucrose 14.00% and anhydrous dextrose 3.00%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Freezing comparison only: sucrose-equivalent added-sugar index = sucrose% + 1.9 × anhydrous dextrose% = 19.70. This is NOT a full PAC/freezing-point prediction: native fruit sugars, lactose, salts, actual solids and overrun also matter. Confirm bite/draw at the real freezer temperatures; never adjust sugar without recalculating solids and rerunning the freeze test.

Numbered production method

1. Confirm the machine accepts this fat/solids range, has a safely refrigerated hopper and a documented cleaning method. Use homogenized pasteurized/UHT dairy, not raw milk.
2. Blend STAB 2000 with part of the sugar; disperse with all other ingredients and hydrate as specified in H02. Follow the verified supplier heat requirements and a validated batch heat process.
3. H02 reference for a qualified batch vessel is at least 69°C throughout for 30 minutes. Record the cold point and hold continuously, then cool rapidly to 4°C or below; never rely on the pot display alone.
4. Age covered at 2–4°C for 4–12 hours as a trial texture window. Reblend gently without air, load cold into a cleaned and sanitized machine following its manual.
5. Trial draw near -5 to -7°C only if the manufacturer supports it. Weigh an equal-volume mix and draw sample to measure overrun; compare 25–40% trial overrun without changing machine internals.
6. Portion by mass; serve immediately. The 2% transfer/purge allowance must be replaced with actual start-up and closing waste. Do not top up a dirty hopper or carry mix overnight without a validated manufacturer process.

Quality, defects and controlled variants

Icy → low solids/slow freezing/heat shock. Greasy → fat destabilization or unsuitable machine. Weak peak → over-soft draw/incorrect sugars. Sanitation failure → stop sale, quarantine, clean and investigate.

Cocoa trial is a variant, not a second shortlist item: replace 30 g milk with 30 g cocoa per 2 kg pilot, then re-balance and verify viscosity. Do not add acid fruit directly to hopper dairy.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Portion into food-grade cups/cones at immediate service; packaging is a separate cost. Mix remains under machine/manual cold controls; no ambient retail transport or unvalidated overnight carry. Evaluate draw, purge, sanitation and holding limits before sale.

S-D02 — Scoopable vanilla ice cream

22 scoops by mass from 2.2 kg pilot; no cones are included in mass or ingredients.

Stage: Investment phase • Fit class 3 • Pilot 22 units • Production example ×5 = 110 units.

Commercial case: Premium scoop counter; higher capex and cold-chain burden than soft serve. Develop technically, defer launch until throughput/demand justify equipment.

Technical basis: [S09], [S10], [S11], [S12], [S13], [S14]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dairy mix

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	1,170	53.182%	5,850
Cream 35%	520	23.636%	2,600
Skim milk powder	90	4.091%	450
Sugar	280	12.727%	1,400
Dextrose anhydrous	60	2.727%	300
STAB 2000	6	0.273%	30
Vanilla	8	0.364%	40
Water	66	3.000%	330
TOTAL	2,200	100.000%	11,000

Mass route, pilot: 2,200 g input − 44 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 2,156 g retained. Allocation: 98 g finished component per unit. Production retained mass: 10,780 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,200 g; modeled losses 44 g; net edible output 2,156 g = 22 × 98 g expected finished units, with no modeled remainder. Production example: 11,000 g input → 10,780 g output across 110 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 10.17%, milk solids-not-fat 10.04%, total solids 35.95%, unfermented lactose 5.41%; added sucrose 12.73% and anhydrous dextrose 2.73%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Freezing comparison only: sucrose-equivalent added-sugar index = sucrose% + 1.9 × anhydrous dextrose% = 17.91. This is NOT a full PAC/freezing-point prediction: native fruit sugars, lactose, salts, actual solids and overrun also matter. Confirm bite/draw at the real freezer temperatures; never adjust sugar without recalculating solids and rerunning the freeze test.

Numbered production method

1. This product requires a batch freezer plus rapid hardening and storage: it is not automatically made by the soft-serve machine.
2. Prepare the listed mix under H02 using specified dairy and STAB 2000; heat-treatment validation and rapid cooling apply. Age at 2–4°C for 4–12 hours.
3. Batch-freeze at the manufacturer's permitted fill volume. Trial extraction around -6 to -8°C and 25–40% overrun; do not use the mixer rating to choose freezer load.
4. Pack immediately in shallow lidded food tubs; harden in an appropriately rated hardener until core is -18°C or colder. Trial hardener air around -30 to -35°C only within its manual.
5. Store at -18°C or colder, minimizing door opening. A warmer display temperature, initially about -12 to -14°C for texture testing, needs its own controlled time/temperature plan. Do not repeatedly melt and refreeze returns.

Quality, defects and controlled variants

Coarse ice → slow hardening/heat shock. Sandy → excess lactose/solids or storage. Buttery → overchurning. Shrinkage → unstable air structure; verify mix and draw.

This dairy product has over 10% modeled fat, unlike the lower-fat soft-serve mix. Legal product naming still requires local confirmation.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Wrap sticks/bars in freezer-grade film or lid scoop tubs immediately, with batch/net mass/allergens. Store and distribute at -18°C or colder under measured control. No reheating, thaw/refreeze sales or assumed month-long life. Study days 0/7/14/28, crystal size, melt, seal and microbial plan under H05.

S-I01 — Pineapple-ginger fruit popsicle

25 × 78.4 g theoretical net sticks; check mold volume at actual mix density before filling.

Stage: Cold-line phase • Fit class 3 • Pilot 25 units • Production example ×5 = 125 units.

Commercial case: Pineapple and ginger echo an urban beverage format; frozen demand and local fruit procurement still need validation.

Technical basis: [S15], [S24], [S31]. All exact formulas and operating adaptations here require controlled trials.

Formula — Fruit mix

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Pasteurized pineapple juice 12 Brix	1,400	70.000%	7,000
Water	290	14.500%	1,450
Sugar	250	12.500%	1,250
Dextrose anhydrous	40	2.000%	200
Treated ginger juice	10	0.500%	50
Pasteurized lime juice	4	0.200%	20
Super Neutrose	6	0.300%	30
TOTAL	2,000	100.000%	10,000

Mass route, pilot: 2,000 g input − 40 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 1,960 g retained. Allocation: 78.4 g finished component per unit. Production retained mass: 9,800 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,000 g; modeled losses 40 g; net edible output 1,960 g = 25 × 78.4 g expected finished units, with no modeled remainder. Production example: 10,000 g input → 9,800 g output across 125 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 0.00%, milk solids-not-fat 0.00%, total solids 23.23%, unfermented lactose 0.00%; added sucrose 12.50% and anhydrous dextrose 2.00%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Freezing comparison only: sucrose-equivalent added-sugar index = sucrose% + 1.9 × anhydrous dextrose% = 16.30. This is NOT a full PAC/freezing-point prediction: native fruit sugars, lactose, salts, actual solids and overrun also matter. Confirm bite/draw at the real freezer temperatures; never adjust sugar without recalculating solids and rerunning the freeze test.

Numbered production method

1. Use documented pasteurized juice or a separately validated local fruit process; do not freeze untreated juice and call it safe. Weigh solids and verify juice °Brix.
2. Premix Super Neutrose with sugar, disperse in formula liquids, rest 15 minutes and heat according to the supplier's hydration instructions. Any whole-mix pathogen-control step must be validated for actual pH, pulp and equipment (H03).
3. Cool rapidly to 4°C or below. Hold covered 2–4 hours for a trial hydration period. Measure °Brix at calibration temperature; the formula predicts roughly 23%, but fruit sugar composition is variable.
4. Fill clean molds by mass, respecting expansion headspace. Insert food-grade sticks centered and deep enough for support. Freeze using the actual machine manual, not an assumed 20-minute cycle.

5. Demold only once cores are fully frozen and sticks hold. Use controlled potable-water release without water entering molds; bag promptly and maintain -18°C or colder.

Quality, defects and controlled variants

Hard/icy → insufficient sugar or heat shock. Sticky/soft → too much sugar or warm freezer. Loose stick → early extraction. Burn from ginger → dose/strength variation.

Mango/passionfruit are future variants only after Brix, acidity and solids rebalancing; same weight does not mean same freezing behavior.

Allergens, packing, life study and costing

Allergens/ingredients to declare: No major allergen intentionally added; verify all labels and shared equipment.

Wrap sticks/bars in freezer-grade film or lid scoop tubs immediately, with batch/net mass/allergens. Store and distribute at -18°C or colder under measured control. No reheating, thaw/refreeze sales or assumed month-long life. Study days 0/7/14/28, crystal size, melt, seal and microbial plan under H05.

S-I02 — Bissap-ginger-lime ice stick

25 × 78.4 g net sticks; the old 65°Brix syrup mismatch is removed by direct weighed sugars.

Stage: Cold-line phase • Fit class 3 • Pilot 25 units • Production example ×5 = 125 units.

Commercial case: Bissap beverage familiarity is supported by S29; ice-stick acceptance and seasonality are hypotheses.

Technical basis: [S15], [S24], [S29]. All exact formulas and operating adaptations here require controlled trials.

Formula — Infusion mix

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Validated hibiscus infusion	1,500	75.000%	7,500
Water	44	2.200%	220
Sugar	380	19.000%	1,900
Dextrose anhydrous	50	2.500%	250
Treated ginger juice	10	0.500%	50
Pasteurized lime juice	10	0.500%	50
Super Neutrose	6	0.300%	30
TOTAL	2,000	100.000%	10,000

Mass route, pilot: 2,000 g input − 40 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 1,960 g retained. Allocation: 78.4 g finished component per unit. Production retained mass: 9,800 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,000 g; modeled losses 40 g; net edible output 1,960 g = 25 × 78.4 g expected finished units, with no modeled remainder. Production example: 10,000 g input → 9,800 g output across 125 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 0.00%, milk solids-not-fat 0.00%, total solids 21.85%, unfermented lactose 0.00%; added sucrose 19.00% and anhydrous dextrose 2.50%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Freezing comparison only: sucrose-equivalent added-sugar index = sucrose% + 1.9 × anhydrous dextrose% = 23.75. This is NOT a full PAC/freezing-point prediction: native fruit sugars, lactose, salts, actual solids and overrun also matter. Confirm bite/draw at the real freezer temperatures; never adjust sugar without recalculating solids and rerunning the freeze test.

Numbered production method

1. For the infusion process input, use 30 g food-grade dried hibiscus and 1,600 g potable water. Boil/simmer as a flavour extraction trial for 5–10 minutes, strain hygienically and weigh 1,500 g infusion. Record retained liquid, spent solids and vapor separately; extraction alone is not a validated commercial kill step.
2. Use only a validated treated infusion for commercial manufacture. Cost its actual dry flowers, water, labor and extraction losses as one input; never treat 1,500 g infusion as 1,500 g dry hibiscus.
3. Premix stabilizer and sugar, combine with all liquids and hydrate according to H03. Verify a suitable whole-mix treatment with the process specialist. Measure pH and °Brix, initially about 21.5% added sugars before native extract solids.
4. Cool rapidly to 4°C or below, rest 2–4 hours covered and fill molds by mass. Freeze and demold using the machine-specific checks in S-I01.
5. Wrap against dehydration, record stick and film lots and store at -18°C or colder. Compare ginger heat and acidity with customers in a controlled trial.

Quality, defects and controlled variants

Weak color → old flowers/dilution. Harsh astringency → too much extraction. Soft stick → sugar excess. Ice crystals → slow freeze/storage cycles.

No claim that acidity eliminates pathogens. A ready pasteurized hibiscus base is a lower-complexity commissioning option if documented and affordable.

Allergens, packing, life study and costing

Allergens/ingredients to declare: No major allergen intentionally added; verify all labels and shared equipment.

Wrap sticks/bars in freezer-grade film or lid scoop tubs immediately, with batch/net mass/allergens. Store and distribute at -18°C or colder under measured control. No reheating, thaw/refreeze sales or assumed month-long life. Study days 0/7/14/28, crystal size, melt, seal and microbial plan under H05.

S-I03 — Cocoa milk popsicle

25 × 78.4 g modeled sticks, no overrun intended.

Stage: Cold-line phase • Fit class 3 • Pilot 25 units • Production example ×5 = 125 units.

Commercial case: Lower equipment complexity than aerated coated bars once the cold chain exists. Cocoa provenance and price require supplier evidence.

Technical basis: [S09], [S14]. All exact formulas and operating adaptations here require controlled trials.

Formula — Cocoa mix

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	1,450	72.500%	7,250
Water	150	7.500%	750
Sugar	270	13.500%	1,350
Skim milk powder	60	3.000%	300
Cocoa powder	60	3.000%	300
STAB 2000	8	0.400%	40
Vanilla	2	0.100%	10
TOTAL	2,000	100.000%	10,000

Mass route, pilot: 2,000 g input − 40 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 1,960 g retained. Allocation: 78.4 g finished component per unit. Production retained mass: 9,800 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,000 g; modeled losses 40 g; net edible output 1,960 g = 25 × 78.4 g expected finished units, with no modeled remainder. Production example: 10,000 g input → 9,800 g output across 125 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 2.90%, milk solids-not-fat 9.38%, total solids 28.75%, unfermented lactose 5.04%; added sucrose 13.50% and anhydrous dextrose 0.00%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Freezing comparison only: sucrose-equivalent added-sugar index = sucrose% + 1.9 × anhydrous dextrose% = 13.50. This is NOT a full PAC/freezing-point prediction: native fruit sugars, lactose, salts, actual solids and overrun also matter. Confirm bite/draw at the real freezer temperatures; never adjust sugar without recalculating solids and rerunning the freeze test.

Numbered production method

1. Use fine unsweetened cocoa with stated fat content and pasteurized/UHT milk. Premix cocoa, sugar, milk powder and STAB 2000, then disperse completely with the liquids.
2. Apply H02 validated mix heating/cooling. A stick blender can disperse particles but is not a pressure homogenizer. Check the bottom for scorching.
3. Age covered 4–12 hours at 2–4°C; reblend gently. Fill clean molds by mass, leaving tested headspace and inserting centered sticks.
4. Freeze by actual machine endpoint, not merely elapsed time. Demold fully hardened sticks, wrap promptly and store at -18°C or colder.
5. Assess bite from the actual freezer: target firm but not tooth-hard. Measure top/bottom cocoa uniformity and no gritty sediment.

Quality, defects and controlled variants

Grit → poor dispersion/cocoa grade. Sediment → insufficient hydration. Ice hardness → sugar/solids imbalance. Gumminess → excessive stabilizer.

Do not replace 8 g STAB with 8 g starch; they are different systems. Coconut milk version requires separate fat/emulsion balance and label.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Wrap sticks/bars in freezer-grade film or lid scoop tubs immediately, with batch/net mass/allergens. Store and distribute at -18°C or colder under measured control. No reheating, thaw/refreeze sales or assumed month-long life. Study days 0/7/14/28, crystal size, melt, seal and microbial plan under H05.

S-I04 — Own-brand premium chocolate-coated ice-cream bar

28 bars: 77 g center + 13.5 g shell = 90.5 g theoretical net. No external brand/logo reproduced.

Stage: Investment phase • Fit class 3 • Pilot 28 units • Production example ×5 = 140 units.

Commercial case: Gift/treat premium hypothesis; cocoa-butter coating and frozen distribution may dominate cost. Preorders first, no impulse-volume promise.

Technical basis: [S09], [S13], [S16], [S17]. All exact formulas and operating adaptations here require controlled trials.

Formula — Dairy mix

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	1,170	53.182%	5,850
Cream 35%	520	23.636%	2,600
Skim milk powder	90	4.091%	450
Sugar	280	12.727%	1,400

Ingredient / specification	Pilot g	Percent	Production g
Dextrose anhydrous	60	2.727%	300
STAB 2000	6	0.273%	30
Vanilla	8	0.364%	40
Water	66	3.000%	330
TOTAL	2,200	100.000%	11,000

Mass route, pilot: 2,200 g input − 44 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 2,156 g retained. Allocation: 77 g finished component per unit. Production retained mass: 10,780 g. All losses are unmeasured planning allowances.

Formula — Ice-chocolate shell

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Callebaut Ice Chocolate White	420	100.000%	2,100
TOTAL	420	100.000%	2,100

Mass route, pilot: 420 g input − 42 g handling/unretained (10%) − 0 g process loss (0% of post-handling mass) = 378 g retained. Allocation: 13.5 g finished component per unit. Production retained mass: 1,890 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,620 g; modeled losses 86 g; net edible output 2,534 g = 28 × 90.5 g expected finished units, with no modeled remainder. Production example: 13,100 g input → 12,670 g output across 140 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (center only; coating excluded): fat 10.17%, milk solids-not-fat 10.04%, total solids 35.95%, unfermented lactose 5.41%; added sucrose 12.73% and anhydrous dextrose 2.73%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Freezing comparison only: sucrose-equivalent added-sugar index = sucrose% + 1.9 × anhydrous dextrose% = 17.91. This is NOT a full PAC/freezing-point prediction: native fruit sugars, lactose, salts, actual solids and overrun also matter. Confirm bite/draw at the real freezer temperatures; never adjust sugar without recalculating solids and rerunning the freeze test.

Numbered production method

1. Make the complete S-D02 center mix under H02, batch-freeze at a controlled low overrun (trial 20–30%) and immediately fill sanitized bar molds by mass.
2. Allocate the usable center evenly to 28 molds; insert sticks to a reproducible depth with at least roughly two-thirds of center length supported. Actual mold geometry must permit this without protrusion.

3. Harden to a measured core of -18°C or colder in a rated hardener. Demold small batches and keep remaining bars frozen; ordinary popsicle equipment is not assumed to aerate/fill this center.
4. Melt the specified Callebaut Ice Chocolate White to 35–40°C according to its product guidance. It is a special ice coating; no conventional tempering is required for this product. Keep water/condensation out.
5. Dip briefly, drain and weigh shell pickup. Model 420 g working shell input, 42 g unretained/unusable and 378 g on bars (13.5 g each). A deeper dip vessel may require extra reusable working stock, tracked separately as inventory.
6. Let the shell set while the core remains frozen, wrap individually in freezer-grade film and return immediately to -18°C or colder. Check stick pullout, shell cracks and package seal.
7. Do not use Magnum branding or imply its proprietary recipe. A own-brand dark/compound version needs its exact supplier's melt instructions, label, pickup and allergen trial.

Quality, defects and controlled variants

Thick shell → cold/high-viscosity coating. Sliding shell → surface ice/condensation. Cracks → warm core or thermal shock. Loose stick → incomplete hardening.

Ordinary couverture follows its own tempering curve; compound coating follows its fat-system instructions. Neither can inherit this special ice-chocolate temperature blindly.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Wrap sticks/bars in freezer-grade film or lid scoop tubs immediately, with batch/net mass/allergens. Store and distribute at -18°C or colder under measured control. No reheating, thaw/refreeze sales or assumed month-long life. Study days 0/7/14/28, crystal size, melt, seal and microbial plan under H05.

S-Y01 — Plain set yogurt

20 cups, theoretical 104.174 g net each. Package by verified net mass; do not imply 100 ml equals 100 g.

Stage: Cold-line phase • Fit class 3 • Pilot 20 units • Production example ×10 = 200 units.

Commercial case: Plain yogurt supports family breakfasts and ingredient use; no probiotic/health claim. A 20 L milk run is conditional, not a confirmed 50 L vessel load.

Technical basis: [S18], [S19], [S20]. All exact formulas and operating adaptations here require controlled trials.

Formula — Yogurt base

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	2,060	96.896%	20,600
Skim milk powder	60	2.822%	600
Yogourmet Original 3g sachet	6	0.282%	60
TOTAL	2,126	100.000%	21,260

Mass route, pilot: 2,126 g input − 42.52 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 2,083.48 g retained. Allocation: 104.174 g finished component per unit. Production retained mass: 20,834.8 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,126 g; modeled losses 42.52 g; net edible output 2,083.48 g = 20 × 104.174 g expected finished units, with no modeled remainder. Production example: 21,260 g input → 20,834.8 g output across 200 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 3.42%, milk solids-not-fat 11.40%, total solids 15.10%, unfermented lactose 6.12%; added sucrose 0.00% and anhydrous dextrose 0.00%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Lactose is the pre-fermentation estimate; acidification changes sugars/acids. These pilot formulas intentionally use no added stabilizer. Culture performance, pH curve and separation are checked, not assumed from a yogurt name.

Numbered production method

1. Use documented homogenized pasteurized/UHT whole milk and skim milk powder. Milk basis is 2 L assumed at 1.030 kg/L, giving 2,060 g; verify actual density and sachet label.
2. Disperse milk powder without foam and heat under H04 to 90–95°C for 5 minutes throughout as a technical yogurt-milk treatment. Validate cold point and covered evaporation; this is not proof from the stirrer display.
3. Cool promptly to 42–44°C, add two complete current 3 g Yogourmet Original sachets, dispersing in a little treated milk then mixing back. Do not use old 5 g-pack instructions or backslop retail yogurt.
4. Fill sanitized food cups and cover before incubation. Incubate undisturbed at 42–44°C; take hygienic sample cups for pH measurement. Start checking after 3 hours; about 5 hours is a supplier timing guide, not release criteria.
5. End incubation at pH 4.5 or the qualified culture-specific endpoint; initiate rapid cooling. H04 requires monitored cooling to 4°C or below in a mapped small-cup process.
6. Refrigerate and retain samples; do not ship at ambient temperature. Verify pH again after cooling, clean mild aroma and intact gel without whey pooling.

Quality, defects and controlled variants

No acidification → inactive culture/inhibitors/wrong temperature: quarantine, do not extend indefinitely. Whey → excessive heat movement/overacidification. Grainy → solids poor dispersion.

Commercial DVS culture may be more economical at scale, but choose a named supplier with units-per-batch instructions and rewrite the dose; no gram-for-gram replacement.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Seal in sanitized food-grade cups/bottles with tamper evidence; transport at 4°C or below with logs. No reheating for ordinary yogurt. Study days 0/1/3/7 with a qualified laboratory, not an assigned 7-day sale life. Do

not taste gas-producing or suspect samples.

Culture check: the retained 6 g entry assumes two 3 g sachets for 2 L milk. Verify sachet mass and current regional supplier instructions before manufacture; a different culture is not substituted gram for gram.

S-Y02 — Stirred mango yogurt

Twenty stirred fruit cups; fruit treatment is an explicit gate. Model fruit evaporation as zero in covered processing; measure and revise.

Stage: Cold-line phase • Fit class 3 • Pilot 20 units • Production example ×10 = 200 units.

Commercial case: Fruit sweetness may widen appeal; compare plain and mango in paid trials. Fruit availability, imported-puree cost and cold-chain supply are unresolved.

Technical basis: [S18], [S19], [S20], [S24]. All exact formulas and operating adaptations here require controlled trials.

Formula — Yogurt base

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	2,060	92.543%	20,600
Skim milk powder	60	2.695%	600
Yogourmet Original 3g sachet	6	0.270%	60
Sugar	100	4.492%	1,000
TOTAL	2,226	100.000%	22,260

Mass route, pilot: 2,226 g input − 44.52 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 2,181.48 g retained. Allocation: 109.074 g finished component per unit. Production retained mass: 21,814.8 g. All losses are unmeasured planning allowances.

Formula — Fruit preparation

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Pasteurized mango puree 15 Brix	200	71.429%	2,000
Sugar	80	28.571%	800
TOTAL	280	100.000%	2,800

Mass route, pilot: 280 g input − 0 g handling/unretained (0%) − 0 g process loss (0% of post-handling mass) = 280 g retained. Allocation: 14 g finished component per unit. Production retained mass: 2,800 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,506 g; modeled losses 44.52 g; net edible output 2,461.48 g = 20 × 123.074 g expected finished units, with no modeled remainder. Production example: 25,060 g input → 24,614.8 g output across 200 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 2.90%, milk solids-not-fat 9.67%, total solids 21.19%, unfermented lactose 5.19%; added sucrose 7.18% and anhydrous dextrose 0.00%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Lactose is the pre-fermentation estimate; acidification changes sugars/acids. These pilot formulas intentionally use no added stabilizer. Culture performance, pH curve and separation are checked, not assumed from a yogurt name.

Numbered production method

1. Make the listed sweet yogurt base with S-Y01 heating and culture controls, but incubate in a sanitized covered fermentation vessel rather than final cups. Obtain supplier confirmation for the sweetened, fortified adaptation.
2. Ferment at 42–44°C and measure pH in hygienically drawn samples. Stop at pH 4.5 by prompt cooling; gently break the gel, avoiding froth.
3. Use an aseptic ready-to-eat mango puree. Dissolve the fruit-stage sugar hygienically in a portion of heated puree and apply a separately validated fruit preparation treatment; do not add unvalidated raw fruit after fermentation.
4. Cool fruit preparation to 4°C or below. Gently blend all 280 g into the cooled yogurt base; record loss, pH and fruit distribution. Do not dilute to replace missing dairy yield.
5. Fill sanitized cups, seal and keep at 4°C or below. Run pH, yeast/mold and sensory trials before setting any commercial storage life.

Quality, defects and controlled variants

Whey separation → overacidification/aggressive stirring. Fermented fruit odor/gas → contamination: reject and investigate. Uneven color → poor gentle blending.

Pineapple is not a direct swap: fruit enzyme, acidity and heat treatment differ. Use separately treated fruit and rebalance.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Seal in sanitized food-grade cups/bottles with tamper evidence; transport at 4°C or below with logs. No reheating for ordinary yogurt. Study days 0/1/3/7 with a qualified laboratory, not an assigned 7-day sale life. Do not taste gas-producing or suspect samples.

Culture check: the retained 6 g entry assumes two 3 g sachets for 2 L milk. Verify sachet mass and current regional supplier instructions before manufacture; a different culture is not substituted gram for gram.

S-Y03 — Vanilla drinking yogurt

Ten bottles, theoretical 216.58 g each. Bottle capacity needs headspace and measured density.

Stage: Cold-line phase • Fit class 3 • Pilot 10 units • Production example ×10 = 100 units.

Commercial case: Portable breakfast drink; compete on taste/portion/handling, not unsupported gut-health claims. S29 kossam is adjacent evidence, not identical product identity.

Technical basis: [S18], [S19], [S20], [S29]. All exact formulas and operating adaptations here require controlled trials.

Formula — Yogurt base

Percent basis: this component input = 100% (not whole assembled product).

Ingredient / specification	Pilot g	Percent	Production g
Whole milk 3.5%	2,060	93.213%	20,600
Yogourmet Original 3g sachet	6	0.271%	60
Sugar	140	6.335%	1,400
Vanilla	4	0.181%	40
TOTAL	2,210	100.000%	22,100

Mass route, pilot: 2,210 g input − 44.2 g handling/unretained (2%) − 0 g process loss (0% of post-handling mass) = 2,165.8 g retained. Allocation: 216.58 g finished component per unit. Production retained mass: 21,658 g. All losses are unmeasured planning allowances.

Batch yield and portion control

Total ingredient input 2,210 g; modeled losses 44.2 g; net edible output 2,165.8 g = 10 × 216.58 g expected finished units, with no modeled remainder. Production example: 22,100 g input → 21,658 g output across 100 units. Display rounding is not label-weight verification.

Composition and freezing/fermentation controls

Development composition estimate (combined input before processing): fat 3.26%, milk solids-not-fat 8.39%, total solids 18.26%, unfermented lactose 4.47%; added sucrose 6.33% and anhydrous dextrose 0.00%. These are formulation estimates, not nutrition analysis or a legal identity certificate.

Lactose is the pre-fermentation estimate; acidification changes sugars/acids. These pilot formulas intentionally use no added stabilizer. Culture performance, pH curve and separation are checked, not assumed from a yogurt name.

Numbered production method

1. Prepare the listed milk, sugar and vanilla using H04 heat treatment; the base deliberately omits added milk powder to reduce body, not milk safety controls.
2. Cool to 42–44°C and add two current 3 g Original sachets per 2 L milk basis. Confirm this sweetened use with supplier; do not split industrial culture sachets without dosing instructions.

3. Incubate in a covered sanitary vessel; track pH and temperature. Stop at pH 4.5 with prompt cooling, then shear gently with the cleaned stick blender in short pulses until pourable.
4. Do not add raw water after fermentation. Fill ten sanitized food bottles by measured net mass and seal with tamper evidence; keep at 4°C or below.
5. Check pourability and separation after 24 and 48 hours, plus pH and laboratory plan. A “shake gently” instruction may address natural separation but cannot excuse microbial gas, spoilage or unsafe storage.

Quality, defects and controlled variants

Too thick → culture/body system or inadequate controlled shear. Sediment/whey → weak stability. Swollen bottle → reject, no tasting; investigate hygiene.

Do not add a random pectin/CMC dose. A stabilized long-distribution version needs a selected acid-dairy stabilizer, supplier process and a new formula.

Allergens, packing, life study and costing

Allergens/ingredients to declare: milk.

Seal in sanitized food-grade cups/bottles with tamper evidence; transport at 4°C or below with logs. No reheating for ordinary yogurt. Study days 0/1/3/7 with a qualified laboratory, not an assigned 7-day sale life. Do not taste gas-producing or suspect samples.

Culture check: the retained 6 g entry assumes two 3 g sachets for 2 L milk. Verify sachet mass and current regional supplier instructions before manufacture; a different culture is not substituted gram for gram.

9. Fruit juice and yogurt preservation

For the first commercial trials, use documented treated juice or purée and pasteurized/UHT dairy, make modest quantities, pack hygienically and maintain the cold chain. Add a validated in-house treatment only when the equipment and process have been qualified. The best method depends on the product, pack, sales rate, power reliability and desired life.

Fruit juices: select the preservation route

Route	Practical use	Conditions and limits
Refrigerated treated juice	Small daily batches; immediate-service drinks	Use treated inputs or a validated final process, hygienic filling and storage at 4°C or below. Refrigeration slows growth; it does not make untreated juice safe.
Pasteurization followed by chilling	A repeatable packaged juice line	Have a specialist establish treatment for the exact juice, pH, soluble solids, pulp, batch size and equipment. Record the coldest product temperature and hold, then cool and fill under the approved procedure.
Freezing	Seasonal fruit purchasing and controlled stock	Use a safe treated product, portion into freezer-compatible containers with expansion space, freeze promptly and hold at −18°C or colder. Thaw under refrigeration. Freezing is not a kill step.
Contract high-pressure processing	A potential later fresh-taste line	Assess an available processor, validated product/package compatibility, transport and cold chain. Do not assume ambient stability or local availability.

Route	Practical use	Conditions and limits
Validated hot-fill or aseptic processing	A later product designed for ambient distribution	Requires a product-specific scheduled process and compatible package/closure. A sealed bottle, hot filling alone or added lemon does not establish shelf stability.

Untreated juice has caused illness. Wash sound whole fruit under potable running water before cutting, remove damaged areas and reject rotten fruit; cleaning is not a substitute for an effective treatment. Do not use soap on fruit. Keep raw fruit handling separate from treated juice and sanitized bottles. [N01]

There is no one safe time–temperature recipe for all pineapple, mango, citrus, avocado blends or hibiscus drinks. Acidity, solids and pulp change the process. FDA's five-log pathogen-reduction framework is a useful technical benchmark from the United States, not a claim about Cameroon's legal requirements. Have the actual product and process assessed. [N02]

For freezing, use containers intended for the freezer, allow appropriate expansion headspace and label each portion. Freezing preserves stock but does not sterilize it. Use potable ice only if it is a declared formula ingredient; adding ice or water after treatment can recontaminate the drink and changes the formula. [N03]

A practical juice production sequence

1. Define one product: fruit variety, supplier, juice/purée proportion, pulp size, water, sugar and all other ingredients. Record fruit condition and lot. Keep pineapple juice, mango nectar and avocado blends as separate specifications.
2. Measure and record pH and °Brix with calibrated instruments. These characterize the batch; neither reading alone proves safety. Record the measurement temperature and instrument ID.
3. Use the established treatment for the whole product, or documented treated inputs under an approved hygienic assembly process. Do not borrow a dairy heating schedule for juice.
4. Clean and sanitize bottles, closures, tools and contact surfaces using their supplier instructions. Use packaging suited to the actual fill temperature and cooling method. Record closure/seal checks.
5. Chill through the qualified process, keep at 4°C or below, and transport in monitored insulated/refrigerated equipment. Put out small service quantities and replace them with fresh sealed stock; do not top up a partly used dispenser.
6. Assign the production lot and the storage instruction immediately. Set the use-by period only from a product-specific study that includes the intended distribution and foreseeable temperature variation.

Lemon, ginger, sugar, ascorbic acid and stabilizers may affect flavour, colour, acidity or texture. They do not independently validate preservation. Antioxidants that reduce browning do not make avocado or banana purée shelf-stable. Do not improvise preservative doses; any future additive system needs an identified food-grade ingredient, a permitted use, supplier specification and a validated formulation.

Yogurt: preserve the fermented product

Method	Best use	Key control
Controlled fermentation, rapid cooling and sealed chilled storage	Set, stirred and drinking yogurt	Follow H04, the actual culture instructions and a verified acidification curve; cool promptly and maintain 4°C or below.

Method	Best use	Key control
Small individual cups or bottles	Retail and transport	Reduces repeated opening and portion handling; validate sealing, cleanliness and cooling of the filled pack.
Separate treated fruit preparation	Mango yogurt and fruit layers	Cool the fruit preparation hygienically, combine under a controlled process and reassess the final product's pH and life.
Freezing	Frozen-yogurt bars, smoothies or baking ingredients	Freeze only sound, safely handled product. Expect texture separation on thawing; freezing does not rescue spoiled yogurt.
Heat treatment after fermentation	A different, professionally developed product	Changes live-culture status and texture. Ambient stability requires a validated process and package; it is not achieved by ordinary reheating and bottling.

For the three yogurt recipes S-Y01–S-Y03, use the shared milk-treatment, culture and cooling plan in H04. The retained 6 g starter amount assumes a particular sachet mass. The current Yogourmet page specifies one sachet per litre and 42–44°C incubation; verify the actual sachet and regional instructions. Do not confuse the dry starter's shelf life with finished yogurt's life. [N05]

The source's pH 4.5 cooling trigger is a development target. Record the time taken to reach it, not simply an overnight incubation time. A failed acidification curve must be investigated; adding more culture late or leaving the batch warm indefinitely is not an approved recovery. A fermented product still needs hygienic handling, cooling and storage. The heat-treatment and fermentation stages work together. [N06]

For set yogurt, ferment in the final suitable cups and minimize disturbance of the gel. For stirred yogurt, use a controlled stirring step after fermentation. For drinking yogurt, adjust the formulation deliberately rather than diluting finished yogurt with untreated water. Add treated fruit or flavouring with sanitized tools and record the lot and amount. Sugar and thickener improve formulation properties; they do not replace refrigeration.

Keep yogurt covered and use clean serving tools. Discard the entire container if mould develops. Freezing can change yogurt's texture, making it more suitable for smoothies or baking than for a smooth spoonable product after thawing. Assess your actual product before choosing that route. [N07]

Power loss, transport and service

Place independent thermometers or loggers in storage and transport. Keep doors closed during interruptions, record the start time and temperatures, and prioritize cold storage in the power plan. Do not judge safety by taste or smell. Isolate stock with unknown temperature history and obtain a disposition under the food-safety plan.

Use 4°C or below as the house refrigeration target. FDA consumer guidance uses a two-hour limit for perishables at room temperature, reduced to one hour above 32°C; this is an action threshold, not a validated commercial shelf life or a substitute for logged cold-chain control. Keep service quantities small and minimize exposure. [N09]

Preservation trial and release sheet

Record	Required observation
Product identity	Recipe version, fruit/milk/culture lots, ingredients and actual masses

Record	Required observation
Process specification	Equipment, batch size, treatment program or treated-input evidence
Measurements	pH, °Brix where relevant, temperature/hold readings and calibrated instrument IDs
Yogurt fermentation	Inoculation temperature, culture identity/dose, timed pH readings, endpoint
Cooling and filling	Start/end times, product temperatures, container type, seal result
Storage and delivery	Continuous or scheduled temperature records, outages, dispatch/receipt
Shelf-life study	Agreed sampling dates, microbiological assessment, sensory/physical changes, packaging integrity
Decision	Approved process and life, rejected lot or further trial, reviewer and date

Start with three repeat process trials before a broader life study. Agree microbiological testing and acceptance criteria with a competent food laboratory for the product. Test the finished packed product, not just incoming milk or fruit. Include the expected transport conditions. No automatic 7-, 14-, 30- or 90-day selling period is assigned here. Sampling on days 0, 1, 3 and 7 is a study design option, not permission to sell through day 7.

Preservation should support flavour as well as safety: compare aroma, colour, separation, texture and customer acceptance while recording refrigeration, packaging and spoilage cost per sold portion. This is especially useful when deciding whether to sell chilled juice, frozen juice portions or yogurt bars from the same seasonal fruit supply.

10. Cake-donut cracking: your controlled trial

T01 — Rolled and cut cake donuts

Source: your September discussion “Reduce cracks in cake donuts”. These are your formula quantities, not a new validated commercial recipe.

Ingredient	Original batch g	1 kg flour pilot g
Flour	6,000	1,000
Sugar	1,000	166.67
Butter	1,000	166.67
Baking powder	80	13.33
Egg, edible mass	1,500	250
Water	1,500	250
EDC, identity to confirm	10	1.67
Total	11,090	1,848.34 rounded

EDC was described as viscous, but its identity, concentration and supplier dose were not confirmed. Do not assume EDC-95 or increase its dose. Obtain the label before continued use; if it cannot be identified, conduct the development control without it and document the change.

1. Record the existing mixing sequence, flour type, dough temperature, rolling thickness, cutter size and piece mass. Hold these constant between comparable tests.
2. Mix briefly after flour addition. Rest covered about 10 minutes as a trial, use minimal bench flour and avoid repeated rolling of scraps.
3. Compare a control at 250 g water per 1 kg flour with 275 g only if the control is stiff or tears. Keep all other identified ingredients unchanged. If already soft, do not add more water simply to remove cracks.
4. Run any EDC omission comparison separately from the water change. One variable at a time makes the result interpretable.
5. Trial frying at 180–185°C with measured oil recovery and a small load. Let the first surface set before turning; determine cook time using actual thickness and a set center. Oil temperature alone is not proof of doneness.
6. Cool samples and compare cracks, greasy texture, interior moisture, handling breakage and eating quality. Record at least ten pieces when possible. Select the formulation from repeated results, not a single attractive piece.

No portion count, rolling thickness, frying duration or shelf life was established in the discussion. Set these during the trial before issuing a production card. Declare wheat, egg and milk plus any emulsifier-carrier allergens.

11. Earlier formulas and development variants

These development records preserve distinct products and important formula differences from earlier drafts. They are not production instructions. Choose a complete primary card where available; otherwise resolve missing specifications, make a small controlled trial, measure yield and issue a full approved card. Original units and ranges remain visible so uncertainty is not hidden by invented conversions.

Older versions of the same product have been consolidated by family. Exact duplicates and superseded generic guidance are accounted for in the editorial record. Marketing claims about months of life, guaranteed profit, zero fuel cost or machinery performance have not been carried into working instructions.

V01 — Agege bread and tangzhong milk bread

Source SRC03/SRC04. Agege, flour basis: flour 100%; water 55–58%; sugar 10–12%; fat 8–10%; instant yeast 1.5–2%; salt 1.5%; improver 0.5% as an unverified original entry. Use the actual improver supplier dose, not the generic percentage. Original starting method: mix 3 minutes low, 8–10 minutes faster; proof to expansion; bake near 180°C for 25–30 minutes. Piece mass, tin size, dough temperature and finished yield are missing. Retain the dense, stretchy style as a separate trial from S-B02.

Shokupan original: tangzhong flour 7% plus liquid 35%; main dough flour 93%, milk 50%, sugar 10%, butter 10%, egg 10%, instant yeast 1.5%, salt 1.2%. The roux liquid plus main milk totals 85% of flour before egg if both are additional. It is unclear whether the 50% milk includes the roux liquid. Resolve this before mixing; do not silently reduce liquid or present the formula as verified. Baking, portion and pan specifications are also absent. Softness claims are not shelf-life evidence.

V02 — Specialist bread and pastry archive

These formula records come from SRC16 and the related SRC17/SRC19 drafts; SRC05/SRC14 later summarized them. Amounts below are original batch quantities unless percentages are stated. Butter, margarine and shortening are distinct ingredients requiring an explicit choice.

Product	Retained formula or distinctive variant	Development decision
Plain butter biscuits	Flour 18 kg; powdered sugar 7.2 kg; butter/margarine 9 kg; egg 1.8 kg; milk powder 540 g; vanilla 90 g	Distinct from S-P04 coconut biscuits. Confirm fat, portion, mixing and water activity; reject six-month life claim.
Croissant	Flour 18 kg; water/milk 9.9 kg with split unspecified; sugar 1.98 kg; dough butter 720 g; salt 360 g; yeast 270 g; roll-in butter 4.5 kg	Other drafts use 9.72 kg water, 900 g dough fat and different roll-in percentages. Select one formula; do not combine them. Confirm lamination equipment, fat plasticity and proof conditions.
Pain au chocolat	Croissant base; 10–12 g chocolate per pastry in one version; two batons in another	Define baton mass and dough weight. Pilot 4 mm sheet and 8 × 12 cm rectangles only after the base is fixed.
Pound cake	Flour 18 kg; sugar 18 kg; fat 18 kg; egg 16.2 kg; baking powder 270 g; salt 90 g; flavour 100 ml	Another draft has 18 kg egg and 360 g baking powder. Select a fixed version, edible egg mass, tin fill and mixing method; use a creaming-capable mixer.
Hard dough loaf	Flour 18 kg; water 8.64 kg; sugar 1.26 kg; shortening 1.08 kg; salt 360 g; instant yeast 108 g	Other versions differ materially in water, sugar and fat. Validate dense-dough load and moulder limits; do not force tight rollers.
Pain aux raisins	Croissant dough; 2.5 kg raisins per 18 kg flour; pastry cream unspecified	Incomplete: define cream recipe, quantity, treatment and chilled handling.
Vendéenne-style braid	Flour 18 kg; sugar 3.24 kg; butter 4.5 kg; egg 5.4 kg; water 1.8 kg; salt 360 g; yeast 360 g	A 25%-butter variant, distinct from S-B06 50%-butter brioche. Regional naming needs care; validate cold fermentation and shaping.
Long sandwich roll	Flour 18 kg; water 10.8 kg; sugar 720 g; shortening 720 g; salt 360 g; instant yeast 144 g; malt 54 g	Define malt type and activity. Original 150 g dough piece is a pilot size, not final label mass.
Coconut crunch bread	Flour 18 kg; water 9.36 kg; sugar 2.7 kg; margarine 1.44 kg; coconut 1.8 kg; salt 270 g; yeast 180 g	Topping quantity absent; specify coconut moisture and topping before costing.
Whole-wheat loaf	Whole-wheat flour 9 kg + white flour 9 kg; water 11.7 kg; honey/molasses 900 g; oil 720 g; salt 360 g; yeast 126 g	Select honey or molasses. Optional oats/flax not quantified. Compare with V04's 60% whole-wheat formula.
Garlic-herb-cheese fougasse	Flour 18 kg; water 11.16 kg; oil 1.08 kg; garlic 90 g; herbs 54 g; salt 360 g; yeast 90 g	Garlic-oil and cheese topping amounts absent. Define yeast type and topping yield.
Laminated sausage roll	Flour 18 kg; lamination fat 9 kg; meat filling 15 kg	Pastry liquid, seasoning and binding missing. O-P08 is a complete shortcrust alternative, not an equivalent laminated formula.
Brownie	Sugar 14.4 kg; ambiguous "cocoa margarine" 9 kg; cocoa 3.6 kg; egg 7.2 kg; flour 5.4 kg; baking powder 54 g; vanilla 50 ml	Identify fat and tray depth; validate center set and cooled cut yield. Do not approve from surface colour alone.

Product	Retained formula or distinctive variant	Development decision
Pain viennoise	Flour 18 kg; milk 9.9 kg; sugar 1.44 kg; butter 1.8 kg; salt 360 g; yeast 180 g; optional chocolate 1.8 kg	Distinct from S-B07 pain au lait. Cost and label the chocolate version separately.
Levain boule	Flour 18 kg; water 11.7 kg; levain 3.6 kg; salt 360 g; instant yeast 36 g	Levain hydration, flour contribution and maturity missing; total hydration cannot be established.
Vanilla gelato	Milk 12 kg; cream 3 kg; sugar 3 kg; skim powder 750 g; glucose/dextrose 300 g; vanilla 60 g; stabilizer 90 g	Ingredient types/solids and stabilizer unidentified. Prefer S-D02; do not assume a soft-serve machine is a batch freezer.
Pineapple-ginger pop	Juice 14 kg; water 3.5 kg; 65°Brix syrup 2 kg; ginger 450 g; lime 150 g	Fruit composition and treatment missing. S-I01 is the more specified trial.
Cocoa cream pop	Milk 11 kg; water 3 kg; cocoa 1.5 kg; sugar 3.5 kg; “cornstarch/stabilizer” 120 g; vanilla 30 ml	Starch and stabilizer are not interchangeable. Prefer S-I03 or develop a separate specified starch formula.
Cocoa-banana gelato	Milk 10 kg; banana 4 kg; cocoa 1.5 kg; sugar 2.2 kg; cream 2 kg (SRC15)	Determine fat/solids, fruit treatment, freezing behaviour, equipment and portions.

Baguette, Kumba, Pullman, burger buns, pain beurre and pain au lait from the same archive map to S-B01–S-B05 and S-B07. Their earlier ingredient ranges were development versions, not independent approved products. The exact original files remain identified in the source register; the complete S cards are the working specifications.

V03 — Additional September concepts

Source SRC05/SRC14. These additions retain ideas not fully covered by the later cards.

Product	Formula retained	What must be resolved
Cocoa-groundnut breakfast loaf	Bread flour 17.1 kg; roasted groundnut flour 900 g; cocoa 540 g; water or milk 10.8 kg; sugar 1.44 kg; oil 900 g; milk powder 540 g; salt 324 g; instant yeast 180 g; optional chocolate 1.8 kg	Select water or milk; test absorption, nut quality, allergens and portion yield. Initial 250–300 g dough, 190°C/18–22 minutes are trial starting points.
Chicken-vegetable pocket	Dough: flour 10 kg, water 5.5 kg, sugar 500 g, oil 400 g, salt 180 g, instant yeast 100 g. Filling: cooked chicken 5 kg, cabbage 1 kg, carrot 800 g, onion 800 g, oil 250 g, seasoning 100 g, breadcrumbs or cooked potato 500 g	Select binder and seasoning; quantify egg wash and cooked filling loss. Original assembly 60–70 g dough + 40 g filling; bake trial 195–205°C/15–18 minutes with verified core cooking.
Mango “ginger” soft serve	Milk 5.5 kg; cream 1.2 kg; sugar 1.4 kg; skim powder 600 g; dextrose/glucose 250 g; mango 1 kg; lime 50 g; stabilizer about 50 g	No ginger amount is supplied despite the name. Total is 10.25 kg before adjustments. Select dry sugar type and specified stabilizer; qualify treated fruit additions and machine compatibility.
Early bissap pop	Water 6.5 kg; infusion 1.5 kg; 65°Brix syrup 1.8 kg; ginger 150 g; lime 50 g; stabilizer 20–30 g	Syrup contributes only 1.17 kg sugar, so the original 18–22°Brix assertion is not supported by the listed sugars. Use S-I02 as the revised trial.
Shawarma baguette	Baguette 250 g; cooked chicken 80–100 g; fries 100–120 g; sauce 30 g; salad 30–40 g	A portion-range concept. Fix exact weights and costs using O-B18/O-B20/O-B22/O-B24 and S-B01 before sale.

Product	Formula retained	What must be resolved
Spring rolls	Cabbage 3 kg; carrot 1.2 kg; onion 700 g; optional cooked chicken/beef 2 kg; oil 250 g; garlic/ginger 120 g; seasoning/salt 120 g; optional breadcrumbs/cooked potato 350 g	Original target about 100 pieces, with 45–50 g filling each, purchased wrappers and 170–175°C frying trial. Define wrapper mass/count, split seasonings, sealing paste, absorbed oil and cooked filling yield; validate the center rather than relying on the original 3–4-minute time.
Focaccia slab; coconut nondairy frozen dessert; mango/passionfruit /banana fruit trio	Selection concepts without fully resolved formulas	Keep on the development list; do not imply a complete recipe was supplied. Rebalance fruit sugar/acid and coconut fat/solids separately.

V04 — October blueprint alternatives

Source SRC22; SRC23 is an exact duplicate. Values are original commercial batches. These preserve formula alternatives and new concepts without carrying over unsupported machine or life claims. Meat uses H01, dairy H02/H04 and fruit H03; no raw-meat massaging in the dairy tank.

Original product	Retained ingredients	Harmonized decision
Milk sandwich bread	Flour 50 kg; water 28 L; milk powder 2.5 kg; margarine 2 kg; sugar 3 kg; salt 900 g; yeast 600 g	Older milk-powder alternative to O-P01/S-B04; do not assume four-day life.
Hard-crust baguette	Flour 50 kg; water 31 L; salt 1 kg; yeast 500 g; ascorbic acid 10 g	Improver specification and supplier dose required; S-B01 is the clearer working card.
Chin-chin	Flour 25 kg; sugar 6 kg; margarine 5 kg; milk powder 1.5 kg; nutmeg/vanilla 200 g combined; eggs 30; water 5 L	Weigh eggs, split flavour quantities, specify dough thickness and fry endpoint. Three-month life unsupported.
Dual-leavened beignets	Flour 20 kg; sugar 4 kg; yeast 400 g; baking powder 150 g; salt 200 g; water 13 L; flavour 100 ml	Distinct from yeast puff-puff and cake beignets. Fix leavener types and test; no guarantee of hollow or oil-free texture.
Powder-milk yogurt	Full-cream powder 5 kg; water 22 L; sugar 2 kg; YF-L811 culture 5 g; flavour 50 ml	Culture units, supplier dose, total solids and working tank volume unresolved; do not inherit S-Y culture dosing.
Chicken marinade	Chicken 20 kg; yogurt 2 L; oil 1 L; garlic 500 g; spices 300 g; lemon 400 ml; salt 250 g	Weigh liquids and identify spices; marination is refrigerated and does not replace cooking.
Yogurt-mayo sauce	Mayonnaise 5.5 L; yogurt 3.5 L; garlic 500 g; lemon 250 ml; vinegar 100 ml; salt/pepper 55 g combined	SRC26 splits seasoning into 40 g salt + 15 g pepper and adds mint. Keep refrigerated; final mass/volume unmeasured.
Meat-pie filling	Beef 4 kg; potato 5 kg; carrot 3 kg; onion 2 kg; starch 500 g; “seasoning/water” 6 L	Seasoning versus water unresolved. O-B09 provides a complete alternative; no unsupported 40% saving claim.
Pie crust	Flour 15 kg; cold margarine 7.5 kg; salt 200 g; cold water 4.5 L	Distinct fat ratio from O-B07; validate short mixing and chilled handling.
Koki corn	Corn mash 15 kg; palm oil 1.5 L; pepper 100 g; salt 120 g; water 2 L	Specify corn solids, grinding, wrap size and validated steaming endpoint. Do not assume scraper aeration.

Original product	Retained ingredients	Harmonized decision
Ndjansang chilli sauce	Pepper 5 kg; onion 2 kg; toasted ndjansang 300 g; oil 2 L; garlic/ginger 500 g combined; salt/bouillon 200 g combined	Split all combined ingredients. Hot bottling and oil separation do not establish ambient safety.
Coconut tarts	Coconut 6 kg; sugar syrup 4 L; vanilla 50 ml; shortcrust shells 100	Specify syrup solids and shell mass/formula. Measure oven temperature throughout baking; residual heat is not automatically adequate.
Brioche poulet	Rich brioche dough 10 kg; cooked chicken filling 5 kg; egg wash 200 ml; sesame 50 g	Original assembly 80 g dough + 40 g filling gives 125 units before losses; define the dough/filling versions and actual yield.
Caramel topping	Sugar 10 kg; glucose 3 kg; water 4 L; cream 2 L; salt 30 g	The original 120°C caramelization claim is inadequate. Use a qualified controlled caramel method and appropriate vessel; dairy sauce is not automatically shelf-stable.
Sugar ring donuts	Flour 20 kg; sugar 3.5 kg; margarine 1.5 kg; yeast 350 g; water 9.5 L; nutmeg 100 g	Salt absent in original; resolve deliberate omission and formula before trial. Sugar coating unquantified.
Ganache rings	Donut rings 200; dark chocolate 4 kg; cream 3 L; butter 500 g	Define cream mass/fat, chocolate type, pickup and chilled handling. No all-day warm holding claim.
60% whole-wheat loaf	Whole-wheat flour 30 kg; white flour 20 kg; water 33 L; brown sugar 2 kg; margarine 1.5 kg; salt 900 g; yeast 600 g	Distinct from V02 50% version. Hydration is 66%; adjust by measured absorption, not an automatic extra 10%.
Avocado-banana cream	Avocado 8 kg; banana 4 kg; lime 200 ml; ascorbic acid 15 g; icing sugar 2 kg	Antibrowning ingredients do not confer shelf stability. Treat as perishable with a qualified process.
Savoury puff-puff	Flour 15 kg; leeks 1.5 kg; onions 1 kg; yeast 300 g; salt 250 g; water 10 L	Sauté/cool vegetables under control, define batter mass/portion and fry endpoint; cooking oil allowance absent.
Akara	Peeled black-eyed-bean mash 12 kg; onion 1.5 kg; salt 100 g; white pepper 30 g; water 1.5 L	Specify mash solids, milling and hydration; validate cooking and aeration with suitable equipment.

V05 — High-value recipe sheet differences

Source SRC20. These are alternative formulas, not automatic improvements to V04.

Product	Distinct original quantities	Gap or correction
Corn/makabo paste	Corn or grated makabo 15 kg; palm/vegetable oil 2.5 L; pepper 150 g; water 3 L; salt/bouillon 120 g	Corn and makabo are not interchangeable raw materials; makabo needs a separately established preparation/cooking process. Frying or baking cannot be assumed equivalent to steamed koki.
Chin-chin	Flour 12 kg; sugar 3.5 kg; margarine 2.5 kg; eggs 30; nutmeg 80 g; vanilla 100 ml; water about 2.5 L	Record edible egg mass and final hydration. Life requires water-activity, packaging and microbial assessment.
Brioche poulet	Flour 10 kg; fat 1.5 kg; sugar 1.2 kg; yeast 120 g; salt 150 g; cooked chicken filling 6 kg	Dough liquid/egg missing. Original 80 g dough + 30 g filling differs from V04; do not blend counts.
Drinking yogurt	Milk powder 4.5 kg; water 22 L; sugar 2.2 kg; starter/plain yogurt 500 g; gelatin/pectin 50 g	Starter alternatives and gelling agents are not interchangeable. Sugar/additive post-treatment introduction and chilling need redesign.

Product	Distinct original quantities	Gap or correction
Vinegar chilli sauce	Pepper 4 kg; onion 2 kg; garlic 500 g; ndjansang 100 g; oil 1.5 L; vinegar 1 L; salt 80 g	Vinegar acidity, final pH and thermal process unspecified; refrigerate under approved process, not an ambient claim.
Sweet milk loaf	Flour 15 kg; milk powder 600 g; sugar 1.5 kg; yeast 150 g; improver 30 g; water 8.5 L; margarine 800 g	Salt missing. Improver label required. Extra softness is not validated life extension.
Meat pie	Flour 10 kg; shortening 4.5 kg; beef 4 kg; carrot 2 kg; potato 2 kg; spices 150 g	Pastry water/salt and filling process loss omitted. Prefer O-P07 until completed.
Dual-leavened beignets	Flour 12 kg; sugar 2.8 kg; baking powder 100 g; yeast 80 g; vanilla 50 g; nutmeg 30 g; water 7.5 L	Compare with V04, with each leavener fixed and measurable. Salt omitted in original.
Caramel	Sugar 8 kg; water 2 L; glucose 1.5 kg; “cream/margarine” 2.5 L; salt 40 g	Cream and margarine are not interchangeable, and margarine volume is ambiguous. Use heat-safe equipment; adding dairy to caramel causes vigorous bubbling.
Avocado-banana cream	Avocado 10 kg; banana 4 kg; ascorbic acid 20 g; 65°Brix syrup 2 L; lime 300 ml	Weigh syrup and specify solids; original 48-hour colour claim is unverified and not a safety period.

V06 — Condensed-milk collection

Source SRC25. Preserve the condensed-milk concept while specifying brand composition, ingredient masses, treatment and machine compatibility. The primary O-series provides more complete trial formulations for most of these formats.

Product	Original batch or assembly	Decision
Soft-serve mix	Condensed milk 4.5 kg; milk 6 L; cream 3.5 L; water 6 L; sugar 1 kg; stabilizer 80 g; vanilla 60 ml	Weigh dairy; identify stabilizer and final solids. O-FBASE1 is the more specified trial.
Yogurt/frozen yogurt	Milk 12 L; condensed milk 5 kg; water 4 L; starter 200 g; vanilla 40 ml	Culture identity/dose and fermentation curve absent. Not automatically compatible with a soft-serve machine.
Milky pops	Condensed milk 6 kg; water 12 L; coconut milk or mango purée 3 L; cornstarch 350 g; salt 15 g	Coconut and mango are distinct variants; treatment and composition differ.
Pita	Flour 5 kg; water 2.8 L; sugar 120 g; salt 80 g; oil 150 ml; active dry yeast 60 g	Active dry yeast requires its own handling; not gram-for-gram instant yeast. Moulder flattening capability unverified.
Meat marinade	Meat 10 kg; yogurt 1.5 kg; garlic 250 g; vinegar 200 ml; oil 150 ml; salt 90 g; spice 150 g	Beef and chicken require separate specifications and cooking records.
Refrigerator pickle concept	Cucumber 5 kg; water 3 L; vinegar 2 L; sugar 250 g; salt 200 g; garlic 10 cloves	Vinegar acidity and garlic mass absent; sealed buckets are not an ambient preservation process. Use O-B23.
Ndjansang sauce	Pepper 2 kg; ndjansang 150 g; garlic 100 g; ginger 100 g; oil 1 L; salt 40 g	No acid control claimed. Treat as a refrigerated perishable sauce.
Condensed-milk burger buns	Flour 6 kg; condensed milk 800 g; water 2.8 L; butter 500 g; eggs 6; salt 90 g; instant yeast 80 g; sesame unspecified	Weigh egg and sesame/egg wash; use O-B25 as working alternative.

Product	Original batch or assembly	Decision
Smash burger	One bun, raw beef 120 g, cheese, sauce, pickles and onion	Toppings unquantified. Original 90-second cook is not a safety endpoint; follow O-P16/H01.
Pain chargé beef filling	Beef 4 kg; onion 2 kg; garlic/ginger 150 g; oil 300 ml; cabbage 3 kg; pepper/seasoning unspecified	Separate combined ingredients; weigh cooked output and portion.
Chips and pain chargé	Initial 140°C/5-minute fry, final 180°C/2–3-minute fry proposed; sandwich adds beef filling 150 g plus chips/sauce unspecified	Potato and plantain need separate trials. Use O-B24 for fries; define bread and all portions.
Condensed-milk sandwich loaf	Flour 10 kg; condensed milk 1.2 kg; water 5.4 L; shortening 400 g; salt 150 g; yeast 100 g	Working alternatives O-P01/O-P02; qualify pan load and finished mass.
Condensed-milk baguette	Flour 10 kg; condensed milk 300 g; water 5.8 L; salt 180 g; yeast 90 g	Distinct enriched bread, not S-B01 lean baguette.
Condensed-milk chin-chin	Flour 8 kg; margarine 1.5 kg; condensed milk 2 kg; eggs 12; nutmeg 40 g; frying oil unspecified	Edible egg, hydration, oil use and finished yield unresolved.
Condensed-milk puff-puff	Flour 5 kg; condensed milk 800 g; water 3.2 L; sugar 400 g; yeast 60 g	Trial sweetness/proof/fry behaviour; “oil-free” claim rejected.
Koki corn	Corn mash 10 kg; palm oil 1.2 L; pepper 50 g; water 1.5 L; salt 80 g	Specify steaming and mash consistency; a scraper does not guarantee aeration.
Condensed-milk caramel	Sugar 4 kg; condensed milk 2 kg; glucose 1.5 kg; water 1 L; butter 250 g	Qualify vessel, heat range, controlled dairy addition, chilling and life; no ambient claim.
Coconut pastry slices	Coconut 2 kg + condensed milk 1.5 kg; pastry sheets unspecified	Specify sheet formula, mass, pan depth and measured bake program.

V07 — Garlic sauce variants

Source SRC26. Toum original: peeled garlic 1.5 kg, oil 8 L, lemon juice 450 ml, water 250 ml, salt 70 g. Claimed 10 L yield is unmeasured; weigh inputs, specify oil, blend/emulsify using suitable equipment and determine usable output. A slow scraper is not assumed to make this emulsion. Garlic-in-oil sauce is a perishable product; no ambient storage is authorized.

Yogurt-mayo original: commercial mayonnaise 5.5 L, yogurt 3.5 L, garlic 500 g, lemon 250 ml, vinegar 100 ml, salt 40 g, white pepper 15 g and optional mint 2 tablespoons. Weigh ingredients and mint; verify the mayonnaise and yogurt supply specifications. Keep the cold preparation and service controlled. O-B20 and S-F04 offer smaller fully quantified alternatives with different flavour profiles.

V08 — Brioche format options retained

SRC05’s extended brioche uses the same 50%-butter base concept as S-B06 and adds useful alternative shapes. A 5 kg flour batch has 11,450 g raw dough; at an assumed 2% handling loss, 11,221 g remains. Alternative allocations are 187 × 60 g with 1 g remainder, 140 × 80 g with 21 g remainder, 37 × 300 g with 121 g remainder, or 22 × 500 g with 221 g remainder. These are alternatives, not simultaneous outputs.

For a 500 g loaf, arrange five 100 g balls; for a three-strand braid, use 167 + 167 + 166 g. A brioche à tête may reserve about one fifth for its head. Trial small pieces at 170–180°C for 14–20 minutes, 300 g loaves at 165–

175°C for 22–30 minutes, and 500 g loaves at 165–175°C for 30–40 minutes. Confirm crumb set and measured bake loss; the 90–94°C crumb target is a quality check, not a substitute for a validated food-safety process. Raw egg wash is applied only before baking.

Cool plain brioche completely before bagging. A frozen-quality trial at −18°C may compare two- and four-week samples; this does not establish a selling period or validate freezing raw dough. Keep butter, egg wash, refrigeration, rejects and unsold portions in the cost model.

12. Costing, production and a staged launch

Revision note: the original product tables remain baseline recipe estimates. For the added preparation-ahead route, chapter 6 provides extra reheating-loss allowances and new O-A01–O-A04 costs; update cold handling, transport, energy and labour before relying on these margins.

The following are transparent planning estimates, not quotations, current local prices or profit promises. Replace every ingredient and packaging input with a dated delivered supplier quote before using the plan to purchase or price stock. Whole milk powder and skim milk powder are different inputs. Edible-weight conversion matters: a 1,000 FCFA/kg gross ingredient with 20% trim loss costs 1,250 FCFA/kg usable, before labour.

Editable ingredient price assumptions

Ingredient	FCFA/kg	Ingredient	FCFA/kg
Baking powder	4,000	Lettuce trimmed	1,800
Baking soda	2,500	Mango puree pasteurized	1,800
Banana peeled	800	Margarine	1,800
Beef boneless	4,500	Mayonnaise pasteurized	2,500
Beef mince	4,500	Mozzarella	6,500
Black pepper	9,000	Mustard	3,000
Bread flour	650	Nutmeg	10,000
Breadcrumbs	1,000	Oil	1,600
Butter	7,000	Onion trimmed	800
Cabbage trimmed	600	Oregano	12,000
Canned fish drained	4,000	Paprika	8,000
Carrot trimmed	900	Peanut roasted	1,800
Cheese slices	6,500	Pepper trimmed	1,200
Chicken boneless	4,000	Pineapple puree pasteurized	1,500
Chili trimmed	2,000	Plain flour	650
Cinnamon	8,000	Potato peeled	900

Ingredient	FCFA/kg	Ingredient	FCFA/kg
Cocoa	7,000	Salt	400
Compound chocolate coating	4,500	Sausage meat	4,000
Condensed milk	3,000	Sesame	3,000
Coriander	8,000	Skim milk powder	5,000
Corn starch	1,800	Sugar	800
Cream 35%	6,500	Sweet potato flour	1,000
Cucumber trimmed	800	Tahini	7,000
Cumin	8,000	Thyme	10,000
Curry powder	6,000	Tomato trimmed	800
Egg	2,200	UHT milk	1,200
Garlic peeled	3,000	Vanilla	12,000
Ginger peeled	2,000	Vinegar 5%	900
Glucose powder	2,500	Water	5
Ice cream stabilizer blend	18,000	Whole milk powder	4,500
Icing sugar	1,400	Yeast	6,000
Ketchup	1,800	Yogurt plain	2,000
Lemon juice	1,800		

Egg example: a 50 g edible egg at the assumed 2,200 FCFA/kg costs 110 FCFA. This is a price assumption, not a local market observation. Quotes for milk sold by litre must be converted with measured density; do not silently treat oil litres as kilograms. Packaging assumptions include the named container, with a food-safe stick for popsicles and a spoon/straw where relevant.

How the model works

1. Batch ingredient cost = sum of ingredient grams × FCFA/kg ÷ 1,000. For a prepared component, recursively include all its raw ingredients and divide by its usable output, not raw input mass.
2. Ingredient cost per saleable unit = costed batch / stated saleable units. Explicit pastry/dough QC allowances are charged to the product. Full unused filling batches are not automatically charged twice: each assembly card charges only its required usable share, so unused safe stock must be recorded and eventually sold or written off.
3. Variable cost per unit = ingredient cost + packaging + other variable allowance. Contribution = selling price − variable cost. Contribution margin = contribution / selling price. Contribution does not pay the owner until overhead and other obligations are met.
4. Other-variable allowances are placeholders for production energy/diesel, incremental cleaning/water and variable handling/delivery. Provisionally allocate 60% to energy/diesel, 20% to incremental

cleaning/water and 20% to variable handling/delivery, then replace with actual logs. Paid shift wages and base route transport are in overhead below; do not charge the same expense in both places.

5. Sensitivity at 20% ingredient inflation: contribution falls by 0.20 × ingredient cost per unit. At 80% sell-through, assume all planned variable cost was incurred, including packaging: effective contribution per sold unit = price − full planned variable cost / 0.80. This conservatively charges unsold units to those sold.

6. All prices and calculations are gross cash planning values without a tax determination. If an applicable tax must be remitted from the displayed selling price, remove it from revenue before computing contribution. Obtain the business-specific treatment locally; no Cameroon tax status is assumed.

Core product contribution table

Product	Ingredients/unit	Pack + other	Test price	Contribution	Margin
O-P01 Soft family bread	278.4	30 + 55	650	286.6	44.09%
O-P02 Condensed milk breakfast buns	92.6	10 + 20	250	127.4	50.95%
O-P03 Nkambe sweet potato milk loaf	436.2	35 + 60	800	268.8	33.6%
O-P04 Sugar ring doughnuts	73.3	10 + 25	250	141.7	56.67%
O-P05 Puff puff	18.9	3 + 8	75	45.1	60.13%
O-P06 Cameroon style fish rolls	139.4	10 + 30	300	120.6	40.19%
O-P07 Beef meat pies	190.5	20 + 35	500	254.5	50.9%
O-P08 Sausage rolls	205.3	15 + 35	450	194.7	43.26%
O-P09 Creamy chicken pies	229.6	20 + 40	650	360.4	55.44%
O-P10 Whole egg dough rolls	197.2	15 + 35	500	252.8	50.56%
O-P11 Banana condensed milk cake slices	153.6	20 + 25	400	201.4	50.36%
O-P12 Personal chicken pizza	780.1	90 + 80	1,800	849.9	47.22%
O-P13 Complete chicken shawarma	957.5	80 + 60	2,000	902.5	45.12%
O-P14 Complete beef shawarma	1,059.7	80 + 65	2,400	1,195.3	49.8%
O-P15 Crispy chicken burger	660.2	70 + 65	1,800	1,004.8	55.82%
O-P16 Beef cheeseburger	780.7	70 + 60	1,900	989.3	52.07%
O-P17 Crispy chicken and fries box	1,272.3	100 + 100	2,500	1,027.7	41.11%
O-P18 Chicken vegetable samosas	106.3	10 + 25	300	158.7	52.9%
O-P19 Loaded chicken shawarma fries	895.9	90 + 60	1,800	754.1	41.89%
O-P20 Condensed milk soaked cake portions	334.2	45 + 35	650	235.8	36.28%

Frozen dessert contribution table

Product	Batch cost / units	Ingredients/unit	Pack + other	Test price	Contribution
O-F01 Vanilla soft serve cup or cone	2,100 / 10	210.0	35 + 30	600	325.0
O-F02 Chocolate peanut sundae	1,560 / 5	312.0	60 + 35	1,000	593.0
O-F03 Chocolate milkshake	2,515 / 5	502.9	75 + 40	1,200	582.1
O-F04 Chocolate cookie ice cream sandwich	2,145 / 10	214.5	25 + 35	700	425.5
O-F05 Condensed milk popsicles	2,057 / 17	121.0	20 + 20	300	139.0
O-F06 Pineapple fruit ice pops	1,017 / 17	59.9	20 + 18	250	152.2
O-F07 Mango yogurt frozen bar	1,902 / 12	158.5	25 + 25	450	241.5
O-F08 Chocolate dipped peanut milk bar	2,290 / 10	228.9	30 + 35	600	306.1
F05L Condensed milk popsicles large example	2,057 / 8	257.1	25 + 25	500	192.9
F06L Pineapple fruit ice pops large example	1,017 / 8	127.2	25 + 22	400	225.8

The large pop is twice the small fill volume, but the selling price need not be twice the small price because one stick/bag and part of the service effort are shared per unit. The lower large-milk-pop margin here also reflects conservative pilot remainders. Do not infer a steady production margin from a small batch that leaves most of a cavity unused.

Recipe and energy cost reality checks

Bread uses approximately 278 FCFA ingredients per loaf at the assumed flour price, before packaging and energy. Its 650 FCFA test price is not automatically affordable locally; test against actual competing loaf weights. O-P03 sweet potato bread and O-P20 soaked cake have lower example contribution margins than several snacks. Their roles are differentiation and basket value, not guaranteed volume or margin leadership.

The family loaf's 55 FCFA other-variable allowance assigns 33 FCFA to energy/diesel provisionally. If an oven load produces 48 loaves, the allocated energy budget is 1,584 FCFA per load. Compare that with measured diesel litres × delivered diesel price plus electricity/battery-related costs. If the actual energy cost per loaf is higher, replace the allowance; do not declare the loaf profitable from ingredient cost alone.

Example daily fixed/standing cost item	FCFA/day assumption	Measurement needed
Paid production and sales shifts including owner replacement labour	18000	Actual staffing hours, wages and employer costs
Rent and standing facility expenses	3500	Both premises monthly costs / actual trading days
Base transport route	2500	Vehicle/fuel/driver allocation; incremental deliveries above excluded
Standing refrigeration and backup power costs	2500	Metered overnight/standby power and backup cost

Example daily fixed/standing cost item	FCFA/day assumption	Measurement needed
Equipment and solar/battery depreciation or replacement reserve	2500	Asset price, expected life and maintenance plan
Routine deep cleaning and pest control	1000	Base cleaning labour/consumables not already variable
Administration and communications	1000	Actual fixed expenses
Total illustrative daily overhead	31000	Excludes financing, income tax and unexpected losses

Daily basket illustration and break even

Product	Units prepared	Price	Revenue if all sell	Planned variable cost
O-P01 Soft family bread	30	650	19,500	10,902
O-P02 Condensed milk breakfast buns	50	250	12,500	6,132
O-P04 Sugar ring doughnuts	40	250	10,000	4,333
O-P05 Puff puff	60	75	4,500	1,794
O-P06 Cameroon style fish rolls	30	300	9,000	5,383
O-P07 Beef meat pies	24	500	12,000	5,892
O-P13 Complete chicken shawarma	12	2000	24,000	13,170
O-F01 Vanilla soft serve cup or cone	20	600	12,000	5,500
O-F05 Condensed milk popsicles	17	300	5,100	2,737

This illustration prepares 283 units in the stated mix, not a demand forecast. Revenue at 100% sale is 108,600 FCFA; planned variable cost 55,844; contribution 52,756. After 31,000 FCFA illustrative overhead, operating surplus is 21,756 FCFA before financing, income tax and exceptional losses. Filling and mix start-up stock can require more cash than these consumed-portion costs.

Scenario	Revenue	Variable cost incurred	Overhead	Operating result
100% sold, base prices	108,600	55,844	31,000	21,756
80% sold, all production costs incurred	86,880	55,844	31,000	36
60% sold, all production costs incurred	65,160	55,844	31,000	-21,684
100% sold, ingredient costs +20%	108,600	64,555	31,000	13,045
80% sold and ingredient costs +20%	86,880	64,555	31,000	-8,675

At full sell-through and unchanged mix, average contribution is 186.4 FCFA/unit; break-even volume is about 167 equivalent units/day, or about 63,814 FCFA sales. If the entire illustrated basket is already produced, its cash break-even sell-through is 80.0%. These are different questions. Changing the product mix, waste rate or staffing changes both thresholds.

Price ladder and simple combinations

Use visible single-item prices and portion descriptions. Entry: 75 FCFA puff puff, 250 FCFA bun/doughnut, 250–300 FCFA small pop. Everyday: 300 FCFA fish roll, 500 FCFA beef pie, 650 FCFA family bread. Treat: 600 FCFA soft serve or 650 FCFA milk cake. Meal: 1,800–2,400 FCFA pizza, burgers and wraps. These are test points; an expensive meal should not displace the affordable traffic range.

Two milk buns: example bundle price 450 FCFA; ingredient/pack/other variable cost 245.3; contribution 204.7 before overhead. Use only when it adds purchases rather than discounting an order the customer would have made anyway.

Beef pie and small milk pop: example bundle price 750 FCFA; ingredient/pack/other variable cost 406.5; contribution 343.5 before overhead. Use only when it adds purchases rather than discounting an order the customer would have made anyway.

Chicken shawarma and small fruit pop: example bundle price 2,200 FCFA; ingredient/pack/other variable cost 1,195.4; contribution 1,004.6 before overhead. Use only when it adds purchases rather than discounting an order the customer would have made anyway.

For the first week, capture actual supplier cost, produced units, rejects, samples, sales, discounts, unsold units and customer comments daily. Update the CSV inputs and re-run the calculations before changing the menu. The supplied CSVs are editable data tables, not a live automatically recalculating spreadsheet; the accompanying Python builder regenerates the report from changed inputs.

Large-scale baguette production: oven-led scheduling

This expanded operating section was prepared after consulting the Commercial Baguette Production Guide (SRC21), Staggered Baguette Production Schedule (SRC27), the five-batch operations manual (SRC17), and the September corrected manual and equipment trial plan (SRC05/SRC07). It replaces their contradictory timing instructions with explicit calculations and a worked scheduling example. The timetable below is an editorial planning model, not an observed factory result.

Choose the loaf specification before choosing the batch

Use S-B01 as the lean baguette formula: flour 100%, water 66%, salt 2%, instant yeast 0.5%. Its total formula factor is 1.685. Keep this formula distinct from the condensed-milk baguette variant. Fix the flour, yeast, dough temperature and proof method through trials before increasing rack volume.

Confirm that five loaves of the selected length and spacing actually fit each 40 × 60 cm tray, and that sixteen loaded trays bake evenly. Five per tray gives 80 loaves per rack. Neither tray area nor a nominal mixer rating alone establishes usable production capacity.

Calculation	Result	Operational meaning
Source guide ingredients: 14.20 kg flour + 9.37 kg water + 284 g salt + 71 g yeast	23.925 kg input	This is not 24 kg. It cannot produce 80 × 300 g even before handling loss.
Same source batch, assuming 2% handling loss	23.4465 kg usable = 78 × 300 g + 46.5 g remainder	Replace the loss assumption with measured bowl, bench and dividing losses.
300 g raw piece with 15% baking/cooling loss	255 g cooled loaf	Do not label the result 250 g merely because the original report did.

Calculation	Result	Operational meaning
Raw piece for a 250 g cooled target at 15% loss	250 ÷ 0.85 = 294.12 g	A theoretical mean target, not a guarantee that every loaf meets declared net weight. Establish the actual distribution and label-control procedure.
Original 18 kg flour batch at 66% water, 2% salt and 0.5% yeast	30.33 kg input; 29.7234 kg after assumed 2% handling loss	About 99 × 300 g plus 23.4 g, not one 80-loaf rack. Plan the surplus rack or resize the mix.

For N loaves, raw piece mass w grams and measured handling-loss fraction h:

Flour required (g) = N × w ÷ [(1 − h) × 1.685]. Water = flour × 0.66; salt = flour × 0.02; instant yeast = flour × 0.005. Ice is part of the water amount, never an extra ingredient.

One-rack planning example: 80 pieces at 300 g raw

Ingredient	Calculated mass g, rounded to 0.1 g
Bread flour	14,534.0
Total water, including any ice	9,592.4
Salt	290.7
Instant dry yeast	72.7
Total input	24,489.8

At the illustrative 2% handling loss, this provides approximately 24,000 g usable dough. Expect about 255 g per cooled loaf if the separate baking/cooling loss is 15%. These fine decimals explain the calculation; use suitable scales and a documented practical rounding/overage policy after measuring real variation. Do not reclaim dirty bench scraps to reach the theoretical count. Reserve separate dough for destructive samples if all 80 pieces must be saleable.

For a 250 g cooled target under the same assumptions, flour is approximately 14,249.0 g and raw pieces approximately 294.12 g; recalculate the remaining ingredients with the formula above. Do not change the dough weight without updating the cost and label specification.

Check minimum and maximum lean-dough load before running this 24.49 kg input example. The older reports conflict on mixer rating. A two-rack mix would be approximately 48.98 kg input and would create extra handling and fermentation queues; it is not automatically permitted by a “50 kg” machine description. Start with one validated rack-sized mix if the machine's minimum load allows it.

Measure the complete oven cycle

Time rack loading, approved steam operation, baking, unloading and recovery to the next approved loading condition. Some steps overlap; record the actual start-to-start interval rather than adding overlapping observations twice. The source reports disagree between three minutes of turnaround and a supposedly mandatory 23-minute recovery. Neither recovery allowance has been established for this oven.

Scenario	Planning capacity at 80 loaves/rack	Requirement
22-minute bake + 3-minute complete turnaround	192 loaves/hour at 25-minute rack starts	Only if repeat loaded trials demonstrate recovery, handling and upstream supply at that pace.
22-minute bake + 8-minute complete turnaround	160 loaves/hour at 30-minute rack starts	Worked example below; every station must sustain this rate.
One rack supplied every 45 minutes	106.7 loaves/hour maximum sustained supply	A faster oven does not remove the upstream gap.
Dividing or cooling/packing limited to 100 loaves/hour	100 loaves/hour; at least 48 minutes per 80-loaf batch	This was a September workbook assumption, not a measured rate. If confirmed, it governs the line until improved.

The line's rack-start interval must be at least the largest of: measured complete oven cycle; mixer occupied time per rack batch; dividing/shaping time per rack; rack-equivalent packing/cooling constraint; and final-proof duration divided by the number of usable proof positions. Also check shared workers, trays, dough bins and rack movements; capacity cannot be counted twice when the same person serves two stations.

At a 22-minute bake, productive baking occupies 88% of a 25-minute interval, 73.3% of a 30-minute interval, and 48.9% of a 45-minute interval. These are time ratios, not burner-efficiency or fuel-saving measurements. Record diesel use and electricity separately.

Corrected five-rack worked schedule

Assumptions for this example: mixing plus the source's short rest occupies 18 minutes; bulk fermentation is 45 minutes; dividing, preshaping, bench rest, moulding and tray loading occupy 27 minutes; final proof is about 60 minutes; oven slots start every 30 minutes with a 22-minute bake and eight minutes for complete turnaround. All times are elapsed from the first mix. Dough readiness overrides the clock.

The 18-minute source sequence includes yeast before a short rest, so it is not a strict flour-and-water-only autolyse. S-B01 uses a different mixing/rest sequence. The timetable is a scheduling exercise: select and validate one mixing program, measure its true occupied time, and shift the downstream times accordingly. Do not silently splice the two processes together.

Rack batch	Mix/rest	Bulk fermentation	Divide, rest, mould, load	Final proof / ready window	Oven in–out
1	00:00–00:18	00:18–01:03	01:03–01:30	01:30–02:30	02:30–02:52
2	00:30–00:48	00:48–01:33	01:33–02:00	02:00–03:00	03:00–03:22
3	01:00–01:18	01:18–02:03	02:03–02:30	02:30–03:30	03:30–03:52
4	01:30–01:48	01:48–02:33	02:33–03:00	03:00–04:00	04:00–04:22
5	02:00–02:18	02:18–03:03	03:03–03:30	03:30–04:30	04:30–04:52

This plans 400 pieces before bake rejects and samples. The last rack exits 4 hours 52 minutes after the first mix; cooling, packing, transport and cleaning continue afterwards. The 160 loaves/hour figure is steady-state slot capacity, not the average output from shift start. Arrange scoring and transfer within the measured handover allowance and readiness window; increase the interval if the actual work cannot fit.

The original schedule placed Rack 1 into proof at 01:30 but into the oven at 02:15 while calling that 60 minutes; it is 45 minutes. Its Batch 2 oven times also disagreed with the stated stagger. The corrected example keeps

every proof window at 60 minutes and every rack start 30 minutes apart. A fixed 45-minute stagger can still be chosen if it suits measured staffing and demand; it simply does not fill a 25-minute oven cycle continuously.

For a chosen interval C minutes, shift every stage of rack n by (n − 1) × C from rack 1, then check that no resource overlaps beyond its capacity. If a station cannot process 80 loaves within the chosen interval, lengthen C or qualify a specific improvement before starting a larger campaign.

Proofers, racks, cooling space and staffing

At 60-minute final proof and 30-minute arrivals, two 16-tray rack-equivalents are in proof simultaneously. Provide handover space and a contingency plan for a delayed oven; do not count an occupied passage as a proof position. The earlier inventory lists two 32-tray proofers, but cabinet tray slots do not prove wheeled oven racks fit. Check actual rack geometry, airflow, door opening and transfer time.

Plan at least two rack-equivalents in proof, one in the oven and one being loaded at peak overlap, plus separate cooling capacity. This can be achieved with the appropriate combination of racks and trays; verify the actual transfer method before purchasing. If bread cools on oven racks, those racks remain unavailable until safely unloaded. With an illustrative 45-minute cooling occupancy and 30-minute output interval, allow at least two rack-equivalents of cooling space, then verify actual cooling and packing time.

Role	Required handoff	Efficiency check
Mixer operator	Label batch/bin with mix end, dough temperature and bulk-ready time	Preweigh the next batch while the mixer is free; maintain enough chilled formula water without adding uncounted ice.
Divider/moulder operator or team	Deliver a correctly counted and identified rack with consistent piece weights	Time all weighing, bench rest and tray handling. The example's 27-minute block must be demonstrated, not assumed.
Oven/proof operator	Accept ready dough, score/load, log program and release baked rack	Control oven slots and proof readiness together; avoid simultaneous duties that leave the oven or proofer unattended.
Cooling/packing/dispatch support	Receive hot racks, cool, count, pack and record dispatch	Keep this work from blocking the oven operator or occupying the next empty rack. Assign a trained runner during peaks if needed.

These are responsibilities, not a guarantee that three people can perform all jobs at the target rate. Time the actual staffing arrangement. Measure units per paid labour hour and avoid moving a bottleneck from the mixer to packing.

Oven operating sequence and fuel discipline

1. Before mixing, confirm the day's orders, approved loaf weight, rack count, tray availability, water, ingredients, fuel, electrical auxiliaries and cooling space. Plan the last partial batch from demand; do not create unsold bread to improve apparent oven utilization.
2. Use the oven manufacturer's startup, combustion and exhaust checks. Start preheating from measured cold-start time so the oven is ready for the first proofed rack. Avoid a long empty hot hold caused by starting the oven far earlier than the dough.
3. Follow the selected baguette trial program. S-B01 starts around 230°C loading and 215–220°C baking for 20–25 minutes; the older guide proposed 240°C then 220°C. These are alternative starting trials, not universal settings. Map colour, crumb set and cooled weight across a full rack before fixing one program.
4. Apply early steam only through the approved system and dose established for this model. The old 15–20-second steam instruction cannot be transferred blindly. Do not add water to an oven floor or improvise a steam method. Score consistently using the S-B01 technique.

5. Keep loading and unloading orderly and doors open only as long as safe handling requires. Do not bypass guards or rush around hot racks to save seconds. Verify the recovery condition before admitting the next rack.
6. Use approved venting near the end of the bake to achieve the tested crust. Assess representative top, middle and bottom trays. Do not release underbaked centers merely because the exterior is brown.
7. Run compatible baguette batches together. Schedule lower-temperature products afterwards only when their validated time, temperature, steam and hygiene requirements fit. Residual heat is useful only if measured conditions complete the product's bake; it does not make fuel cost zero.
8. Transfer baked bread to protected ventilated cooling. Do not seal hot baguettes in airtight transport bins. Fully cool before any packaging that traps moisture. Treat par-baking or frozen-dough distribution as separate development projects with their own cooling, storage and finishing specifications.

When the timetable slips

Observation	Immediate operating response	Adjustment before the next campaign
Oven is ready but the next rack is not proofed	Wait for dough readiness; record the idle interval	Recheck upstream timing and proof conditions. Do not compensate with an unrecorded yeast increase.
Dough is ready but oven recovery is late	Apply the approved delay/hold procedure and identify the affected batch; withhold unacceptable overproofed product	Lengthen the rack interval or resolve recovery with the service technician. Do not overheat the next bake to catch up.
Dividing or loading falls behind	Slow or pause future mix starts while keeping existing batches controlled	Measure task times, revise crew allocation and only then revise batch cadence.
No empty rack, tray or cooling position	Stop releasing additional batches into the bottleneck	Recalculate equipment circulation and cooling space; separate dispatch from oven handoffs.
Bake colour differs across trays	Segregate the affected output and inspect the rack pattern	Check load spacing, approved airflow/rotation and maintenance; do not improvise unsafe rack manipulation.
Demand is below the planned output	Reduce or cancel unstarted batches	Use orders and sell-through to set campaign length; a full rack of unsold bread is not an efficiency gain.

Do not use an unqualified refrigerator as an emergency dough-retarder or extend proof indefinitely. Any hold/retard method must be tested with the actual formula and equipment.

Campaign record and scale-up decision

For each rack record: batch ID; input/usable dough mass; raw piece count and weights; dough temperature; mix/bulk/shape/proof timestamps; oven load/unload/next-ready times; steam/vent program; rack positions sampled; cooled weights; rejects; cooling/packing release; dispatch count; fuel and electrical meter readings where available; labour and stoppage reasons.

Calculate observed rack interval, good loaves/hour, diesel litres per 100 good loaves, electricity kWh per 100 good loaves, paid labour minutes per 100 good loaves, reject rate and sell-through. Include warm-up and shutdown in campaign energy costs; do not allocate only the minutes when bread is inside the oven.

Qualify the formula and one loaded rack first, then run two consecutive racks to expose overlaps, followed by the five-rack campaign only when proof space, handoffs, recovery and cooling remain stable. Repeat the

successful arrangement on three runs before treating its capacity as dependable. Replace every assumed time and loss in this example with measured results and issue the resulting station schedule.

Two-site production, staffing and energy

Site responsibilities and transport

Proposed allocation, subject to the actual installed locations: production handles dough, pastry, bread, buns, cakes, cooked fillings and validated frozen-mix preparation. Sales handles shawarma final cooking, burgers, fries, soft-serve dispensing and final assembly. Popsicle production belongs at whichever site has suitable sanitation, rapid hardening and enough power headroom; its unknown rating prevents a firm allocation.

Use separate clean food-safe crates for cooled bread, insulated chilled containers for perishable fillings/mix, and insulated frozen transport for bars. Raw meat travels sealed and segregated below ready-to-eat food. Record dispatch and receipt time, batch, quantity and product temperature. Chilled food must meet H01–H05; hot transport must maintain the house hot standard; frozen products should remain frozen at the operating target. Route duration and container capacity require a measured trial.

The receiver checks seal condition, label, batch identity and temperature before accepting. Reject unknown histories and investigate failed deliveries. Do not put newly hot food alongside chilled dairy, or close hot bread into airtight plastic. Return empty crates through a cleaning route, separate from new food.

Illustrative daily sequence

Window	Production site	Sales site	Main constraint
05:30–06:00	Check water, cold logs, battery state, diesel and scales	Receive opening stock/check cabinets when staff arrive	Early electrical work may be battery-fed
06:00–07:00	Mix and proof first bread/buns; stagger spiral and cooker	Set chilled garnish station; sanitize machine	Do not assume sunrise PV covers mixing
07:00–08:30	Bake first wave; cooling racks; start pie filling only with cooling space	Sell prior prepared fresh-delivery stock; small fry batches	Separate hot bread from chilled transport
08:30–10:30	Pie and snack batches; second bread wave if ordered	Morning snack sales; log demand and run safe replenishment	Fryer recovery and hand assembly
10:30–12:00	Dispatch controlled lunch components; prepare limited frozen stock	Prepare limited shawarma cone/pan portions	No oversized cone based only on oven capacity
12:00–14:00	Smaller second batch only against sell-through	Lunch service; stagger fryer, hot plate and machine loads	15 kW sales inverter and compressor surges
14:00–16:00	Mix/dessert prep during measured solar surplus if available	Frozen dessert and snack peak hypothesis to test	Hardening capacity and battery recharge
16:00–18:30	Targeted evening bread dispatch; finish early enough to cool	Evening wraps and frozen service	Use forecast, not automatic full batches
Close	Waste log, cleaning, cold checks and next-day quantities	Dispose per policy, machine cleaning and safe overnight cold loads	Protect cold storage battery reserve

Staffing and work design

Opening working model: one trained lead baker; one prep/finishing assistant with cold-chain responsibility; one sales cook who owns raw-meat controls; one service/cashier person. A relief/delivery role may be needed during peaks and cleaning. These are roles, not proof that four people can cover all hours or all 28 products. Observe labour minutes for the limited menu before adding items. The overhead allowance must be replaced by actual paid hours, including owner work.

Label each weighed kit before mixing. Maintain a batch sheet for dough, filling and finished units. Share O-B07 pastry across three pies/rolls; O-B18 chicken across wraps, pizza and loaded fries; O-B20/O-B28 sauces across defined products; O-B25 buns across burgers; O-FBASE1 across cups, sundaes, shakes and sandwiches. Sharing reduces ingredient variety but never permits unsafe carryover or cross-contamination.

Demand and replenishment method

Start with customer commitments and small trials. During the first three days, prepare at most one economical equipment-minimum batch per launch product, split between opening and later waves where fermentation allows. Record sales in hourly blocks. Days 4–14: forecast from comparable trading days plus paid orders; prepare about 60% of the forecast first, then release a second batch only when remaining sales and production lead time justify it. These percentages are management trial rules, not demand evidence.

Use a practical reorder trigger: expected sales during production plus delivery lead time, plus a small measured buffer, minus saleable stock already present. Do not replenish perishable stock close to closing simply to make a counter look full. Record unsold stock by cause: forecast error, poor quality, temperature failure, transport damage, portion rejects or customer preference.

Electrical loads and daily energy

Power in kW is simultaneous demand. Energy in kWh is power × operating hours × duty fraction. Thirty kWh nominal battery energy does not mean 30 kW continuous output. Two separate battery banks cannot share load unless specifically interconnected. The following site allocation, hours and duty fractions are illustrations, not measurements. It assumes one appliance at each stated rating.

Production illustrative electrical energy

Load	Rated/assumed kW	Hours on	Duty fraction	kWh/day
Oven auxiliaries	3	4	1	12.00
Spiral mixer	6.5	1	1	6.50
Moulder	1	0.5	1	0.50
Proofer	1	4	0.5	2.00
Mixing machine	2	0.5	1	1.00
Induction cooker	5	2	0.6	6.00
Slicer unknown assumed	0.5	0.5	1	0.25
Cold storage aggregate assumed	1.2	24	0.5	14.40

Load	Rated/assumed kW	Hours on	Duty fraction	kWh/day
Lights and small loads assumed	0.3	10	1	3.00
Total electrical energy				45.65

Sales illustrative electrical energy

Load	Rated/assumed kW	Hours on	Duty fraction	kWh/day
Ice cream machine A	4	8	0.45	14.40
Ice cream machine B	1.5	8	0.45	5.40
Shawarma	3.5	4	0.7	9.80
One induction fryer	5	4	0.6	12.00
Hot plate/cooker assumed allocation	5	2	0.6	6.00
Cold storage aggregate assumed	1	24	0.5	12.00
Lights and small loads assumed	0.3	10	1	3.00
Total electrical energy				62.60

The popsicle machine, professional pot heating, ventilation, pumps, water heating and any extra fryers/proofers are excluded because their ratings or quantities are unknown. They must be added before this is an operational load budget. Oven diesel is excluded from electrical kWh and must be measured separately. Startup and cleaning cycles can exceed the assumed running duty fractions.

Simultaneous load examples

Scenario	Illustrative total	Implication
Production oven + spiral + proofer + cold + small loads	3 + 6.5 + 1 + 1.2 + 0.3 = 12.0 kW	Below 20 kW nominal before surges/unknown loads
Add production cooker + mixer + moulder + assumed slicer	12 + 5 + 2 + 1 + 0.5 = 20.5 kW	Exceeds 20 kW; schedule these apart
Sales both ice machines + shawarma + fryer + cold + small loads	4 + 1.5 + 3.5 + 5 + 1 + 0.3 = 15.3 kW	Exceeds 15 kW even before startup surge
Add sales hot plate/cooker	15.3 + 5 = 20.3 kW	Not an acceptable all-on schedule
Sales A + shawarma + cold + lights	4 + 3.5 + 1 + 0.3 = 8.8 kW	One possible low-load service block, still verify phase/surge
Sales A + fryer + cold + lights	4 + 5 + 1 + 0.3 = 10.3 kW	Another block; do not add uncounted hot plate/popsicle load

Turning a machine off is not a food-safety control. Maintain refrigeration and any active cone's continuous cooking requirements. If the safe process requires simultaneous loads beyond the inverter's verified ability, reduce the menu or use a professionally designed supply/load-control change; do not alternate a partially

cooked cone into unsafe warm waiting. Check single-phase/three-phase limits and battery BMS discharge capability with an electrician before adopting any schedule.

Battery runtime illustration

At each site, assume 30 kWh nominal × 80% usable fraction × 90% conversion efficiency = 21.6 kWh delivered AC energy. These 80% and 90% figures are assumptions, not the battery specification. Without solar/grid input, that supports a constant 1 kW load for about 21.6 hours, 2 kW for 10.8 hours, 5 kW for 4.3 hours, or 10 kW for 2.2 hours, subject to discharge power limits. Actual cycling refrigerator loads should be measured over a full day.

A twelve-hour overnight critical load averaging 1 kW needs 12 kWh delivered AC, equivalent to about 16.7 kWh nominal under the same assumptions. Set the evening reserve using measured essential refrigeration plus margin and forecast morning demand. Do not consume the cold-chain reserve on optional fryers or a second frozen-dessert machine.

PV energy cannot be estimated from the inverter size. Use measured site generation or actual installed panel kWp × site-specific equivalent sun hours × system performance ratio. The illustrative daily electrical totals above are about 45.6 kWh at production and 62.6 kWh at sales before omitted loads. Neither is covered merely because a 30 kWh battery exists; daily replenishment and evening reserve matter.

Launch validation and daily controls

Phased launch

Phase	Range and work	Advance only when
Preparation week	Supplier quotes; oven tray map; five core recipe pilots; probes/scales; machine manuals; transport and power tests	Measured yield, cooking/cooling logs and electrical plan are workable
Week 1	O-P01, O-P02, O-P04, O-P05, O-P06, O-P07; small paid O-P03 taste test	Repeatable portions, positive contribution after waste, staff can replenish safely
Week 2	Add O-P13 plus either O-F01 or O-F05 after its dedicated commissioning	Recorded service drill, cold chain and machine cleaning pass
Weeks 3–4	Add selected O-P09/O-P11/O-P18 and a second frozen format; test standard vs large pop demand	Existing range is stable and added work does not create queues/waste
Later development	O-P12, O-P14–O-P17, O-P19–O-P20 and frozen sandwiches/coated bars as justified	Measured incremental demand and margin justify equipment, labour and storage time

Pilot approval record

Test	Record	Proposed acceptance rule
Formula	Ingredient brand, exact mass, water, dough temperature	No unrecorded substitutions; all ingredients traceable
Yield	Raw input, usable component mass, units, cooked mass, rejects	Three successive batches with explained variance; aim within 5% of adopted target

Test	Record	Proposed acceptance rule
Portion	Ten random pieces per batch	Aim within ±3% of target average; no underweight labelled product
Sensory	Blind taste from intended customers, not only staff	At least 20 paid/structured test responses per candidate; agree threshold before testing
Holding quality	Check at service, 1 hour and 2 hours; longer only under safe control	No significant loss of intended texture at the planned sale window
Safety	Cooking, cooling, storage and transport records	All adopted critical limits met; any failure blocks sale
Cost	Actual delivered ingredients, oil and packaging	Positive contribution with measured waste; overhead coverage assessed
Service	Timed assembly and queue length	Meet practical service target without bypassing hygiene
Market	Units produced, sold, samples and repeats	Track 7–14 days before claiming repeat demand
Frozen bars	Mould ml, fill g, density, hardening time, seal/stick integrity	No unverified volume claim; stable texture and intact packaging

The numerical business acceptance targets are proposed management rules, not regulatory standards. Adjust after baseline measurements, but do not lower food-safety limits simply to pass a commercial test. Taste tests do not establish microbial safety. Quantities in this report are recipes for controlled trials until the team demonstrates repeatability.

Allergens and cleaning

Create a matrix from actual supplier labels. Wheat is present in breads, pastry, doughnuts, wrappers, buns, cake and cookies; milk in enriched doughs, creamy fillings, cheese, many sauces and frozen dairy; egg in egg wash, cakes, egg rolls and many commercial mayonnaise products; fish in O-P06; sesame in O-B35/tahini and optional bun topping; peanuts in O-F02 and O-F08; mustard in O-B28. Margarine, sausage meat, coatings, chocolate and stabilizer blends can introduce soy, milk or other allergens. Supplier labels override this preliminary list.

Use dedicated fish frying equipment or a documented separation plan; shared oil can transfer allergens and flavours to doughnuts and fries. Never label a product allergen-free merely because its formula omits the allergen. Separate raw meat tools, cooked-food tools, bakery utensils and frozen-dessert tools. Clean and sanitize according to the chemical label; do not mix chemicals or invent sanitizer concentrations.

Power interruption and cold-chain response

1. Record outage start, cold cabinet temperatures and battery state immediately. Keep doors shut and reserve power for food-critical refrigeration.
2. Pause new high-risk preparation if safe cooling or storage cannot be maintained. Use temperature-verified backup storage/transport if available.
3. Record product temperatures and duration, not just the freezer display. Quarantine food with an unknown history; do not assume refreezing restores safety.
4. Restart machines only after safe equipment checks and the manufacturer’s sanitation/recovery procedure. Discard questionable hopper mix and record the loss.

5. Review the event, reserve setting and load schedule before resuming the full menu.

Daily opening and closing sheet

Opening	During service	Closing
Probe/scale check; chilled/frozen logs; potable water; handwash; battery reserve; diesel; clean equipment	Batch IDs; cook/cool checks; hourly stock; portions; machine temperatures; delivery acceptance; actual load peaks	Unsold stock by cause; safe disposal; cash/sales reconciliation; machine strip-clean; freezer log; reserve confirmation; next-day batch plan

Keep purchase invoices, ingredient lot numbers, preparation dates, temperatures and sales batch links. Avoid statements such as “safe for three days” or “high margin” on staff sheets unless the process and numbers support them. Assign one person to authorize formula changes and update the weighed batch sheet; improvisation should not silently become the standard.

Additional commercial ideas retained from September

Test breakfast BHB (beignets, haricots and bouillie), school/office packs, pain omelette, sandwich preorders, afternoon brioche and seasonal fruit lines. Beans and porridge were concepts, not complete costed formulas in the gathered recipe cards; develop separate controlled recipes before adding them. Urban menu examples from Douala and Yaoundé show availability, not Nkambe demand. Use customer interviews, paid tastings and actual sell-through.

Use paper/ventilated protection for crisp bread and fully cooled sealed packs for soft sliced bread. Compare slice thickness, loaf weight, bag strength, closure, food-contact documentation, printed label space and supplier minimum orders before ordering packaging. The older 10,000–20,000 printed-bag proposal is a purchasing idea; prove the design and turnover with a small run first.

Track contribution per labour minute and bottleneck hour alongside margin per unit. Separate direct costs, overhead and working stock. Do not treat all produced units as sold. Measure food waste, refunds, cold-chain rejects, energy use and repeat purchases. Set replenishment from real time-of-day sales; do not produce every recipe daily.

The September workbook uses low/base/high stress cases: demand multipliers 0.6/1/1.3; ingredient cost 1.2/1/0.95; energy cost 1.3/1/0.95; energy availability 0.6/0.85/0.98; rejects 12%/5%/2%. These are management scenarios, not observed rates. Keep them separate from the October price illustration.

Its capacity example assumes 16 trays, 5 loaves/tray, 30-minute complete oven cycle, 32 working proof trays, 60-minute proof, 16.513 kg usable mixer dough, 330.26 g raw pieces, 30-minute mixer occupancy, 100 units/hour dividing and 100 cooling/packing. The resulting limiting rate is 100 units/hour before availability; a 6-hour shift at 85% availability gives 510 units of illustrative capacity, not demand. Confirm every input through a time study.

Read the workbook’s Formulas, Yields and Batch plan sheets using the S-series suffixes. The source workbook was inspected, including its formulas and unfilled input cells; it was not recalculated or represented as a completed financial forecast. Use the handbook’s cost method to populate verified quotations and actual operating inputs.

13. Quick-reference records and troubleshooting

Daily batch record

Field	Entry
Date / station / operator / recipe version	_____
Batch ID / ingredient lots / actual weights	_____
Mixer or treatment load / equipment ID	_____
Dough temperature or milk/juice pH and °Brix	_____
Proof / cook / treatment readings and times	_____
Cooling start, intermediate and final readings	_____
Raw, cooked and cooled weights / count / leftovers	_____
Packaging lot / allergen label / storage	_____
Dispatch and receipt temperatures / quantity	_____
Release decision / deviation / responsible person	_____

Common defects

Observation	Check first	Next controlled adjustment
Dense bread	Flour strength, yeast, dough development, proof	Correct one measured cause; avoid extra yeast without diagnosis.
Greasy brioche	Butter condition, dough/proofer temperature	Chill and incorporate progressively; do not hide with flour.
Soggy pies or leaking rolls	Filling water, cooling, sealing, pack condensation	Reduce free liquid, weigh filling and verify cooling/ventilation.
Greasy fried snacks	Oil recovery, load, piece thickness, proof	Reduce load and confirm temperature before changing formula.
Icy or soft frozen bars	Sugar/solids, freezing rate, temperature cycling	Measure Brix and storage history; reformulate only in a new trial.
Yogurt whey separation	Milk solids, heat treatment, acidification, shear	Review logs; stabilizer is not a substitute for process control.
Juice separation or browning	Fruit lot, oxygen, formulation, package	Trial smaller sealed portions; distinguish quality from safety.
Gas, swelling, mould, unusual odour	Spoilage or lost process control	Isolate and withhold; do not taste or reprocess for sale.

Working glossary

Baker's percentage: ingredient mass relative to flour. °Brix: refractive soluble-solids estimate, not a safety test. DDT: desired dough temperature. Overrun: percentage volume increase from incorporated air. Pasteurization: validated pathogen-reduction treatment, not sterilization. Proof: final dough fermentation before baking. Syneresis: whey separation. Usable yield: measured output available for the next step. Contribution: revenue minus the defined variable costs; it is not net profit.

14. Sources and edition notes

Source documents are identified by SRC numbers in the separate editorial record. Recipes are inherited trial formulations, not laboratory-tested results. October source links embedded in retained text were supplied with that edition; September references below retain their original access date of 13 September 2026. The preservation references N01–N09 were checked for this edition on 3 October 2026. An inherited citation does not certify every adapted quantity or local market claim.

Preservation and process references checked for this edition

N01 — FDA: What You Need to Know About Juice Safety. Untreated juice hazards and whole-fruit hygiene.

N02 — FDA: Juice HACCP Regulator Training and Juice HACCP questions and answers. Product-specific pathogen reduction; United States framework used as a technical reference only.

N03 — National Center for Home Food Preservation: Freezing and Freezing citrus fruit and juice. Freezing principles and packaging; domestic guidance is not a commercial shelf-life validation.

N04 — University of Guelph: Pasteurization of Mix. Batch ice-cream mix treatment reference, 69°C for 30 minutes; actual equipment/process qualification required.

N05 — Yogourmet: Original Yogurt Starter. Current supplier instructions; verify the product and sachet supplied locally.

N06 — Yogurt Made Simple, Washington State University Extension. Milk treatment, culture and fermentation principles; home-use holding periods are not transferred to this commercial booklet.

N07 — Oregon State University Extension: Yogurt. Covered refrigeration, mould disposal and texture changes after freezing.

N08 — UK government: Cooking safely in your business. Cooking/reheating reference of 75°C for 30 seconds and hot holding at 63°C. These are house operational choices, not asserted Cameroonian statutory limits.

N09 — FDA: Refrigerator Thermometers—Cold Facts about Food Safety. Refrigeration, freezing and time outside cold storage.

S01 — King Arthur Baking

Brioche: professional formula. Publication: undated. Accessed 2026-09-13.

Supports: Develop gluten before progressive butter addition; cold handling.

Limits: Technical basis only; published total must be independently summed. Readable page or indexed primary-source extract reviewed.

S02 — King Arthur Baking

Essential baguette tools and techniques. Publication: 2021-03-26. Accessed 2026-09-13.

Supports: Gentle shaping, scoring, early steam and subsequent venting.

Limits: Not a rotary-oven operating manual. Readable page or indexed primary-source extract reviewed.

S03 — Lesaffre

Saf-Instant Gold. Publication: undated. Accessed 2026-09-13.

Supports: Sweet-dough yeast selection; progressive brioche mixing; yeast donut development.

Limits: Regional product specifications vary; confirm actual package. Readable page or indexed primary-source extract reviewed.

S04 — King Arthur Baking School

Bread Baking Basics workbook. Publication: undated. Accessed 2026-09-13.

Supports: Pain de mie method and pan-based production.

Limits: Domestic classroom scale; our formula and rotary settings are adaptations. Readable page or indexed primary-source extract reviewed.

S05 — Precious Core / Precious Nkeih

Kumba Bread Recipe. Publication: 2016-03-28; updated 2017-04-26. Accessed 2026-09-13.

Supports: First-person Cameroonian product identity and fine-crumb loaf familiarity.

Limits: Culinary source, not sales survey or industrial validation. Readable page or indexed primary-source extract reviewed.

S06 — Precious Core / Precious Nkeih

Beignets Souffles. Publication: 2018-02-23; updated 2022-11-26. Accessed 2026-09-13.

Supports: Cameroonian bakery-style egg/milk cake-like fritter uses baking powder.

Limits: Names vary locally; our weights and fryer temperatures are original trial settings. Readable page or indexed primary-source extract reviewed.

S07 — King Arthur Baking

Banana Bread. Publication: undated. Accessed 2026-09-13.

Supports: Chemical leavening and limited mixing in banana quick bread.

Limits: Different original formulation here; not a commercial shelf-life study. Readable page or indexed primary-source extract reviewed.

S08 — King Arthur Baking School

Bread Principles and Practice workbook. Publication: filename November 2024. Accessed 2026-09-13.

Supports: Sheeting, filling and rolling cinnamon dough.

Limits: Our weights and production scale are independent. Readable page or indexed primary-source extract reviewed.

S09 — H. Douglas Goff / University of Guelph

Pasteurization of Mix. Publication: undated. Accessed 2026-09-13.

Supports: Ice-cream mix batch reference: 69°C for 30 minutes; chilled cooling.

Limits: Ontario reference, not Cameroon law; vessel cold-point validation required. Readable page or indexed primary-source extract reviewed.

S10 — Tetra Pak

Ice cream: Dairy Processing Handbook. Publication: undated. Accessed 2026-09-13.

Supports: Mixing, heat treatment, homogenization, ageing, freezing and hardening are distinct operations.

Limits: Industrial guidance does not establish capability of the user’s pot or freezer. Readable page or indexed primary-source extract reviewed.

S11 — H. Douglas Goff / University of Guelph

Ice Cream Mix General Composition. Publication: undated. Accessed 2026-09-13.

Supports: Fat, milk-solids and sugar balancing framework.

Limits: General composition, not a universal legal identity for Cameroon. Readable page or indexed primary-source extract reviewed.

S12 — H. Douglas Goff / University of Guelph

Overrun calculations. Publication: undated. Accessed 2026-09-13.

Supports: Volume expansion and equal-volume weight method.

Limits: Actual mix density and draw overrun must be measured. Readable page or indexed primary-source extract reviewed.

S13 — H. Douglas Goff / University of Guelph

Hardening. Publication: undated. Accessed 2026-09-13.

Supports: Rapid hardening and frozen storage protect crystal size.

Limits: No guarantee a storage freezer can harden a production load. Readable page or indexed primary-source extract reviewed.

S14 — Louis François

STAB 2000. Publication: undated. Accessed 2026-09-13.

Supports: Named ice-cream stabilizer/emulsifier; supplier specifies 0.2–0.5%, differentiated by fat level.

Limits: Obtain current full ingredients, allergens, legal-use and hydration specification. Readable page or indexed primary-source extract reviewed.

S15 — Louis François

Super Neutrose. Publication: undated. Accessed 2026-09-13.

Supports: Named sorbet stabilizer; 0.2–0.5%; sugar premix, hydration and hot use.

Limits: Not interchangeable gram-for-gram with starch or pure gums. Readable page or indexed primary-source extract reviewed.

S16 — Callebaut

Gelato Dipping. Publication: undated. Accessed 2026-09-13.

Supports: Special ice-chocolate products can be used without conventional tempering.

Limits: This exception does not apply to ordinary couverture. Readable page or indexed primary-source extract reviewed.

S17 — Callebaut

Ice Chocolate White, ICE-50-WNV-552. Publication: undated. Accessed 2026-09-13.

Supports: Product-specific dipping guidance: 35–40°C.

Limits: Page has inconsistent dietary tags: use actual label, not web badges, for milk/allergens. Readable page or indexed primary-source extract reviewed.

S18 — Tetra Pak

Fermented Milk Products. Publication: undated. Accessed 2026-09-13.

Supports: Yogurt heat treatment, distinct set/stirred/drinking methods and pH-controlled cooling.

Limits: Industrial process; actual small-batch cooling and incubation require mapping. Readable page or indexed primary-source extract reviewed.

S19 — Yogourmet / Lallemand

Original Yogurt Starter. Publication: undated. Accessed 2026-09-13.

Supports: Culture species and 42–44°C incubation guidance.

Limits: Consumer trial culture, not approval for industrial scale or a shelf-life guarantee. Readable page or indexed primary-source extract reviewed.

S20 — Yogourmet / Lallemand

Sachet weight. Publication: undated. Accessed 2026-09-13.

Supports: Current 3 g sachet replaces historical 5 g; one sachet per litre milk.

Limits: Check purchased version; never copy old 5 g instructions onto concentrated 3 g packs. Readable page or indexed primary-source extract reviewed.

S21 — Taylor Company

Operator manuals and specifications. Publication: undated. Accessed 2026-09-13.

Supports: Sanitation, loading and cleaning are model-specific.

Limits: Example manufacturer only; user machine brand and model unknown. Readable page or indexed primary-source extract reviewed.

S22 — FoodSafety.gov

Safe Minimum Internal Temperatures. Publication: undated. Accessed 2026-09-13.

Supports: Ground meat/egg dishes 71°C; poultry and reheated leftovers 74°C.

Limits: US technical benchmark, not a statement of local statutory limits. Readable page or indexed primary-source extract reviewed.

S23 — US FDA

Food Code 2022. Publication: 2022 edition. Accessed 2026-09-13.

Supports: Sections 3-501.14/.15 cooling; time-temperature and hygiene framework.

Limits: Model food code, not Cameroon law. Readable page or indexed primary-source extract reviewed.

S24 — US FDA

Juice HACCP: Questions and Answers. Publication: 2001. Accessed 2026-09-13.

Supports: Fruit juice treatment must control relevant pathogens; acidity/freezing alone is not treatment.

Limits: US framework; no generic kill time adopted for every fruit recipe. Readable page or indexed primary-source extract reviewed.

S25 — ANOR Cameroon

Sensibilisation des boulangers. Publication: 2023-10-12 event. Accessed 2026-09-13.

Supports: ANOR identifies NC 213:2013 bread and NC 30:2001 food hygiene as mandatory.

Limits: Full current standards and amendments must be obtained from ANOR. Readable page or indexed primary-source extract reviewed.

S26 — ANOR Cameroon

List of mandatory standards. Publication: undated legacy list. Accessed 2026-09-13.

Supports: Lists dairy terminology and yogurt standards including NC 126/127.

Limits: Historical list is not confirmation of all current editions or requirements. Readable page or indexed primary-source extract reviewed.

S27 — MINSANTE DPS

Sous-direction de l’alimentation et de la nutrition. Publication: undated. Accessed 2026-09-13.

Supports: AMC framework for food under Law 2018/020; authority describes three-year authorization.

Limits: Obtain current application scope, inspection and product requirements locally. Indexed official text; direct open returned error/limited text.

S28 — Presidency of Cameroon

Law 2018/020 on Food Safety. Publication: 2018-12-11. Accessed 2026-09-13.

Supports: Existence and date of food-safety framework law.

Limits: Not a complete legal compliance opinion. Indexed official text; direct open returned error/limited text.

S29 — Nice Cream City / Velko menu

Yaoundé catalogue. Publication: undated live menu. Accessed 2026-09-13.

Supports: Sardine omelette and BHB each listed at 1,000 XAF; shawarma, kossam and ice-cream bundles.

Limits: Availability and menu price only; not Nkambe transaction evidence. Readable page or indexed primary-source extract reviewed.

S30 — Chic Meal

Douala Bali menu. Publication: undated live menu. Accessed 2026-09-13.

Supports: Burgers are an identified menu category in Douala.

Limits: Individual burger prices not verified in the readable page this run. Readable page or indexed primary-source extract reviewed.

S31 — Complexe Santa Lucia

Homepage and food menu. Publication: live page; 2026-09-13 menu. Accessed 2026-09-13.

Supports: Bakery/pastry/ice cream/shawarma services; beignets and beans; packaged beverage comparison.

Limits: Dynamic offers differ by branch; template remnants reduce reliability of generic claims. Readable page or indexed primary-source extract reviewed.

S32 — MINMIDT Cameroon

2019 joint inspection schedule. Publication: 2019. Accessed 2026-09-13.

Supports: Historical listing of bread businesses in Nkambe.

Limits: Not evidence these businesses still operate or of current sales/competition. Readable page or indexed primary-source extract reviewed.

S33 — INS Cameroon

Consumer prices, December 2025. Publication: posted 2026-06-16; period 2025. Accessed 2026-09-13.

Supports: 2025 national inflation 3.4%; food 7.0%.

Limits: National historical index, not Nkambe ingredient quotations or household budgets. Readable page or indexed primary-source extract reviewed.

S34 — WHO

Five Keys to Safer Food. Publication: undated. Accessed 2026-09-13.

Supports: Cleaning, separation, thorough cooking, temperature control, safe water/materials.

Limits: General training principles, not product-specific validation. Readable page or indexed primary-source extract reviewed.

Audit of accessible user documents

- Commercial_Baguette_Production_Guide.pdf: reviewed, 2 pages. User-provided draft, not a validation source.
- Staggered_Baguette_Production_Schedule.pdf: reviewed, 2 pages. User-provided draft, not a validation source.
- Cameroon_Bakery_Empire_Master_Guide.pdf: reviewed, 10 pages. User-provided draft, not a validation source.
- Bakery_Empire_Master_Operations_Manual.pdf: reviewed, 4 pages. User-provided draft, not a validation source.
- Bakery_Empire_Master_Operations_Manual (1).pdf: reviewed, 5 pages. User-provided draft, not a validation source.
- Bakery_Empire_Master_Operations_Manual (2).pdf: reviewed, 14 pages. User-provided draft, not a validation source.

Deliberately excluded or restricted evidence

Bamenda internship upload: not current primary data. Urban menus: availability, not transaction volumes. Santa Lucia indexed yogurt promotion did not appear in the opened current page: no yogurt price adopted. Pizzaland price from the former edition was not reverified: no current pizza price asserted. Old claims of six-month biscuit life, 300% basket growth, guaranteed margins and poor competitor hygiene are unsupported and removed.

Remaining research gaps

Current Nkambe ingredient quotes, household budgets, paid product demand, competition, utilities, road/route conditions, fruit seasons, exact machine manuals, current full local standards, license scope and all laboratory/factory trial evidence remain unresolved.